

Instruments Apart: Monetary Policy and Consumption Inequality across U.S. Regions and Income Groups*

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Abstract

This paper examines how three monetary policy instruments—the federal funds rate, forward guidance, and large-scale asset purchases—differ in their distributional effects on U.S. household consumption. Using Consumer Expenditure Survey data for 1992–2019, orthogonal high-frequency shocks, and cell-level local projections, this paper finds that the increase in non-housing consumption following expansionary asset purchases is larger for the lowest income quintile than for the highest, with this differential significant in the Northeast and statistically distinguishable from other regions. This difference is absent in the first year, emerges in the second, and persists into the third. Across housing contracts, conventional interest rate policy produces a larger consumption decline among lower-income renters than among mortgagors with the same income, whereas asset purchases produce a larger decline among mortgagors, consistent with a cash flow channel through refinancing rather than a dominant employment channel. The Northeast’s prominence is consistent with housing’s greater weight in local household budgets. Direct tests show that the three instruments are not interchangeable, and forward guidance has no consistent distributional direction. The choice of instrument itself carries distributional consequences.

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*Queensland University of Technology (PhD, 2026). Email: jiayi.yang.econ@outlook.com. This paper draws on Chapter 6 of my doctoral dissertation Yang (2026). The research question and data source are unchanged, but the empirical methodology has been substantially revised, and the results reported here therefore differ from those in the dissertation. I am grateful to Pedro Gomis-Porqueras, Jayanta Sarkar, and Ruchith Dissanayake for their guidance throughout my doctoral studies, and to Jorge Miranda-Pinto for detailed comments on this paper. I also thank Frank Mathmann and Stephen Thiele, and two anonymous examiners of my dissertation, for helpful comments on earlier versions of this work. This research was supported by QUT Postgraduate Research Award (International) and QUT HDR Tuition Fee Sponsorship. All remaining errors are my own.

1 Introduction

After the 2008 financial crisis, the Federal Reserve lowered the federal funds rate (FFR) to its effective lower bound, leaving conventional interest rate policy with no further room to cut, and turned to two unconventional instruments on a large scale: large-scale asset purchases (LSAP) and forward guidance (FG). Three rounds of quantitative easing (QE) expanded the Federal Reserve's balance sheet from under one trillion dollars before the crisis to approximately 4.5 trillion dollars by 2015. This intervention took place against a backdrop of decades of rising income and consumption inequality in the United States (Piketty and Saez, 2003), so its distributional consequences are not merely a technical matter but bear on the social legitimacy of central bank action. One view of quantitative easing holds that it raises equity and housing prices and primarily benefits wealthy households that hold financial assets (Domanski et al., 2016); another emphasizes that it compresses long-term interest rates and supports investment and employment, favoring lower-income households that depend on wages (Bivens, 2015). Both positions have support, but they rest on different measures and frameworks and rarely engage each other directly. Unconventional instruments are now a standard part of the central bank toolkit, and if different instruments have different distributional consequences, the choice of instrument itself carries distributional implications.

A substantial body of literature addresses this question and can be grouped into several strands. One strand examines the distributional consequences of monetary policy empirically, finding at both cross-country and single-country levels that contractionary policy generally widens inequality (Furceri et al., 2018; Sen and Sensarma, 2025), and documenting differences along dimensions such as race (Albert and Gómez-Fernández, 2024) and the composition of household income and wealth (Ampudia et al., 2018). A second strand identifies monetary policy shocks using high-frequency data, extending from two-factor to multi-factor decompositions and providing the basis for separating distinct instruments (Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018b). On the theoretical side, Heterogeneous Agent New Keynesian (HANK) models show that the distributional effects of monetary policy in general equilibrium operate primarily through labor income and aggregate demand, mattering especially for liquidity-constrained lower-income households, while direct asset price effects fall more on wealthy households (Kaplan et al., 2018; Auclert, 2019). Recent empirical work also provides evidence on the role of these channels. Del Canto et al. (2025) measure the responses of U.S. households to oil and monetary policy shocks in terms of monetized welfare changes and find that expansionary monetary policy is progressive along education and life-cycle dimensions: rate cuts raise equity, housing, and bond prices, which hurts middle-aged, highly educated households that are still accumulating these assets, while less educated households are compensated by higher labor income, consistent with the implications of the employment channel. By focusing on differences across regions, this paper also builds on work

examining the regional effects of monetary policy (Carlino and DeFina, 1998; Beraja et al., 2019).

Most of these studies, however, combine the full range of Federal Reserve actions into a single composite shock. Coibion et al. (2017), using Consumer Expenditure Survey (CEX) data since 1980, find that contractionary monetary policy persistently widens consumption and income inequality. The single Romer and Romer (2004) shock they use, however, cannot distinguish the separate roles of LSAP, FG, and conventional interest rate policy, whose transmission channels differ considerably: LSAP primarily affects long-term asset prices and borrowing costs, while FG operates on expectations of future interest rates, both distinct from adjustments to the overnight rate. Beyond the choice of instrument, U.S. Census regions differ markedly in industrial structure, housing markets, and labor markets, which should produce different transmission across regions; few distributional studies examine this systematically, and consumption-side evidence covering the full quantitative easing cycle remains limited.

This paper addresses these gaps using CEX Interview survey data from 1992 to 2019. The sample period spans the pre-crisis years, the zero interest rate period, and the subsequent policy normalization, covering the full quantitative easing cycle. The CEX is a standard data source for studying consumption inequality in the United States (Attanasio and Pistaferri, 2016; Aguiar and Bils, 2015). I construct two grouped panels. The first crosses four Census regions with five income quintiles, yielding 20 cells, and is used to identify where distributional effects appear and whether they are general. The second crosses three housing tenure types with five income quintiles, yielding 15 cells, and is used to identify the transmission channel. The dependent variable is non-housing consumption, which is total consumption net of housing expenditure. This excludes the direct effects of interest rates on housing expenditure through rents and refinancing-induced changes in monthly payments, thereby avoiding confounding changes in consumption demand with changes in housing costs. Total consumption and housing consumption are reported for comparison, and the food expenditure ratio provides supplementary evidence on the composition of expenditure.

Identifying the shocks is central, and measuring unconventional policy has long been difficult. The narrative shocks of Romer and Romer (2004) barely vary during the zero interest rate period; the shadow rate of Wu and Xia (2016) compresses the overall policy stance into a single number that masks differences across instruments; event studies of individual announcements (Krishnamurthy and Vissing-Jorgensen, 2011) do not yield a continuous series. I use the high-frequency identification data of Swanson (2021), which extracts three mutually orthogonal factors from a high-frequency window around Federal Open Market Committee (FOMC) announcements, corresponding to FFR, FG, and LSAP. The high-frequency window ensures that these shocks reflect the policy news itself rather than other information released at the same time, and the orthogonality allows consumption responses to be attributed to specific instruments. For estimation, I apply the local projections method of Jordà (2005) to the time series formed by each cell separately; this

method is well suited to externally identified shock variables and facilitates comparison across cells. To improve the readability of the impulse response figures, most plots use the smoothed local projections of Barnichon and Brownlees (2019), while all statistical inference relies on the standard local projections.

The findings fall into three groups. The first group of findings concerns LSAP. Under a within-region relative income ranking, the increase in non-housing consumption following expansionary asset purchases is larger for the lowest income group ($Q1$) than for the highest ($Q5$). This paper refers to this direction, in which lower-income groups benefit more, as progressive. Among the four regions, only the Northeast passes the joint test under this specification. The point estimates for the other three regions are of the opposite sign, but none is statistically significant. The difference between $Q1$ and $Q5$ is indistinguishable from zero during the first year after the shock, emerges in the second year, and persists through the third.

The Northeast is the only region with a significant difference, and a direct test shows that its distributional effect of LSAP is statistically distinguishable from those in the remaining three regions: the pairwise comparisons between the Northeast and each of the other regions all reject equality at the 5% level. The food expenditure ratio provides corroborating evidence from the expenditure composition side: following expansionary LSAP, the food expenditure ratio for $Q1$ in the Northeast falls more, and this direction holds over the large majority of horizons, consistent with an improvement in expenditure composition under Engel's Law.

The second group of findings is that the three instruments do not operate in the same way. When the two instruments are compared directly within the same region and consumption measure, the progressive effect of asset purchases in the Northeast differs significantly from that of forward guidance in the second year, while its difference from conventional interest rate policy is significant at the 10% level. The significant results for forward guidance are scattered across individual regions and periods and show no consistent direction; when the sample is split by tenure, the tenure comparisons within income groups are also all insignificant. The results for conventional interest rate policy are concentrated in housing consumption: following a contractionary shock, the decline in housing consumption for the lowest income group is smaller than that for the highest, with this pattern appearing in the Northeast in the first year and in the South in the third year rather than holding generally across all four regions; total consumption in the South shows the same pattern in the third year.

The third group of findings comes from the estimation that crosses housing contract type with income, which helps explain why the first two sets of results take the form they do. The three instruments reach households through different contracts: short-term rates do not change the monthly payment on existing fixed-rate loans but do pass through to rents, while asset purchases open or close the refinancing channel for existing borrowers by compressing long-term rates and

mortgage spreads. Consistent with this, within the lower income groups, renters show a larger consumption decline than mortgagors at the same income level under conventional interest rate policy, the direction reverses under asset purchases, and there is no difference between the two under forward guidance. In the lowest income group, this reversal is itself statistically significant in a direct test: the difference between renters and mortgagors differs significantly across the two instruments. The income gradient under asset purchases appears within the mortgagor group but not within the renter group, while the gradient in the food expenditure ratio within the same group runs in the same direction and is significant in the full-horizon test. By the identification logic of this paper, the renter group is the only source of evidence for the employment channel, so the evidence points instead to a cash flow channel operating through refinancing. This is consistent with Opazo (2023), who finds that mortgagor households adjust expenditure more than outright owners.

This gradient is concentrated among low-income borrowers, who face the greatest difficulty in meeting the refinancing thresholds; the gradient is close to zero in the first year, widens window by window thereafter, and is most pronounced in the third year, consistent with the time required to complete refinancing and for its budgetary effects to accumulate gradually. The Northeast stands out not because it has a higher share of mortgagor households. The result is consistent with housing carrying the greatest weight in local household budgets there, but this explanation is inferred from the two separate splits by region and tenure. Other studies emphasize the income effects of reduced unemployment (Chang and Schorfheide, 2024; Lee, 2026); this paper cannot rule out that channel, but finds no evidence on the consumption side that it is dominant.

I assess robustness through several checks: replacing the income classification variable, excluding recession and extreme shock quarters, deflating the consumption variables by the price level, and comparing standard with smoothed local projections. Using the national income percentile, the Northeast remains significant in the same time windows as in the baseline, with a slightly larger gap, while the delayed onset of the effect is unchanged. Excluding recession quarters narrows the gap, which is no longer significant, whereas excluding extreme shock quarters widens the gap, which remains significant. The excluded quarters are precisely those with the largest shock variance, so statistically distinguishing this effect requires a sample that contains policy shocks of sufficient magnitude. The results along the tenure dimension likewise depend on these quarters: in both reduced samples, the gradient within the mortgagor group is no longer significant, while after excluding extreme shock quarters, a gradient instead appears within the renter group, which is the pattern required by the employment channel. Details are provided in Section 6.

The paper makes three contributions. First, drawing on the orthogonal decomposition of Swanson (2021), this paper compares the distributional effects of conventional interest rate policy, forward guidance, and asset purchases separately using a unified consumption panel, rather than

combining these policies into a single shock as most of the literature does (Coibion et al., 2017). Even when the sample covers the zero lower bound (ZLB) and quantitative easing periods, evidence based on a single shock cannot answer which instrument the distributional consequences come from. Chang and Schorfheide (2024) find that expansionary shocks raise consumption inequality slightly, but the credible interval spans both directions. Del Canto et al. (2025) find that monetary expansion is progressive in welfare terms, but their shock is measured by unexpected movements in the interest rate path, which cannot distinguish which instrument the progressivity arises from. The decomposition in this paper provides one explanation: in the consumption data, the progressive direction appears only in the asset purchase component and is concentrated in the Northeast, the conventional rate component produces no income gradient in non-housing consumption, and forward guidance has no consistent direction. Within the same income group, conventional interest rate policy and asset purchases affect different households in opposite directions, and a direct test confirms that this difference is statistically significant. Combined into a single shock, these differences dilute or offset one another; since central banks often use several instruments at once, these differences are obscured in aggregate analysis and therefore carry policy implications. Model-based work has already addressed this issue at the instrument level: Lee (2026) decomposes quantitative easing and forward guidance in an estimated heterogeneous-agent New Keynesian model and notes that empirical conclusions on the distributional effects of unconventional policy are not consistent. This paper provides instrument-level empirical evidence from consumption data, with two points of difference: forward guidance has no consistent distributional direction here, and the gains to lower-income households operate mainly through refinancing cash flow, with no indication that the employment channel is dominant.

Second, most of these studies summarize the entire distribution using a statistic such as the Gini coefficient, a quantile ratio, or the top 10% share. Rather than compressing inequality into a single statistic, this paper directly compares impulse responses across region-by-income and tenure-by-income cells. Combining the four measures of non-housing consumption, total consumption, housing consumption, and the food expenditure ratio with income quintiles defined by within-region relative income, this design reveals both which households are affected and through which type of consumption, while separating nominal changes from changes in expenditure composition.

Third, by splitting the sample by housing contract, this paper shows that the differences in distributional effects across the three instruments are consistent with the types of contracts through which they reach households, and provides evidence that asset purchases operate through the cash flow of mortgagor households. Di Maggio et al. (2017) use the reset of adjustable-rate mortgages to show that lower monthly payments substantially increase durable goods purchases, with a higher marginal propensity to consume among low-income borrowers with low housing wealth. Cloyne et al. (2020) find that the aggregate consumption response to conventional policy is driven mainly

by mortgagor households. This paper finds evidence of a cash flow channel operating through mortgagor households under large-scale asset purchases. Under conventional interest rate policy, however, comparing households at the same income level within the lower income groups shows a larger response among renters, which differs from the finding in Cloyne et al. (2020). This paper also provides an explanation for the regional differences in the distributional effects.

The rest of the paper proceeds as follows. Section 2 introduces the data sources and panel construction. Section 3 describes the empirical strategy. Section 4 reports the main results from the regional estimation. Section 5 uses the split by housing contract to explain why the distributional effects of the three instruments differ and why the Northeast stands out. Section 6 presents the robustness checks. Section 7 concludes and discusses policy implications and future directions.

2 Data

2.1 CEX Interview Survey

The Bureau of Labor Statistics (BLS) provides detailed household expenditure data through the Consumer Expenditure Survey (CEX) (U.S. Bureau of Labor Statistics, 2025; Paulin and Krishnamurty, 2019). This paper uses the quarterly Interview Survey as the source for household consumption, income, and related characteristics. The Interview Survey is conducted at a lower frequency than FOMC meetings, but it covers a broad range of consumption categories, including total expenditure, durable goods such as appliances and vehicles, and services, and has a relatively large sample size.

The consumption data cover 1992Q1 to 2019Q4, a total of 112 quarters. The four dependent variables are total consumption expenditure, housing consumption, non-housing consumption, and the food expenditure ratio. Among these, non-housing consumption serves as the primary outcome variable, with the food expenditure ratio providing supplementary analysis, and total consumption and housing consumption used for comparison. Non-housing consumption is constructed as the difference between total consumption expenditure and housing expenditure in the raw dataset. I use this variable as the primary outcome because central bank interest rate policy directly affects lending rates and thus monthly mortgage payments. This creates the possibility that the response of housing consumption may reflect both the direct effect on mortgage payments and the distributional effect of policy operating through consumption demand.

After winsorizing total expenditure and income at the 99.9th percentile within each calendar year, I assign all households to region-income cells defined by Census region and within-region income quintile based on pre-tax household income, as recorded in the BLS interview questionnaire. From 2004 onward, pre-tax income is measured using the BLS-imputed FINCBTXM. For 1992–2003,

no imputed values are available, so I use the respondent-reported FINCBTAX and drop households with incomplete income reports in those years, since their income is zero, missing, or understated, and retaining them would concentrate them in the lowest income quintile. Households with negative income are dropped in all years. This yields 20 groups: four Census regions by five income quintiles. Within each cell, I compute quarterly weighted averages using the sampling weights provided by the CEX, producing a representative value for each cell-quarter observation.

I construct a parallel set of cells stratified by housing tenure to provide complementary evidence on the mechanisms through which monetary policy shocks operate. Following Cloyne et al. (2020), who show that households in different tenure categories—renters, mortgagors, and outright owners—respond differently to monetary policy shocks, I form 15 groups: five income quintiles by three tenure types. In the region-by-income panel, the median number of households per cell per quarter is approximately 220, compared with approximately 270 in the tenure-by-income panel. The standard errors of the cell means are therefore much smaller than the dispersion across individual households. Sample sizes for the region-by-income cells are reported in Appendix A.2.

2.2 Monetary Policy Shocks

Swanson (2021) analyzes changes in asset prices in a 30-minute window around Federal Reserve FOMC announcements from July 1991 to June 2019, applying principal component analysis to the joint responses of federal funds futures, Eurodollar futures, and 2-, 5-, and 10-year Treasury yields. This procedure extracts three orthogonal policy factors: an FFR factor capturing unexpected changes in the current policy rate; an FG factor capturing revisions to the expected path of future policy rates; and an LSAP factor capturing unexpected changes in balance sheet policy. The orthogonality of these three series is central to the analysis, as it allows me to attribute any heterogeneity in the estimated household consumption responses to specific, identifiable policy instruments rather than to undifferentiated shifts in the overall monetary policy stance.

For the FFR and FG factors, negative values indicate expansion, while the LSAP factor has the opposite orientation. Following Swanson (2021), this paper reverses the sign of the LSAP factor so that negative values indicate expansion and positive values indicate contraction for all three series. The shock series are summed within each quarter, aggregating them from meeting-date frequency to quarterly frequency to match the consumption data (Appendix A.4).

Figure 1 plots the three shock series. The FFR factor fluctuates within a narrow band around zero during 2009–2015, reflecting the constraint imposed by the zero lower bound, while the LSAP factor is concentrated almost entirely after 2008. The standard deviations of the three series are 1.03, 1.29, and 0.91, respectively, so a one-unit shock corresponds roughly to one standard deviation. Across the full sample, 15 quarters record shocks with absolute values exceeding two

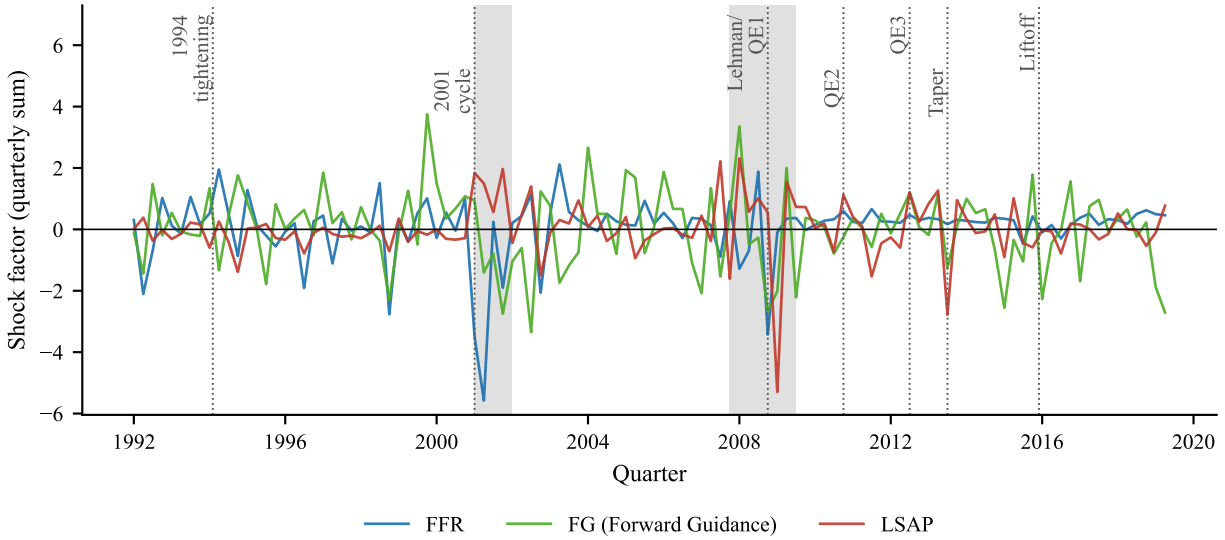


Figure 1: Monetary Policy Shock Factors, 1992Q1–2019Q2

Notes: Quarterly aggregates of FOMC-meeting-level FFR, FG, and LSAP shock factors. All three factors are signed so that positive values denote monetary tightening and negative values denote easing (including QE expansions). Shaded bars indicate National Bureau of Economic Research (NBER) recession quarters.

standard deviations, concentrated mainly around 2001 and at the peak of the 2008–2009 financial crisis; these extreme values may affect the precision of the estimates, and Section 6 returns to this issue.

2.3 Macro Control Variables

Household consumption decisions respond to a broad range of macroeconomic conditions, not only to unexpected monetary policy shocks. To prevent the model from attributing variation driven by domestic or global economic conditions to monetary policy, I include a set of macroeconomic control variables covering consumption, employment, output growth, housing prices, exchange rates, and financial asset prices. I draw ten representative series from the Federal Reserve Economic Data (FRED) database (Federal Reserve Bank of St. Louis, 2026); variable definitions are provided in Table A1 of Appendix A. In selecting these controls, I include the NASDAQ Composite Index to capture movements in financial markets, but exclude financial indicators with direct links to Federal Reserve policy instruments—such as the FFR, Treasury yields, and mortgage rates—as their prices move closely with the policy instruments themselves, and including them risks absorbing part of the variation attributable to monetary policy shocks.

These ten series capture distinct yet interrelated dimensions of macroeconomic conditions and exhibit substantial correlation with one another, making their direct inclusion in the model prone to multicollinearity. To address this and reduce model complexity, I apply principal component

analysis to the ten series as a pre-processing step (Bernanke et al., 2005). The first three principal components explain 68.2 percent of the total variance (see Figure A2 in Appendix A). As these variables serve as controls rather than the primary object of interest, this level of explanatory power is sufficient for the purposes of this paper. The three factors correspond to an economic activity factor (PC1), a price and inflation factor (PC2), and a labor market pressure factor (PC3).

Merging the CEX panel, the monetary policy shock series, and the macroeconomic factor series yields the estimation panel, covering 1992Q1–2019Q2, a total of 110 quarters. The shock variables and macroeconomic controls are common across all cells at any given point in time, while consumption variables, income variables, and household size—used to construct per-capita consumption and income—vary across panel units.

3 Empirical Strategy

This paper uses two panels to address two questions: the region-by-income panel identifies geographic heterogeneity in the distributional effects of monetary policy, while the tenure-by-income panel identifies the transmission channels of the shocks (Cloyne et al., 2020). A panel crossing region, income, and tenure cannot be reliably estimated at the CEX sample size: many cells have too few observations in a non-trivial share of quarters, which amplifies the influence of individual households and increases residual variance. The tenure-by-income panel is therefore a compromise given this constraint, and its estimates should be interpreted as national-level results.

The primary estimation framework is the local projections method of Jordà (2005). Swanson’s monetary policy shock series are identified from high-frequency data and enter the model as exogenous regressors, which is consistent with the local projections setting and reflects current practice in the literature (Plagborg-Møller and Wolf, 2021). This choice is also supported on inferential grounds. Montiel Olea et al. (2026) show that, in a locally misspecified vector autoregression setting, confidence intervals based on local projections retain correct coverage, whereas those based on vector autoregressions with short to medium lag orders can substantially undercover. Since the core conclusions of this paper hinge on assessing the significance of gaps between quantiles, reliable inference under model misspecification is particularly important for the choice of estimator. To improve the interpretability of the impulse response functions, I complement the standard local projections specification with a smoothed variant.

3.1 Local Projection Specification and Identification

The local projections specification used to identify the effects of monetary policy shocks on consumption is given in Equation (1). The dependent variable is specified as $y_{g,t+h} - y_{g,t-1}$,

following Jordà (2005), so that the estimated coefficients at each horizon h can be interpreted as cumulative effects, and the resulting impulse response functions are cumulative impulse response functions. Rather than pooling observations into a single panel estimator—which would prioritize identifying common effects across cells—I estimate the model separately for each cell by OLS. This approach is better suited to the objective of documenting heterogeneity across cells in response to common shocks, and prevents cross-cell dependencies from confounding the cell-level estimates that form the basis of the subsequent comparisons.

$$\begin{aligned}
y_{g,t+h} - y_{g,t-1} = & \alpha_g^h + \beta_g^h \cdot s_t + \sum_{j=1}^{p_s} \delta_j^h \cdot s_{t-j} + \sum_{j=1}^{p_y} \phi_j^h \cdot \Delta y_{g,t-j} \\
& + \sum_{j=1}^{p_x} \Gamma_j^h \cdot \mathbf{PC}_{t-j} + \sum_{q=1}^3 \gamma_q^h D_{q,t} + \varepsilon_{g,t+h}
\end{aligned} \tag{1}$$

where g indexes panel cells (region $\times$ income quintile in the regional estimation, housing tenure $\times$ income quintile in the tenure estimation), t indexes calendar quarters, and $h = 1, \dots, 12$ denotes the forecast horizon. α_g^h is the regression intercept; s_t is one of the three Swanson (2021) monetary policy shock series (FFR, FG, or LSAP), and β_g^h is the coefficient of interest, tracing out the impulse response of cell g 's consumption to the shock at horizon h . The $p_s = 2$ lags of the shock (δ_j^h), $p_y = 3$ lags of consumption growth (ϕ_j^h), and $p_x = 1$ lags of the three macroeconomic principal components ($\Gamma_j^h, \mathbf{PC}_{t-j}$) absorb the dynamic feedback between the shock, macroeconomic conditions, and household consumption. $D_{q,t}$, $q = 1, 2, 3$, are quarter dummies (Q1–Q3, with Q4 as the reference quarter) with coefficients γ_q^h that absorb seasonality in the CEX data, and $\varepsilon_{g,t+h}$ is the regression error. Equation (1) is estimated separately for each cell g and each horizon h by OLS on cell g 's own quarterly time series ($T_g \approx 108$ observations); all coefficients are therefore cell-specific, although the subscript g is suppressed on δ_j^h , ϕ_j^h , Γ_j^h , and γ_q^h for notational economy, and α_g^h is simply that regression's intercept rather than a fixed effect estimated jointly across a pooled panel. Standard errors are computed using the Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) estimator with bandwidth $h + 2$ (Newey and West, 1987).

The maximum horizon is $H = 12$ (three years). The lag orders p_y and p_x are determined by information criteria, while p_s is determined by autocorrelation diagnostics on the shock series and the sensitivity of the point estimates, yielding $(p_s, p_y, p_x) = (2, 3, 1)$. Under the baseline specification, each regression uses 96 anchors and estimates 13 parameters. The selection procedure is described in Appendix B.1.

Consumption variables are measured in nominal per-capita dollars. Price-level movements are common across all cells, so the Q1–Q5 difference, regional comparisons, and tenure comparisons

are affected very little by whether the series are CPI-deflated, though not entirely unaffected because the specification includes each cell’s own lagged dependent variable. The level of each group’s impulse response should be interpreted as nominal consumption growth, which may include a common aggregate price response to the shock shared across groups; the price factor ($PC2$) among the macroeconomic controls partially absorbs this component. Section 6 reports the deflated results, and the assumption that all income groups face the same price index is discussed in Appendix F.4.

Swanson (2021) extracts monetary policy shocks from asset price movements in a window extending from 10 minutes before to 20 minutes after each FOMC announcement. The identifying assumption is that price movements within this window are driven mainly by monetary policy information and that the effects of other information arriving at the same time are negligible. Households cannot adjust their consumption decisions within such a short window, so these shocks can be treated as exogenous with respect to consumption decisions. A potential concern is the Fed information effect associated with FOMC announcements, whereby market reactions may partly reflect information about the economic outlook conveyed by the Federal Reserve rather than changes in the policy stance itself (Nakamura and Steinsson, 2018a; Jarociński and Karadi, 2020). The factor decomposition mitigates but does not fully eliminate this concern, so Section 6 re-estimates the model after excluding the periods most likely to be affected, namely NBER recession quarters and extreme shock quarters.

3.2 Smooth LP

Standard local projections are susceptible to sawtooth impulse response functions due to correlated residuals across horizon-specific regressions, making the pattern of lagged dynamics harder to discern. Following Barnichon and Brownlees (2019), I parameterize the impulse response function using a B-spline basis with K functions, imposing smoothness and continuity across the forecast horizon without changing the identifying assumptions.

I parameterize the impulse response vector using K cubic B-spline basis functions, as follows:

$$\beta_g^h = \sum_{k=1}^K \theta_k^g \cdot B_k(h), \quad h = 1, \dots, H \quad (2)$$

where θ_k^g , $k = 1, \dots, K$, are cell-specific B-spline coefficients to be estimated and $B_k(h)$ are cubic B-spline basis functions evaluated over the forecast horizon $h = 1, \dots, H$. The number of basis functions K is selected from $\{3, 4, 5\}$ by generalized cross-validation (GCV) separately for each cell g .

In practice, all cells select the upper bound $K = 5$, which this paper treats as a fixed regularization choice rather than an unconstrained optimum in the GCV sense (Appendix B.2). The standard

errors of the smoothed local projections rely on a diagonal covariance approximation and may underestimate the true standard errors, so the smoothed specification is used only for graphical display; all Wald tests and significance annotations in the figures are based on the standard local projections. The confidence bands in the figures are obtained by smoothing the bootstrap paths from the standard local projections using the same B-spline projection (Appendix B.2), and a comparison of point estimates between the two specifications is provided in Appendix C.

3.3 Measuring Distributional Effects

Since the primary objective of this paper is to analyze the distributional effects of monetary policy shocks on consumption inequality—rather than their effects on consumption levels alone—the central focus is on how the distributional impact of policy shocks differs across groups. Conventional inequality measures such as the Gini coefficient summarize the entire distribution with a single statistic, which facilitates computation and allows straightforward tracking of inequality over time, but does not reveal which population groups or consumption categories are driving the inequality. I therefore adopt a disaggregated approach, examining each consumption variable separately by income quintile, and characterize inequality through visual comparisons and simple algebraic contrasts across income groups within the same shock-consumption variable combination. This approach preserves more information about the distribution of policy effects and avoids reducing economy-wide inequality to a single index. This paper examines the distributional effects using four tools: fan charts, adjacent-gap charts, gap-path plots, and a joint Wald test comparing $Q1$ and $Q5$.

Fan charts place the impulse responses for $Q1$ through $Q5$ in a single figure, with one panel per region, so that both the within-region income gradient and differences across regions can be assessed from the same figure. $Q1$ and $Q5$ are shown with 68% and 90% confidence bands, while the three middle quintiles show point estimates only, following Cloyne et al. (2020) and Coibion et al. (2017).

The adjacent-gap charts plot the differences in impulse responses between each pair of neighboring income quintiles, indicating where in the income distribution the gap is concentrated; the definition is provided in Appendix D.4.

Gap-path plots show the path of the difference between $Q1$ and $Q5$ itself over time, together with a confidence band for that difference, and mark the boundaries of the three windows and the test result for each window in the figure. In a fan chart, the gap must be judged by eye from the vertical distance between the two curves, while the uncertainty around the gap itself is not visible; gap-path plots instead show directly the quantity on which the inference in this paper actually rests. These plots use the standard local projections, consistent with the estimates underlying all inference

in this paper, and are not smoothed; their confidence bands are obtained from the same bootstrap procedure.

The three graphical tools allow differences in impulse responses to be assessed visually, but do not support formal statistical inference. I therefore use a joint Wald test as the primary inferential tool, testing whether the impulse response sequences for $Q1$ and $Q5$ differ jointly across all horizons $h = 1, \dots, 12$. I restrict the test to $Q1$ and $Q5$ for two reasons. First, testing all pairs of income quintiles would yield up to ten pairwise test statistics, making it difficult to draw clear conclusions. Second, the estimates reveal substantial within-group heterogeneity among $Q2$ – $Q4$, suggesting that consumption differences among the middle 60 percent of the income distribution reflect a range of factors that the model does not fully capture. A joint Wald test on $Q1$ and $Q5$ therefore targets the most salient dimension of consumption inequality: the gap between the bottom and top of the income distribution.

For each cell combination (region/tenure type $\times$ consumption variable $\times$ shock), I extract the standard local projection estimates $\hat{\beta}_{g,Q1}^h$ and $\hat{\beta}_{g,Q5}^h$ and conduct a joint hypothesis test across all 12 horizons:

$$W = \sum_{h=1}^H \left(\frac{\hat{\beta}_{g,Q5}^h - \hat{\beta}_{g,Q1}^h}{\widehat{SE}(\delta^h)} \right)^2 \sim \chi^2(H) \quad (3)$$

where $\hat{\beta}_{g,Q1}^h$ and $\hat{\beta}_{g,Q5}^h$ are the estimated impulse responses of the bottom and top income quintiles at horizon h , each estimated by a separate cell-level regression (1) with its own Newey–West standard error $\widehat{SE}_{Q1}^h$ and $\widehat{SE}_{Q5}^h$, and $\delta^h \equiv \hat{\beta}_{g,Q5}^h - \hat{\beta}_{g,Q1}^h$ is their difference.

Under the null, W is asymptotically distributed as $\chi^2(12)$, but this result relies on two simplifications. The first is cross-cell independence, $\widehat{SE}(\delta^h) = \sqrt{\widehat{SE}_{Q1}^2 + \widehat{SE}_{Q5}^2}$, which assumes that the estimation errors from the $Q1$ and $Q5$ regressions are uncorrelated. The second is a diagonal approximation across horizons, which sums the squared terms over the 12 horizons directly and ignores the serial correlation of δ^h across horizons. A size simulation under a data-generating process with a true coefficient of zero shows that the test based on Newey–West standard errors has rejection rates between 3.5 and 22 times the nominal level at the 10%, 5%, and 1% nominal levels, systematically classifying noise as significant effects. This paper therefore uses a residual moving block bootstrap (Künsch, 1989) as the primary basis for reported significance, with block length $L = 12$ and $B = 5000$ replications, jointly resampling the residuals for $Q1$ and $Q5$ using the same set of block indices. In the same simulation, the rejection rates of the bootstrap are close to the nominal level, and the conclusions are insensitive to the block length (Appendix B.3). The appendix tables report both the Newey–West and bootstrap results, while the main text reports only the bootstrap results.

Equation (3) tests only whether the path as a whole departs from zero and does not identify the

horizons at which the departure occurs. To characterize the timing of the effect, this paper divides the 12 horizons into three yearly windows ($h = 1-4, 5-8, 9-12$), constructs $S_w = \sum_{h \in w} z_h^2$, where $z_h = \delta^h / \widehat{SE}(\delta^h)$, separately within each window, and obtains p -values using the same bootstrap procedure. The full-horizon test spends one degree of freedom on each horizon regardless of whether that horizon carries a signal, so it has limited power for effects concentrated in a short window; the window-based tests mitigate this problem. The two types of test are reported alongside each other: the former assesses whether the path as a whole departs from zero, while the latter identifies the year in which the departure occurs. When the main text cites the result for a given cell, it uses the test for the window in which that result falls. The three windows are defined in advance by year and do not depend on inspection of the estimation results.

This paper follows the reporting conventions set out below. Subsequent results and conclusions use the 10% level as the threshold for determining whether an effect is present, with tables marking significance at the 10%, 5%, and 1% levels using *, **, and ***, respectively. For cells that fail the test, this paper does not report the magnitude of the point estimate, in order to avoid treating an estimate that is statistically indistinguishable from zero as an effect size; when a window mean is cited, the test for that window provides the basis for the inference. The few values reported only to indicate direction are explicitly noted in the text as statistically insignificant.

4 Main Results

All impulse responses in this section are cumulative consumption responses to a +1 unit contractionary shock, and the response to an expansionary shock is equal in magnitude and opposite in sign. A larger consumption decline for $Q1$ than for $Q5$ ($Q1$ lying below $Q5$ in the figures) therefore implies that $Q1$ gains more under an expansionary shock, that is, a progressive effect. The paths in the figures are estimated using smoothed local projections, while the differences, test statistics, and p -values cited in the text all come from the standard local projections; the qualitative conclusions are the same under both specifications (Appendix C).

4.1 LSAP: Progressive Distributional Effects

The literature offers two competing predictions about the distributional effects of unconventional monetary policy. If LSAP operates primarily through the asset price channel (Domanski et al., 2016)—compressing long-term interest rates and pushing up equity and housing valuations—higher-income households, which hold more financial assets, should benefit more, producing a regressive distributional outcome ($Q5 > Q1$). If instead the dominant channel is employment and labor income (Kaplan et al., 2018)—LSAP expanding aggregate demand and boosting job

creation—lower-income households should gain more, as labor income accounts for a larger share of their total income and they are more sensitive to employment conditions, producing a progressive outcome ($Q1 > Q5$). Distinguishing between these two predictions is the starting point for the empirical analysis in this section.

Figure 2 shows the cumulative impulse responses of non-housing consumption to a contractionary LSAP shock across the four Census regions. The direction of the point estimates is not consistent across the four regions. Only in the Northeast is the consumption decline for $Q1$ systematically larger than that for $Q5$, with the path for $Q1$ lying below that for $Q5$ and farther from the zero line; only in the Northeast does the joint test reject the null that the two responses are the same. The evidence for this progressive effect therefore comes entirely from the Northeast, while the other three regions provide neither independent statistical evidence nor corroboration in the same direction. The peak difference in the Northeast is close to 7 percentage points and averages approximately 5 percentage points in the second year, the largest among the four regions. The shapes of the difference paths in the other three regions vary. The differences for the Midwest and West move from negative to positive: the Midwest is negative in the first year and turns positive in the third, while the West is negative in the first two years and turns positive in the third. The full-horizon means for these two regions therefore do not adequately characterize their paths. The difference for the South is negative in most horizons, meaning that consumption for $Q1$ declines less than that for $Q5$, in the opposite direction to the Northeast, but it is insignificant in every window. The full paths for each region are provided in Table D1 of the appendix.

The window-based tests show that the effect emerges with a clear lag, and the timing is consistent across the three consumption measures. In the first year, none of the three is statistically distinguishable from zero. In the second year, non-housing consumption and total consumption both become significant, making this the window with the strongest evidence in the paper. In the third year, both remain significant, although the evidence is weaker. The food expenditure ratio lags by one window, becoming significant only in the third year. The point estimates show the same timing: the $Q1$ – $Q5$ difference in non-housing consumption for the Northeast is below 1 percentage point in the first year, rises to approximately 5 percentage points in the second, and is approximately 4 percentage points in the third. This is not a mechanical artifact of cumulation. Within the same set of cells, the mean magnitude of the difference increases more than fivefold from the first year to the second, while its bootstrap standard deviation remains almost unchanged; across all LSAP tests, only one passes in the first year, fewer than would be expected by chance alone at the 10% level, while six to seven pass in each of the second and third years.

Figure 3 plots the path of the difference δ^h directly, with dotted lines marking the boundaries of the three windows and panel titles reporting the test results for each window.

The adjacent-quintile gap charts in Appendix D show that the progressive effect in the Northeast

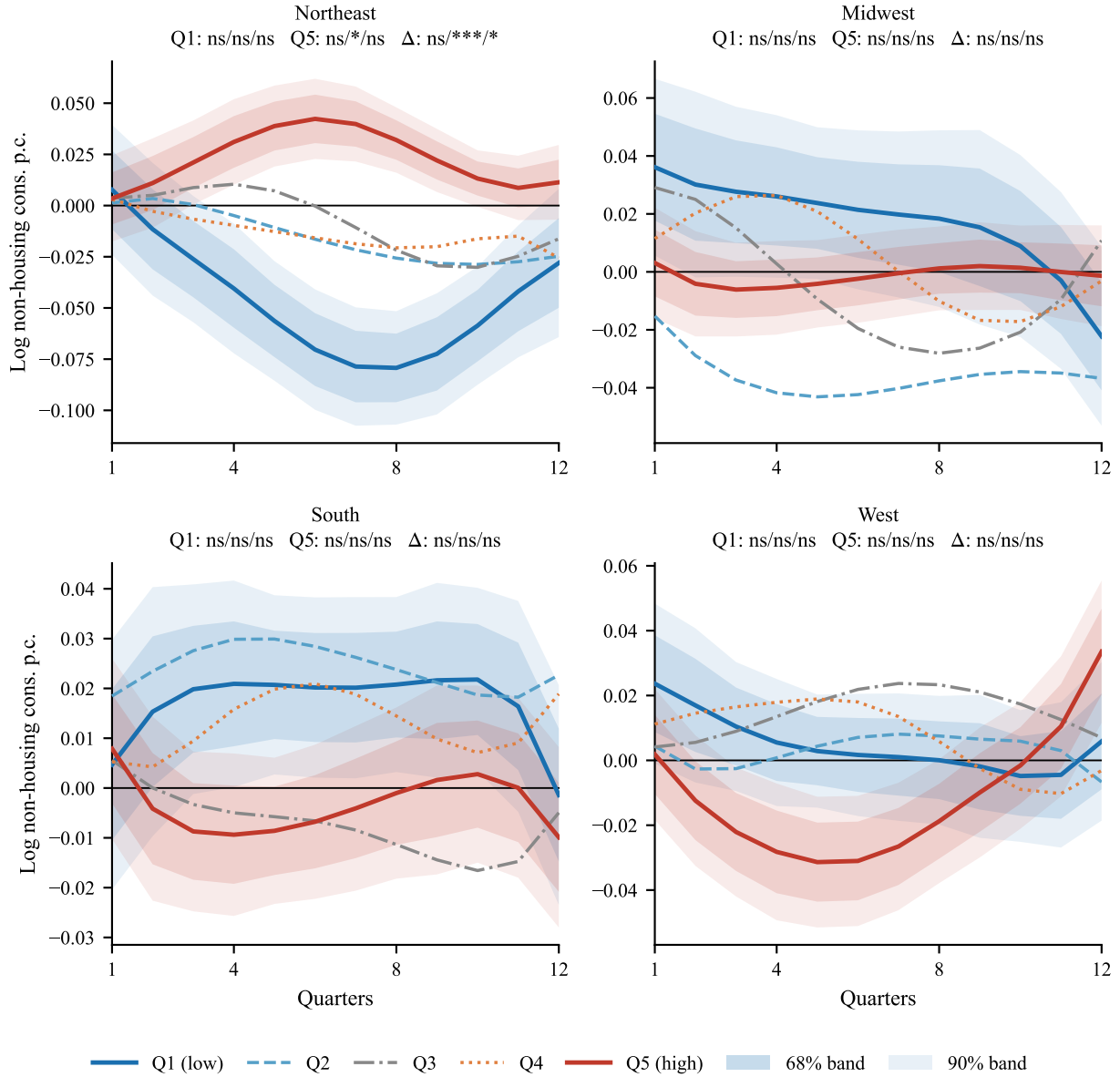


Figure 2: IRF: Log Non-Housing Consumption per Capita — LSAP Shock

Notes: (1) Non-housing consumption is total consumption expenditure minus housing costs (rent and mortgage payments), in log per-capita nominal dollars. (2) Responses to a one-unit contractionary LSAP shock; the response to an expansionary shock of equal magnitude is the mirror image (see Section 2.2). (3) IRF paths are Smooth LP estimates ($K = 5$, GCV-selected; Section 3.2). Confidence bands and significance annotations come from the residual moving block bootstrap on the Standard LP estimates ($B = 5000$, $L = 12$; Section 3.1); no Newey–West band is plotted. (4) Shaded bands: 68% CI (inner, darker) and 90% CI (outer, lighter) for Q1 and Q5; Q2–Q4 show point estimates only. (5) Panel titles report the bootstrap joint tests by horizon window, in the order $h = 1-4 / 5-8 / 9-12$, for Q1 ($H_0: \beta_{Q1}^h = 0$ for all h in the window), for Q5, and for the difference $\Delta \equiv \delta^h$; ns denotes $p \geq 0.10$, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The same tests are tabulated in Table 1. (6) Analysis sample: 1992 Q1–2019 Q2 (110 quarters).

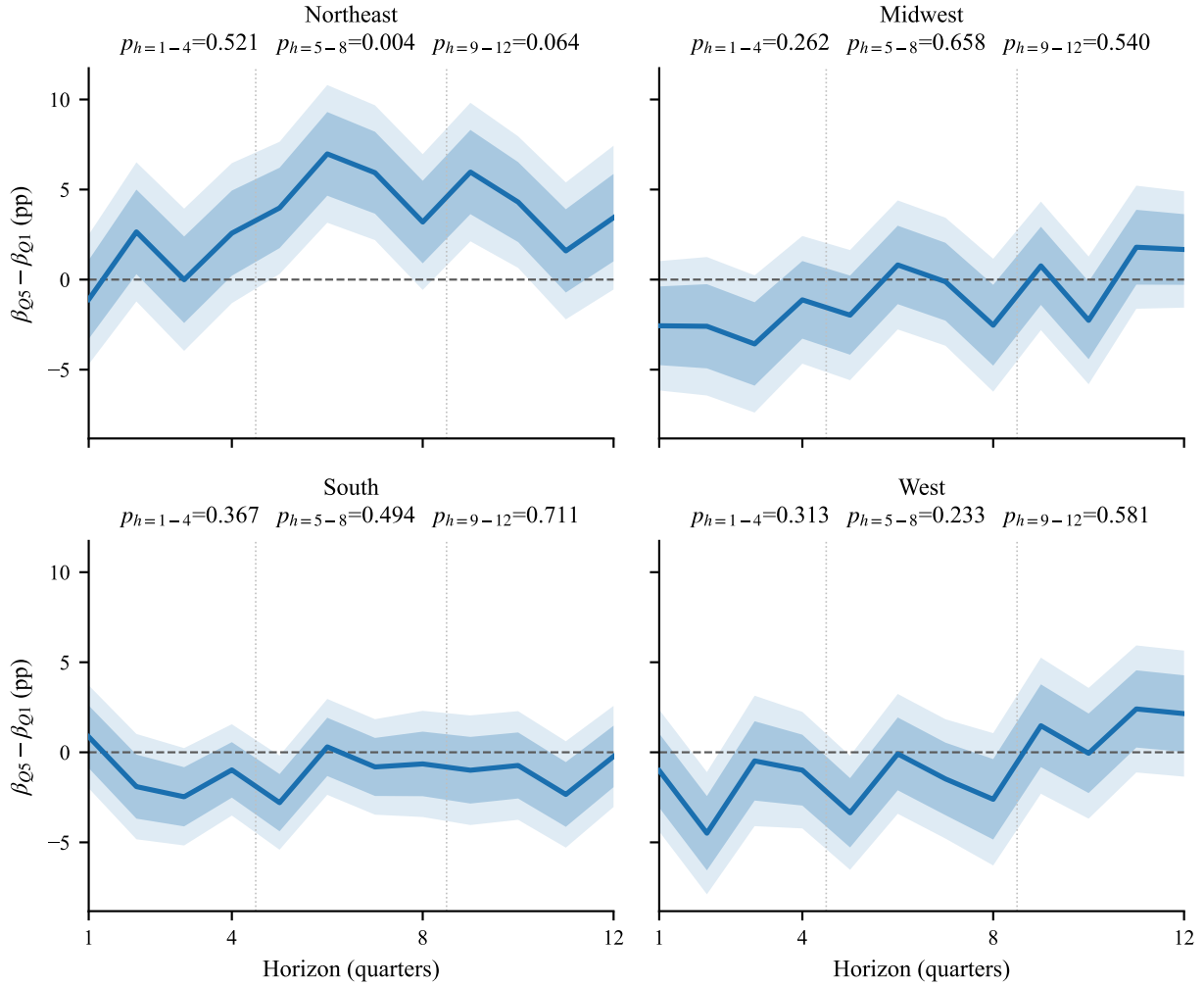


Figure 3: The $Q1$ – $Q5$ Gap in Log Non-Housing Consumption per Capita — LSAP Shock

Notes: (1) Each panel plots the distributional gap itself, $\delta^h \equiv \beta_{Q5}^h - \beta_{Q1}^h$, rather than the two quintiles' separate responses as in Figure 2. A positive value means that $Q1$'s cumulative response lies below $Q5$'s, that is, $Q1$ loses more under a contractionary shock. (2) Estimates are Standard LP, matching the basis for all inference in this paper; no smoothing is applied. (3) Shaded bands are ± 1 and ± 1.645 bootstrap standard deviations (68% and 90%), obtained from the residual moving block bootstrap ($B = 5000$, $L = 12$). (4) The three p -values in each panel title are the sub-window joint tests over $h = 1$ – 4 , 5 – 8 , and 9 – 12 (Section 3.3); dotted vertical lines mark the window boundaries. (5) Analysis sample: 1992 Q1–2019 Q2 (110 quarters).

is not distributed evenly across the income distribution. Of the four adjacent gaps, only those at the bottom (between $Q2$ and $Q1$) and at the top (between $Q5$ and $Q4$) are clearly positive, while the two gaps in the middle of the distribution are both close to zero, with one slightly negative. No joint test is conducted on the adjacent gaps themselves, so they are used here only to indicate where in the distribution the gap is concentrated, not as effect sizes. The overall gap between $Q1$ and $Q5$ therefore arises mainly from the widening of the two tails relative to their neighboring quintiles, rather than from a stepwise progression across the distribution.

By Engel’s Law, the food expenditure ratio can serve as an indirect indicator of changes in consumption composition. In the Northeast, the increase in the food expenditure ratio for $Q1$ following a contractionary LSAP shock is significantly larger than that for $Q5$, with this difference emerging in the third year; the difference is negative over the large majority of horizons and reaches its largest magnitude between the end of the second year and the middle of the third. This runs in the same direction as the non-housing consumption results: $Q1$ not only shows a larger decline in nominal consumption but also a more pronounced deterioration in expenditure composition. The food expenditure ratio provides no robust evidence in the remaining three regions or under FFR and FG shocks (Appendix D).

4.2 Forward Guidance

Compared with LSAP, FG produces no consistent regional pattern. Across the sixteen region-by-consumption-measure combinations, none passes the full-horizon test; under the window-based tests, only three pass—the food expenditure ratio in the Northeast, non-housing consumption in the South, and housing consumption in the South—and these occur in different periods and do not point in a consistent direction. This scattered pattern is sufficient to show that FG transmits differently from LSAP, but not to characterize the distributional direction of the effects of FG itself.

The differences across regions may be related to factors such as industrial structure, but this paper does not incorporate region-by-industry data to test this possibility. Thus, this explanation remains a conjecture to be examined rather than an empirical finding of this paper. Full impulse responses for each region are reported in Appendix D.

In the Northeast and West, non-housing consumption for $Q1$ is higher than that for $Q5$ following a contractionary FG shock, in the opposite direction to LSAP, while this pattern does not hold in the Midwest and South. The evidence from the food expenditure ratio is directionally consistent but weak: the $Q1$ – $Q5$ difference for the Northeast passes the test only in the second-year window and is insignificant over the full horizon, while the direction in the Midwest is not consistent.

The adjacent-gap charts in Figure D6 of the appendix show that the FG differences are close to zero and insignificant in most regions. Taking the two unconventional instruments together, the evidence that the decline for $Q1$ is systematically larger than that for $Q5$ under LSAP is concentrated in the Northeast, while the $Q1$ – $Q5$ ordering under FG varies across regions in both direction and statistical significance. The distributional consequences of quantitative easing and forward guidance therefore cannot be treated as interchangeable forms of “monetary easing”.

4.3 Conventional Monetary Policy (FFR)

The distributional effects of FFR show marked regional differences, with the evidence concentrated in the South: the tests for total consumption and housing consumption pass in the third-year window, with a smaller decline for $Q1$ than for $Q5$, in the opposite direction to LSAP. Non-housing consumption in the South is not statistically significant in any window, so the evidence for FFR in the South is limited to the two measures that include housing, a point that should be kept in mind when interpreting the results. Housing consumption in the Northeast is also significant in the first year, again with $Q1$ less affected. The Midwest and West show no discernible difference under any of the consumption measures.

The rise in nominal consumption for $Q1$ in the South requires cautious interpretation: the food expenditure ratio is insignificant in every window in all four regions and cannot provide supplementary evidence. Housing consumption for $Q1$ is positive in the Northeast, Midwest, and South and negative in the West, consistent with rents rising with interest rates following a contractionary shock, while the monthly payments on existing fixed-rate loans do not respond to short-term rates. This is also why this paper uses non-housing consumption as the primary outcome variable. The differences for each cell are reported in Table 2, while the full regional discussion, fan charts, and adjacent-gap charts are provided in Appendix D.

4.4 Significance Summary

Table 1 summarizes the bootstrap joint $Q1$ – $Q5$ tests, reported separately for the three horizon windows ($h = 1$ – 4 , 5 – 8 , 9 – 12), across the 36 cells formed by three shock types (FFR, FG, LSAP), three primary outcome variables (non-housing consumption, total consumption, and food expenditure ratio), and four Census regions; the full four-outcome tabulation, which additionally includes housing expenditure, is reported in Table D2 of Appendix D.

When the two instruments are compared directly within the same region, consumption measure, and window (Table D4 of the appendix), the progressive effect of asset purchases in the Northeast is statistically distinguishable from that of forward guidance at the 1% level in the second year. Relative to conventional interest rate policy, the corresponding difference is statistically significant at the 10% level for non-housing consumption and at the 5% level for total consumption. Table 2 reports the magnitude of the difference for each cell that passes the joint test, while Table D3 of the appendix provides direct comparisons of the LSAP and FFR differences within the same region, consumption measure, and window.

The evidence for LSAP is concentrated in the Northeast; including housing consumption, 6 of the 16 cells pass the test in at least one window. Outside the Northeast, the only LSAP result that is also significant in the full-horizon test is housing consumption in the South, again with a

Table 1: Joint Test for Distributional Effect by Horizon Window: $H_0 : \beta_{Q1}^h = \beta_{Q5}^h$ for all h in the Window

Outcome	Shock	NE	MW	S	W
Non-Housing Cons.	FFR	ns/ns/ns↑	ns/ns/ns↑	ns/ns/ns↓	ns/ns/ns↑
	FG	ns/ns/ns↓	ns/ns/ns↑	*/ns/ns↑	ns/ns/ns↓
	LSAP	ns/***/*↑	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↓
Total Consumption	FFR	ns/ns/ns↓	ns/ns/ns↑	ns/ns/**↓	ns/ns/ns↑
	FG	ns/ns/ns↓	ns/ns/ns↑	ns/ns/ns↑	ns/ns/ns↓
	LSAP	ns/***/*↑	ns/ns/ns↓	ns/ns/ns↓	ns/*ns↓
Food Share	FFR	ns/ns/ns↑	ns/ns/ns↓	ns/ns/ns↑	ns/ns/ns↓
	FG	ns/**/ns↑	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↑
	LSAP	ns/ns/**↓	ns/ns/ns↑	ns/ns/ns↓	ns/ns/ns↓

Notes: each cell reports the bootstrap significance of the joint $Q1$ – $Q5$ difference test ($H_0 : \delta^h = 0$ for all h in the window) for the three horizon windows in order, $h = 1$ – 4 / 5 – 8 / 9 – 12 , followed by the direction of $\delta^h \equiv \beta_{Q5}^h - \beta_{Q1}^h$ averaged over $h = 1, \dots, 12$. The test is reported by horizon window rather than as a single test over $h = 1, \dots, 12$ because the latter averages over the timing of the response: a distributional effect confined to one year is diluted by the eight horizons in which nothing happens.

↑ denotes $\delta > 0$, i.e. $Q1$'s cumulative response lies below $Q5$'s. All three shock series are oriented so that a positive value is a monetary tightening, so the arrow has the same meaning for every instrument and no instrument-specific sign adjustment is applied. For the three consumption measures ↑ is the progressive case ($Q1$ loses more under a contraction); for the food expenditure share the reading is reversed, since a smaller rise in $Q1$'s food share indicates $Q1$ is relatively better off.

ns denotes $p \geq 0.10$; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Inference is the residual moving block bootstrap ($B = 5000$, $L = 12$) throughout. The Newey–West counterpart, which over-rejects in this design, is reported alongside the bootstrap in Table D2. $(p_s, p_y, p_x) = (2, 3, 1)$.

larger decline for $Q1$, significant in both the first- and third-year windows. Housing consumption is directly affected by interest rates, and this paper does not treat it as evidence on consumption demand; within the same region, however, it is the outcome for which the difference between asset purchases and conventional interest rate policy is most pronounced: in the third-year window, the differences under LSAP and FFR are +2.52 and –2.23 percentage points, respectively, and the gap between the two, +4.76 percentage points, is significant at the 1% level (Table D3 of the appendix). For non-housing consumption, the difference between the two instruments is significant only in the second-year window for the Northeast (a gap of +4.78 percentage points, at the 10% level).

The evidence for FG is the weakest: no cell passes the full-horizon test, and only three pass the window-based tests, scattered across two regions, three consumption measures, and two periods (Section 4.2). The significant results for FFR are concentrated in total consumption and housing consumption in the South in the third year and in housing consumption in the Northeast in the first year (Section 4.3).

Among the three instruments, the $Q1$ – $Q5$ ordering is most consistent for LSAP, although its statistical support is concentrated in a single region; the ordering for FG and FFR varies by region and consumption measure and cannot be classified simply as progressive or regressive.

Table 2: Distributional Gap Magnitudes for Cells Passing the Joint $Q1$ – $Q5$ Test in at Least One Horizon Window

Shock	Region	Outcome	Horizon	$\bar{\delta}_w^a$	Boot.	Reading
LSAP	Northeast	Non-housing cons.	5–8	+4.98	***	$Q1$ loses more
LSAP	Northeast	Non-housing cons.	9–12	+3.76	*	$Q1$ loses more
LSAP	Northeast	Total consumption	5–8	+3.77	***	$Q1$ loses more
LSAP	Northeast	Total consumption	9–12	+2.54	*	$Q1$ loses more
LSAP	Northeast	Food share	9–12	−0.24	**	$Q1$ loses more
LSAP	Northeast	Housing expenditure	9–12	—	*	<i>sign varies</i> ^b
LSAP	South	Housing expenditure	1–4	+1.58	*	$Q1$ loses more
LSAP	South	Housing expenditure	9–12	+2.60	**	$Q1$ loses more
LSAP	West	Total consumption	5–8	−2.42	*	$Q1$ loses less
FFR	Northeast	Housing expenditure	1–4	−2.89	***	$Q1$ loses less
FFR	South	Total consumption	9–12	−1.47	***	$Q1$ loses less
FFR	South	Housing expenditure	9–12	−2.06	***	$Q1$ loses less
FG	Northeast	Food share	5–8	+0.25	**	$Q1$ loses less
FG	South	Non-housing cons.	1–4	+1.44	*	$Q1$ loses more
FG	South	Housing expenditure	5–8	−1.25	**	$Q1$ loses less

^a $\bar{\delta}_w$ is the Standard LP gap $\delta^h \equiv \beta_{Q5}^h - \beta_{Q1}^h$ averaged over the horizons of the stated window, in percentage points, under a one-unit contractionary shock, computed on each horizon’s own estimation sample. $(p_s, p_y, p_x) = (2, 3, 1)$; residual moving block bootstrap, $B = 5000$, $L = 12$. All three shock series are oriented so that a positive value is a tightening, so the gap is formed identically for every instrument and no instrument-specific sign adjustment is applied.

The table lists every cell–window whose joint $Q1$ – $Q5$ test passes at the 10% level, i.e. exactly the magnitudes the paper’s reporting convention (Section 3.3) permits quoting. The unit of selection is the cell–window rather than the cell: a test over all twelve horizons that rejects does not establish that the gap is present throughout, so the average quoted is the average over the window that rejects. 15 cell–windows qualify, spanning 12 distinct cells. The “reading” resolves the sign: for the three consumption measures $\bar{\delta}_w > 0$ means $Q1$ loses more under a tightening; for the food expenditure share the reading inverts, since a larger rise in $Q1$ ’s food share is the case in which $Q1$ ’s expenditure composition deteriorates more.

^b A window average is reported only when it summarises the path. The joint test rejects $\delta^h = 0$ at every h in the window, which a path that changes sign within the window can do while averaging to almost nothing; the average would then be read as an effect size it is not. Where the mean absolute deviation exceeds twice the absolute mean, the average is suppressed. 1 entry is suppressed on this rule. The one suppressed entry that is also statistically significant is Northeast, LSAP, housing expenditure at $h = 9$ –12, whose four horizons are −1.29, −0.82, +3.61 and −3.11 percentage points.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show the exact p -value. Inference is the bootstrap throughout; the Newey–West counterpart, which over-rejects in this design, is reported alongside it in Table D2. Cell–windows failing the joint test are omitted rather than reported with magnitudes.

The differences in distributional effects across the three instruments reflect differences in their transmission channels, rather than differences in the strength of a common “monetary easing” effect.

The geographic concentration of the evidence for a given instrument does not by itself imply that differences exist across regions: when one region is significant and another is not, the difference between the two is not necessarily significant (Gelman and Stern, 2006), and such a contrast may

reflect nothing more than differences in estimation precision. I therefore directly test the null hypothesis that the $Q1$ – $Q5$ difference paths are the same across the four regions.

For non-housing consumption and total consumption under LSAP, the null that the Northeast equals the average of the remaining three regions is rejected (bootstrap p -values of 0.005 and 0.006, respectively); the three pairwise comparisons between the Northeast and the Midwest, South, and West are also all rejected at the 5% level (non-housing consumption $p = 0.024, 0.007, 0.023$; total consumption $p = 0.028, 0.040, 0.012$). By window, these rejections are concentrated in the second year, when the effect is strongest, with second-year window p -values of 0.001 and 0.002 for the comparison of the Northeast with the remaining regions.

The Northeast is the only one of the four regions with a significant difference, so subsequent tests focus on comparisons between the Northeast and the remaining regions; the differences for the other three regions are all insignificant, and comparisons among them are therefore less informative. The joint test that all four regions are equal does not reject the null ($p = 0.20$ and 0.14). This does not contradict the results above: it imposes three times as many restrictions at once and therefore has markedly lower power. What can be established is that the Northeast differs from the remaining regions, but this does not support any claim that the other three regions differ from one another. The conclusion on regional differences in this paper therefore does not rely on the invalid inference that one region being significant and another not means that the two are significantly different, but is instead supported by direct tests.

5 Mechanism Analysis

Section 4 leaves four questions to be explained: of the three instruments, only asset purchases produce a robust income gradient; the gradient is concentrated in the Northeast and is statistically distinguishable from that in the remaining regions; it takes shape only in the second year; and the other two instruments affect housing consumption, with their effects appearing early.

Region as a way of slicing the data cannot answer these questions. Monetary policy does not act on the Northeast; it acts through contracts, changing the cash that must be paid in the current period under a given type of contract or the terms on which that contract can be renegotiated. This section therefore reslices the same sample by housing contract type, crossed with income quintile. This again yields fifteen cells, with the estimation and inference procedures unchanged, and the results are at the national level.

5.1 Housing Contracts and Testable Predictions

Mortgage contracts in the United States are predominantly long-term and fixed-rate, unlike those in most advanced economies (Badarinza et al., 2016). As a result, the monthly payments of households that have already entered into such contracts do not move with market interest rates over the life of the loan, and a rise in short-term rates does not cost them a single dollar more. The contract also embeds a prepayment option, which is the only route through which monetary policy can affect the cash flow of existing borrowers, and whether it is worth exercising depends on long-term mortgage rates rather than the federal funds rate. Leases are the opposite. Their terms are short and repricing is frequent, while both landlords' holding costs and the supply of rental housing are sensitive to the short end. The housing cash flow of renter households is therefore floating, while that of mortgagor households is fixed, and this contractual difference provides the basis for why the same shock produces different consequences.

Conventional interest rate policy moves the short end. It does not change the monthly payment on an existing loan, but it passes through to rents and the price of new credit, so its impact falls on the renter side. Asset purchases move the long end and compress mortgage-backed securities spreads through direct purchases; these are precisely the two prices that determine the value of the prepayment option. LSAP is therefore the only instrument that can open or close the refinancing channel for existing borrowers, and its impact falls on the mortgagor side; a contractionary shock means that this door is closing. Forward guidance moves the expected path of the short end, changing neither this month's rent nor this month's mortgage payment, while expectations alone are not sufficient to trigger refinancing.

Comparing mortgagors and renters within the same income group therefore implies a larger response for renters under conventional interest rate policy, a larger response for mortgagors under asset purchases, and no difference between the two under forward guidance. These three predictions point in different directions, making it difficult for a single alternative explanation to account for all three simultaneously. They also imply two mutually exclusive criteria: the cash flow channel requires the income gradient to be concentrated within the mortgagor group, whereas the employment channel requires a gradient to appear within all three groups simultaneously. Renter households carry no refinanceable debt, so a gradient within the renter group provides the only source of identification for the employment channel.

5.2 Results from the Tenure Estimation

Tenure and income are highly correlated, so comparisons are made within a given income quintile. Mortgagors account for roughly one-ninth of the lowest quintile, making this cell thinner than the other two cells in the same quintile.

The pattern under all three instruments is consistent with the predictions in the previous subsection. Under conventional interest rate policy, within the three lowest income quintiles, renter households show a larger decline in non-housing consumption than mortgagor households at the same income level, with the difference detectable across multiple windows (Table 3). For non-housing consumption, however, no income gradient is evident within any of the three groups. Conventional policy therefore generates differences across tenure rather than across income. This pattern differs from the finding in Cloyne et al. (2020), who report that under conventional policy, the consumption response of mortgagor households is larger than that of renters. One possible explanation is that this paper compares households within the same income group, whereas mortgagors are concentrated at the high-income end of the distribution; without conditioning on income, differences across tenure are confounded with differences across income.

The direction reverses under asset purchases. Again within the lowest income quintile, mortgagor households show a larger decline this time. This reversal can itself be tested directly: in the lowest income group, the renter-minus-mortgagor difference under asset purchases is statistically distinguishable from that under conventional interest rate policy in all three windows, with the full-horizon test rejecting equality at the 1% level; relative to the corresponding difference under forward guidance, the full-horizon test rejects equality at the 5% level (Table E8 of the appendix). The income gradient also shifts to the mortgagor group and widens over time (Figure 4), while no gradient appears within the renter group under any measure or in any window, except that the food expenditure ratio is marginally significant in the first year. Forward guidance produces neither pattern. For outright owners, the significant results are for housing consumption and total consumption, which includes housing consumption, while non-housing consumption is not significant. The former reflects a direct change in holding costs rather than consumption demand (Appendix E); for the main outcome variable, the pattern is clear.

The gradient in non-housing consumption within the mortgagor group passes the test only in the third-year window, while the full-horizon test is not significant. This is consistent with the power issue discussed in Section 3.3: the gradient is close to zero in the first year and widens window by window thereafter, while the full-horizon test counts the periods in the first two years that carry little signal toward its degrees of freedom, thereby diluting its power. This pattern of gradual accumulation is precisely the lag predicted in advance by the refinancing channel and differs in character from the forward guidance window results in Section 4.2, which are scattered across regions and periods and mostly reverse or decay thereafter. The expenditure composition within the same group provides stronger corroborating evidence: following a contractionary asset purchase shock, the food expenditure ratio for low-income mortgagor households rises more than that for high-income mortgagor households, and this gradient is significant in both the second- and third-year windows and in the full-horizon test ($p = 0.004$), in the same direction as the gradient in non-

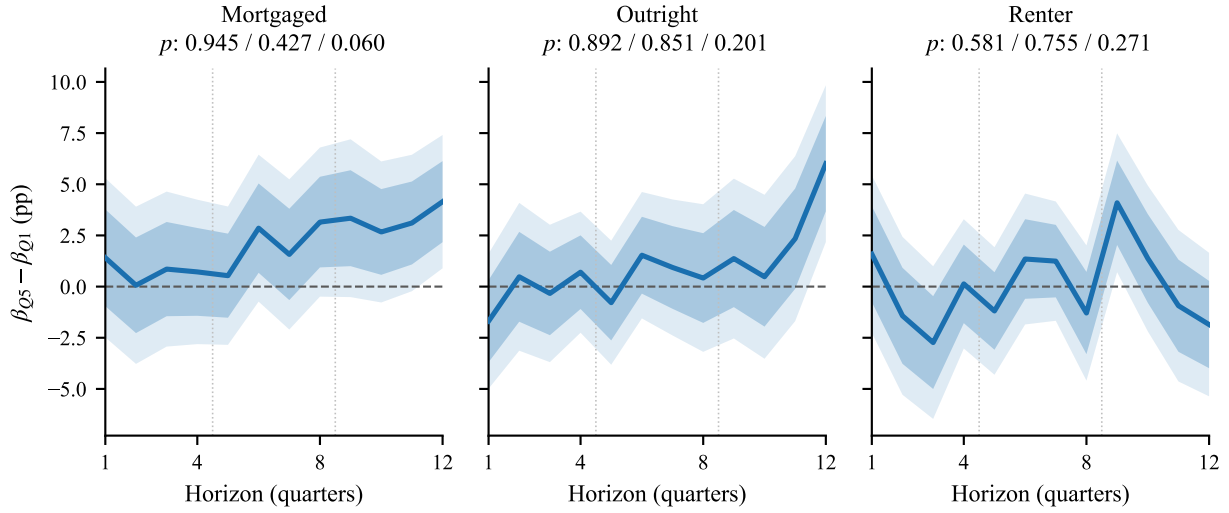


Figure 4: The $Q1$ – $Q5$ Gap in Log Non-Housing Consumption per Capita — LSAP Shock, by Housing Tenure

Notes: (1) Each panel plots the distributional gap itself, $\delta^h \equiv \beta_{Q5}^h - \beta_{Q1}^h$, for one tenure group, rather than the five quintile responses separately as in Figure E1. A positive value means that $Q1$'s cumulative response lies below $Q5$'s, that is, $Q1$ loses more under a contractionary shock. (2) Estimates are Standard LP, matching the basis for all inference in this paper; no smoothing is applied. (3) Shaded bands are ± 1 and ± 1.645 bootstrap standard deviations (68% and 90%), obtained from the residual moving block bootstrap ($B = 5000$, $L = 12$). (4) The three p -values in each panel title are the sub-window joint tests over $h = 1$ –4, 5–8, and 9–12 (Section 3.3); dotted vertical lines mark the window boundaries. The same window tests for all outcomes are reported in Table E6. (5) Analysis sample: 1992 Q1–2019 Q2 (110 quarters).

housing consumption, with low-income borrowers facing greater budgetary pressure (Table E6). For conventional interest rate policy, the expenditure composition is likewise consistent with the contract explanation: following a contractionary shock, the food expenditure ratio for low-income households within the renter group rises more than that for high-income households, with this difference significant in both the first- and third-year windows, consistent with rent increases placing greater pressure on low-income renters; this does not alter the conclusion regarding non-housing consumption set out above.

In short, no gradient appears within the renter group, so the only source of evidence for the employment channel in this design does not hold. The gradient is concentrated within the mortgagor group, and the condition for the cash flow channel holds. Full cell-by-cell test results for all consumption measures are provided in Table E6 of the appendix.

Forward guidance does not reject the null in any pairwise comparison within income groups, consistent with the contractual interpretation: it generates no current-period cash flow and therefore produces no systematic differences across tenure. Following a contractionary forward guidance shock, however, non-housing consumption for the lowest-income households within the outright owner group rises, while it falls in the other two groups. This may be related to the higher share of

older households in this group that have already paid off their mortgages. None of the responses for the lowest-income households in the three groups is statistically significant, so these results indicate direction only. The two unconventional instruments are therefore not interchangeable: asset purchases operate through mortgage contracts, whereas forward guidance operates through no contractual cash flow channel.

Table 3: Mortgagor versus Renter Households within the Same Income Quintile: Log Non-Housing Consumption per Capita

Quintile	FFR			FG			LSAP		
	$h=1-4$	5-8	9-12	$h=1-4$	5-8	9-12	$h=1-4$	5-8	9-12
Q1	-1.68	-2.09	-1.93*	-0.89	-0.03	-0.12	+2.81	+2.88	+3.91**
Q2	-1.58**	— ^a	-1.04	+0.19	+1.01	+1.34	+1.87	-0.61	-1.20
Q3	-1.09	-1.97**	-0.91	+0.10	+0.45	+0.40	+0.10	-0.18	+0.24
Q4	+0.65	+1.25	+0.78	+1.13	+0.73	+0.44	— ^a	-1.76	-0.69
Q5	-0.78	-0.45	-1.65	-1.52	-0.41	+0.17	+1.44	+0.88	+1.26

^a The joint test rejects in this window, but the path changes sign within it, so the window average is not reported as an effect size (mean $|d^h|$ exceeds twice $|\text{mean } d^h|$).

Notes: each cell reports the window average of $d^h \equiv \beta_{\text{Renter}}^h - \beta_{\text{Mortgaged}}^h$ within the stated income quintile, in percentage points, following a one-unit contractionary shock. A positive value means that the consumption of mortgagor households falls more than that of renter households with the same income; a negative value means that renters' consumption falls more. Stars refer to the bootstrap joint test of $H_0 : d^h = 0$ for all h in the window (residual moving block bootstrap, $B = 5000$, $L = 12$): * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Results for the other outcomes are reported in Table E7.

Contracts explain which households are reached but not why those reached differ among themselves. The key point is that every borrower holds a prepayment option, but exercising it is subject to several thresholds: closing costs, verifiable income, sufficient equity, and a credit assessment that must be passed (Beraja et al., 2019). Low-income borrowers are precisely the group for whom these conditions bind most tightly, with monthly payments accounting for the largest share of disposable income and the thinnest financial buffers; higher-income borrowers, whose payments account for a much smaller share of income, are less dependent on this route for relief (Di Maggio et al., 2017). When a contractionary asset purchase shock raises long-term rates and spreads together, the door closes for everyone, but those with the most constrained budgets are the households that relied most on this option for relief.

This also explains the timing. Refinancing must be initiated by the borrower, and application, appraisal, approval, and closing each take time; what matters is not any individual refinancing decision, but the cumulative budgetary effect of refinancing remaining out of reach. If the distributional effect were driven mainly by asset prices and wealth effects, it should emerge soon after the announcement. The observed pattern is the reverse: the gradient within the mortgagor group is close to zero in the first year, widens from one window to the next thereafter, and is largest

in the third year. The timing therefore provides a piece of evidence independent of the tenure decomposition.

5.3 Regional Differences and Limitations

A natural explanation is that the Northeast has a higher share of mortgagor households, but the data do not support this. Tenure composition is very similar across the four regions, with the gap in the mortgagor share between the highest and lowest income quintiles at around 60 percentage points in each region (Table E2 of the appendix). A pure composition difference would, moreover, predict the opposite sign: mortgagors are concentrated at the high-income end, so if asset purchases affected only mortgagor households, the decline should be larger for the high-income group. One possible explanation is that housing expenditure accounts for a slightly larger share of household budgets in the Northeast, so the same change in housing cash flow crowds out more non-housing consumption. The directions of the three measures in the Northeast are also consistent with this explanation, with non-housing consumption declining relatively more and housing consumption and the food expenditure ratio rising relatively more. The data in this paper are not sufficient to test this explanation.

The two estimations provide two ways of slicing the same sample rather than a joint estimation; the number of cells does not permit the data to be partitioned along all three dimensions simultaneously. The claim that the effect in the Northeast is driven by low-income borrowers is therefore an inference: the tenure estimation does not distinguish regions, while the regional estimation does not distinguish contracts. The employment channel (Kaplan et al., 2018) is likewise not ruled out, but the design of this paper cannot identify its contribution separately: the income gradient within the renter group is the only evidence that would support the employment channel, yet no such gradient is present in this sample. The employment channel therefore cannot be regarded as dominant.

The source of identification should also be noted. After excluding either recession quarters or extreme shock quarters, the gradient within the mortgagor group is no longer statistically significant. After excluding extreme shock quarters, a significant gradient emerges within the renter group in the third year, which is the pattern required by the employment channel criterion. The direct tests for all three groups fail to reject the null, however, so no shift in the path of any group can be established. The excluded quarters are precisely those in which the unconventional instruments were deployed on a large scale. The distinctions along the tenure dimension, including the absence of a gradient within the renter group, are therefore identified from crisis-period variation, as is the main result (Appendix F).

6 Robustness

This section reports four robustness checks to assess the sensitivity of the main findings to the estimation method, income classification, and sample selection. Table 4 summarizes how the two main results perform under each check, and full results for each check are provided in Appendix F.

Consistency between standard and smoothed local projections. In the regional estimation, the relative ordering of $Q1$ and $Q5$ is the same under both specifications across all regions and consumption variables, and the significance judgments are identical. In the tenure estimation, the paths for some tenure groups under the LSAP shock show visible magnitude shifts, reflecting bias introduced by the B-spline parameterization under weak-signal conditions, which is why Figure 4 uses the standard local projections for display. Full comparisons are provided in Appendix C.

Alternative income classification variable. The baseline specification assigns households to quintiles based on their relative position within the annual regional income distribution; as an alternative, this paper uses the BLS INC_RANK variable, which gives each household's percentile in the national income distribution. $Q1$ in the baseline consists of households that are relatively poorest within each region, while $Q1$ under INC_RANK consists of households that are poorest nationally; households in high-cost-of-living regions that are relatively poor locally but rank in the middle nationally are assigned to different groups under the two schemes. The two classifications use exactly the same sample, so their significance levels can be compared directly. The results show that the Northeast is insignificant in the first year and significant in the second and third years under both classifications, with a slightly larger difference under INC_RANK, while the remaining three regions are insignificant under both (Table 4 and Table F1 of the appendix). The core finding of this paper therefore does not depend on the classification based on within-region relative income.

Excluding recession and extreme shock quarters. I re-estimate the model after separately dropping the recession quarters defined by the National Bureau of Economic Research (2001Q1–2001Q4 and 2007Q4–2009Q2, reducing the available anchors from 96 to 85) and the extreme quarters in which any shock series exceeds two standard deviations in absolute value (15 quarters, leaving 81 anchors). The following discussion concerns the income difference in non-housing consumption under the LSAP shock in the Northeast. The two exclusions push the point estimate in opposite directions: excluding extreme shock quarters widens the difference, whereas excluding recession quarters narrows it (Table 4). In both cases, the direct test of whether the difference paths in the two samples are the same fails to reject the null over the full horizon, and the same holds for the cells in the mechanism analysis. By window, the third-year window rejects the null at the 10% level after excluding extreme shock quarters, but the difference widens rather than narrows. The resolution of the test is nonetheless comparable to the magnitude of the effect being tested, so it can rule out a large shift but not necessarily a moderate attenuation. The excluded quarters are

precisely those with the largest shock variance and the main source of information for identifying the coefficient, so the standard errors rise by approximately 70% and 100%, respectively. After excluding recession quarters, all three window tests are clearly insignificant, rather than merely close to the threshold without crossing it; after excluding extreme shock quarters, the results remain significant. Whether this effect can be statistically distinguished therefore depends on whether the sample contains policy shocks of sufficient magnitude. Details are provided in Tables F2 and F3 of the appendix.

Deflating consumption by the price level. Because the specification includes each cell's own lagged dependent variable, deflation does not affect the estimates for the two quintiles identically, so this check is not mechanical. After deflating non-housing consumption, total consumption, and housing expenditure using CPIAUCSL from FRED and re-estimating the models, the conclusions for the Northeast under the LSAP shock and for the South under the FFR shock are unchanged, indicating that the distributional effects reported in this paper are not driven by price-level movements but reflect differences in real consumption (Table F4 of the appendix).

Table 4: Robustness of the $Q1$ – $Q5$ Gap: Regional and Mechanism Results

Panel A: Northeast, LSAP, non-housing consumption (regional estimation)					
Specification	$\bar{\delta}$ (pp) ^a	p (1–4)	p (5–8)	p (9–12)	p (stability) ^b
Baseline	+3.22	0.521	0.004	0.064	—
INC_RANK income classification	+4.05	0.513	0.009	0.023	0.235
Excluding extreme-shock quarters	+4.39	0.758	0.087	0.042	0.372
Excluding recession quarters	+2.29	0.843	0.327	0.283	0.385
Panel B: Mortgaged owners, LSAP, non-housing consumption (tenure estimation)					
Specification	$\bar{\delta}$ (pp) ^a	p (1–4)	p (5–8)	p (9–12)	p (stability) ^b
Baseline	+2.12	0.945	0.427	0.060	—
INC_RANK income classification	+2.24	0.706	0.350	0.100	0.791
Excluding extreme-shock quarters	+1.59	0.610	0.781	0.531	0.960
Excluding recession quarters	+1.45	0.939	0.612	0.150	0.639

^a $\bar{\delta}$ is the gap $\beta_{Q5}^h - \beta_{Q1}^h$ averaged over the 12 horizons, in percentage points; all p -values are bootstrap-corrected ($B = 5000$; $B = 2000$ for INC_RANK).

^b Tests whether the gap path under the alternative specification equals the baseline path at every horizon. The construction of each subsample, the resolution of this test, and the results for all 84 cells are reported in Appendix F.

7 Conclusion

7.1 Main Findings

This paper uses the three-factor monetary policy shock series of Swanson (2021) and the CEX consumption panel to estimate the distributional effects of FFR, FG, and LSAP on household consumption, with direct tests showing that the distributional effect of asset purchases is statistically distinguishable from those of the other two instruments. In the Northeast, expansionary quantitative easing produces a larger consumption boost for the lowest income quintile than for the highest, and a direct test shows that this effect is statistically distinguishable from those in the remaining three regions. Although the statistical evidence is concentrated in a single region, LSAP remains the instrument with the clearest direction and the most complete chain of evidence among the three. The distributional effects of FG are scattered across regions, consumption measures, and periods, with no consistent pattern. The results for conventional interest rate policy are concentrated in housing consumption and cannot be interpreted directly as an improvement in welfare.

The food expenditure ratio helps distinguish nominal consumption changes from changes in expenditure composition. Under the FFR shock, the $Q1$ – $Q5$ difference in the food expenditure ratio is insignificant in every window across all four regions; the smaller declines in nominal housing consumption and total consumption for $Q1$ partly reflect rent increases driven by higher interest rates, consistent with the result in Section 5.2 that renters show a larger consumption decline within the same income group. Under the LSAP shock, the difference in the food expenditure ratio in the Northeast is significant in the third-year window ($p = 0.041$), indicating that the progressive effect in non-housing consumption in this region reflects a genuine change in expenditure composition rather than a purely nominal or price effect.

The estimation split by housing contract shows that the three instruments reach households through different types of contracts. Within the same income group, renters show a larger consumption decline under conventional interest rate policy, mortgagors decline more under asset purchases, and there is no difference between the two under forward guidance; in the third-year window, the consumption decline for low-income mortgagors is approximately 4 percentage points larger than that for renters at the same income level. This reversal between asset purchases and conventional interest rate policy is confirmed by a direct test in the lowest income group. The income gradient under asset purchases appears only within the mortgagor group, which is inconsistent with an explanation in which the employment channel operates alone. The gradient is concentrated among low-income borrowers, who face the greatest difficulty in meeting the refinancing thresholds, and appears only after the second year, consistent with the time required for refinancing, whereas asset price effects should appear immediately. The prominence of the Northeast is consistent with housing carrying the greatest weight in local household budgets rather than with a higher share of

mortgagor households; this explanation is inferred separately from the regional and tenure splits. This conclusion does not conflict with the HANK framework of Kaplan et al. (2018), in which cash flow effects on borrowers are also present.

Two points are worth setting out regarding the scope of this progressive effect. First, it does not depend on the income classification: under the national income percentile, the results for the Northeast are almost identical to those in the baseline, and the timing is unchanged. Second, identifying this effect depends on whether the sample contains policy shocks of sufficient magnitude: excluding extreme shock quarters widens the point estimate, while excluding recession quarters narrows it by approximately 30%; in the latter case, the full-horizon test and all window-based tests fall well short of conventional significance levels. The results along the tenure dimension likewise depend on these quarters.

7.2 Policy Implications

These results provide direct consumption-side evidence for the debate on the distributional consequences of QE. Wealth-based critiques such as Domanski et al. (2016) argue that QE raises asset prices and benefits wealthy households that hold financial assets. This paper does not dispute the existence of the wealth channel but shows that, on the consumption side, the net effect runs in the opposite direction: expansionary LSAP produces a larger consumption boost for lower-income households than for higher-income households, and this effect operates mainly through borrowers' cash flow. The statistical evidence for this conclusion is concentrated in the Northeast, though it does not hinge on the specific income classification. The distributional consequences of monetary policy therefore depend on which side of the balance sheet is considered: wealth effects on the asset side favor wealthier households, whereas mortgage cash flow on the liability side favors low-income borrowers. At the same time, the difference in distributional outcomes across the three instruments is itself a policy-relevant fact: in an environment where the ZLB renders the FFR ineffective, LSAP and FG are not interchangeable forms of "easing" and differ materially in their capacity to improve outcomes for lower-income households. The evidence for the three instruments is unevenly distributed geographically, with that for LSAP concentrated in the Northeast; this pattern is visible only when the estimation is grouped by region. The direct test in this paper further shows that the distributional effect of LSAP in the Northeast is statistically distinguishable from that in the remaining three regions, indicating that a uniform federal monetary policy can produce different distributional consequences across regions.

7.3 Limitations and Future Research

Several limitations apply. The rotating panel structure of the CEX tracks individual households for at most five quarters; this paper therefore constructs time series from grouped cell means, bypassing the short-panel constraint at the cost of being unable to observe consumption dynamics at the household level. The three-factor decomposition of Swanson (2021) mitigates but does not eliminate the Fed information effect: the signal about the economic outlook embedded in FOMC announcements cannot be fully separated from the monetary policy stance itself, and the robustness checks excluding recession and extreme shock quarters reduce but do not resolve this concern.

The evidence from the mechanism analysis is consistent with the mortgage cash flow channel and inconsistent with an explanation in which the employment channel operates alone, but the design of this paper cannot separately quantify the contributions of the two channels, so a substantive role for the employment channel cannot be ruled out. The three limitations discussed in Section 5 also apply: the explanation that regional differences stem from the weight of housing in household budgets is inferred separately from the two splits, the distinctions along the tenure dimension depend on crisis-period shock variation, and the low-income mortgagor cell is small, so its estimates are imprecise. The difference between the Northeast and the other regions is supported by direct tests, but the sample length is insufficient to determine whether the remaining three regions also differ from one another.

These limitations point to several directions for future research. Incorporating quarterly labor income microdata such as the Survey of Income and Program Participation (SIPP) would allow the employment channel to be tested directly, rather than relying on the indirect inference from the tenure estimation. Similarly, combining loan-level refinancing data would allow the cash flow channel to be tested directly, as well as whether regional differences are indeed driven by the weight of housing in household budgets, without relying on indirect inference from the two separate slices. Whether the distributional effect varies with shock size could be tested with a formal nonlinear specification, such as a threshold or breakpoint regression, rather than the sample-split approach used here. Sample splitting drops the quarters with the largest shock variance, which are precisely the main source of identification for the effect. A nonlinear specification may address this question while retaining all observations. Extending the same framework to other advanced economies that have implemented asset purchase programs, such as the European Central Bank's Public Sector Purchase Programme (PSPP), would allow the generalizability of these findings to be assessed and clarify the meaning of the within-region relative income measure in a cross-country context.

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A Data Description and Diagnostics

A.1 CEX Interview Survey: Variable Construction

Household pre-tax income is taken from FINCBTXM, the BLS multiply imputed total family income before taxes; for 1992–2003, before this variable is available, the reported (non-imputed) measure FINCBTAX is used instead. For 1992–2003, households with incomplete income reports (RESPSTAT = 2) are dropped. FINCBTAX is not imputed in those years, and the income of such households is either zero or missing (13.9% of records in this period) or positive but understated (a further 5.8%). Retaining them would mechanically assign them to lower income quintiles, with the first group falling entirely into the lowest quintile. From 2004 onward, FINCBTXM imputes incomplete reports, so all households are retained in those years. Households with negative income are dropped in all years. Income quintiles are constructed within each calendar year $\times$ Census region cell, ranking households by their position in that year-region income distribution; this within-year, within-region ranking is the baseline income classification used throughout the paper, in contrast to the national ranking (INC_RANK) used in the robustness check of Appendix F.1.

Consumption variables are constructed from CEX Interview Survey expenditure totals. Total consumption is TOTEXPCQ; non-housing consumption subtracts housing-related expenditure, HOUSCQ (which includes rent and mortgage payments), from TOTEXPCQ, so that interest-rate-driven changes in monthly housing payments do not mechanically enter the measure of consumption demand. All consumption variables are converted to a per-capita basis by dividing by household size (FAM_SIZE) and are expressed in nominal dollars (see the note to Table A2 below). Household-level survey weights (FINLWT21) are applied throughout when computing cell means, so that the resulting group averages are nationally representative rather than artifacts of the CEX’s rotating sample design. Total expenditure and income are winsorized at the 99.9th percentile within each calendar year to limit the influence of extreme outliers on cell-level averages.

Housing tenure status is coded from CUTENURE into three categories: mortgaged owners (households with an outstanding mortgage), outright owners (households who own their home with no mortgage), and renters. This three-way classification, crossed with the same five income quintiles, forms the tenure-by-income panel used in Section 5.

A.2 Sample Coverage and Adequacy

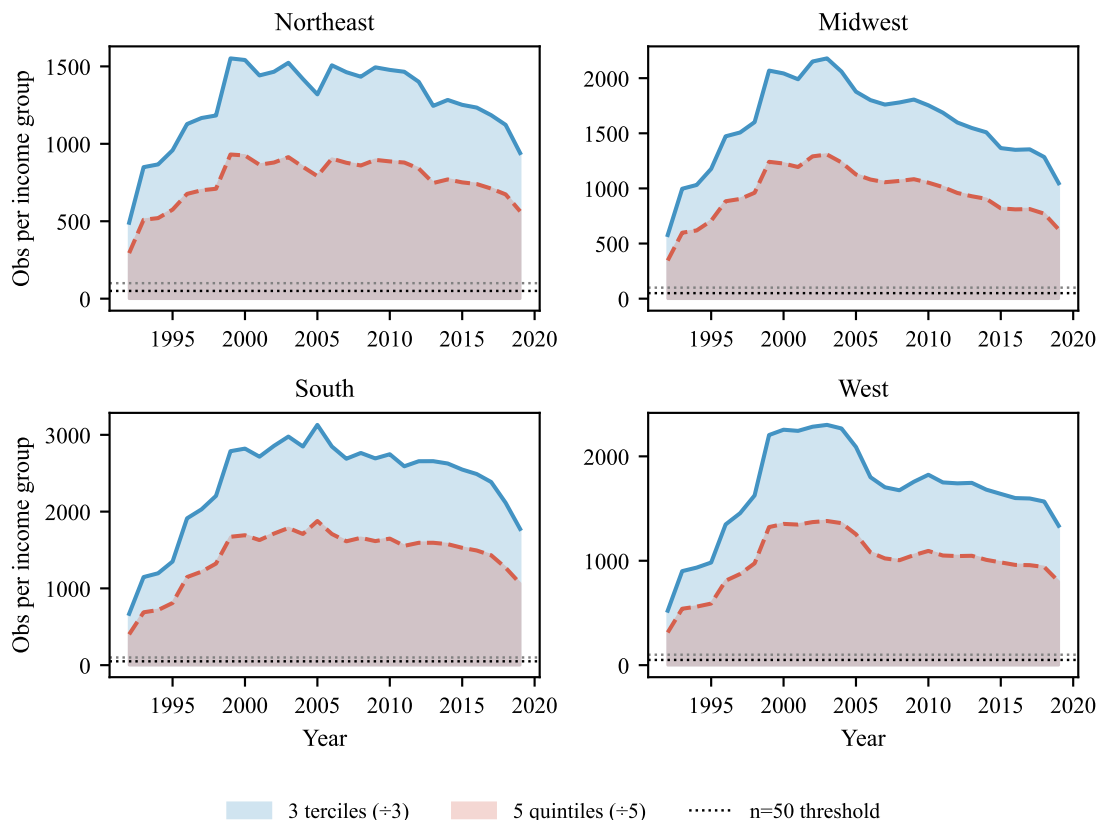


Figure A1: Annual Observations per Income Group within Region

Notes: For each Census region, the panel plots the annual number of CEX Interview Survey household observations per income group (summed across the four quarters of each calendar year), under two alternative partitions: terciles (3 income groups, shaded blue) and quintiles (5 income groups, shaded red, the partition used in the main analysis). Dotted lines mark reference thresholds of $n = 50$ and $n = 100$ observations. The counts plotted are raw CEX Interview Survey observations, before the sample restrictions described in Section 2.1; the estimation sample is smaller, and more so before 2004, where the income-variable correction removes incomplete income reporters. The conclusion is unaffected either way: on the estimation panel itself the smallest region $\times$ quintile cell contains 240 observations in a year (median 926), so the cells used to construct the quarterly averages are not constrained by small-sample concerns.

A.3 PCA: Macro Control Variables

Table A1 lists the ten FRED series used to construct the macroeconomic control factors described in Section 2.3. Four interest rate series — the FFR, the 3-month Treasury bill rate, the 10-year Treasury yield, and the 30-year mortgage rate — are deliberately excluded from the principal component analysis, since including them would let the resulting factors partially absorb variation in the monetary policy shocks themselves, attenuating the estimated impulse responses.

Standardizing the remaining ten variables and extracting principal components yields three factors with eigenvalues exceeding one under the Kaiser criterion (3.77, 1.82, and 1.29), jointly explaining 68.2% of the total variance (37.4%, 18.0%, and 12.8% individually). Figure A2 plots

Table A1: Macro Control Variables: Data Sources and Transformations

FRED Mnemonic	Variable	Source Agency	Frequency	Transformation
CPIAUCSL	Consumer Price Index for All Urban Consumers	U.S. Bureau of Labor Statistics	Monthly	100× log-difference
GDPC1	Real Gross Domestic Product	U.S. Bureau of Economic Analysis	Quarterly	100× log-difference
UNRATE	Unemployment Rate	U.S. Bureau of Labor Statistics	Monthly	Level
INDPRO	Industrial Production: Total Index	Board of Governors of the Federal Reserve System	Monthly	100× log-difference
PCEPI	Personal Consumption Expenditures: Chain-type Price Index	U.S. Bureau of Economic Analysis	Monthly	100× log-difference
PAYEMS	All Employees, Total Nonfarm	U.S. Bureau of Labor Statistics	Monthly	100× log-difference
HOUST	New Privately-Owned Housing Units Started	U.S. Census Bureau / U.S. Dept. of Housing and Urban Development	Monthly	100× log-difference
CSUSHPISA	S&P/Case-Shiller U.S. National Home Price Index	S&P Dow Jones Indices LLC	Monthly	100× log-difference
TWEXMMTH	Trade Weighted U.S. Dollar Index: Broad	Board of Governors of the Federal Reserve System	Monthly	100× log-difference
NASDAQCOM	NASDAQ Composite Index	Nasdaq OMX Group, Inc.	Daily	100× log-difference

Notes: All series are retrieved from the FRED database maintained by the Federal Reserve Bank of St. Louis; the table reports each series' FRED mnemonic and the agency that originally compiles the underlying data. All series are resampled to quarterly frequency (mean within quarter for monthly/weekly/daily series; last value for already-quarterly series), log-differenced ($\times 100$, to approximate percentage changes) where appropriate, standardized, and restricted to 1992Q1–2019Q2 before entering the principal component analysis described in the text.

the full eigenvalue spectrum. The first factor loads most heavily on real activity measures (real GDP growth, industrial production, nonfarm payrolls) and is interpreted as an economic activity factor; the second loads on price measures (CPI and PCE inflation) and is interpreted as a price and inflation factor; the third loads on labor market slack (the unemployment rate) and is interpreted as a labor market pressure factor. These three factors enter the local projection specification as $\mathbf{PC}_t$ in Equation (1).

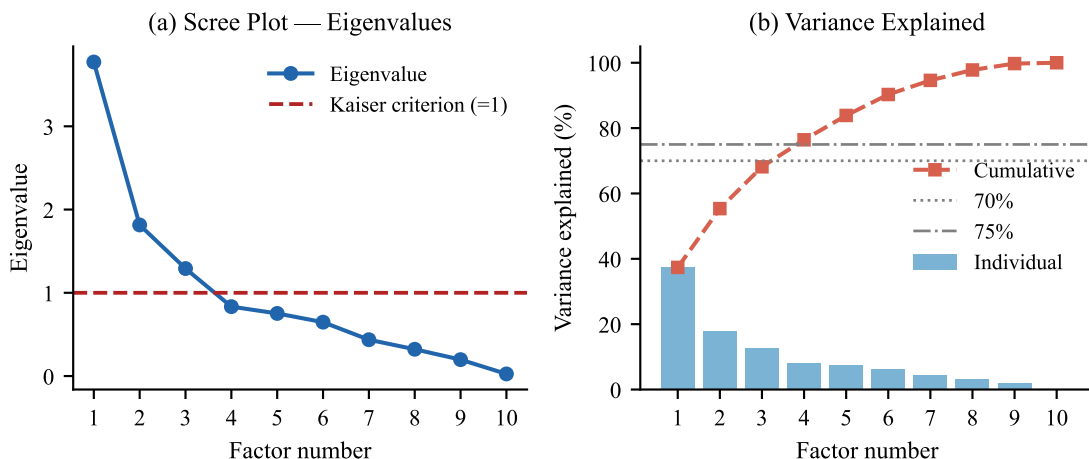


Figure A2: PCA Scree Analysis — Macro Control Variables

Notes: Eigenvalues from the principal component analysis of the ten macroeconomic control variables in Table A1. The dashed line marks the Kaiser threshold (eigenvalue = 1); the first three components lie above this threshold and are retained, jointly explaining 68.2% of total variance.

A.4 Summary Statistics

Table A2 reports summary statistics for the region-by-income panel underlying the regional estimation in Section 4: mean observations per quarter, mean consumption, and food share for each of the twenty region $\times$ income-quintile cells. Consumption figures are in nominal per-capita dollars (see Section A.1); since the distributional comparisons throughout the paper are differences between quintiles within the same quarter, deflation affects them very little, as the deflation check in Appendix F.4 confirms. Table A3 reports descriptive statistics for the three Swanson (2021) shock series — mean, standard deviation, and extreme observations — over the full sample period, complementing the discussion of shock timing in Section 2. The shock series are summed within each quarter, aggregating them from meeting-date frequency to quarterly frequency. This ensures consistency with the consumption dataset and avoids the mechanical zeros that would otherwise arise in roughly one-third of monthly observations due to the irregular spacing of FOMC meetings.

Table A2: Summary Statistics: CEX Interview Survey Panel (1992–2019)

Region	Income	Quarters with data	Mean obs per quarter	Non-housing cons. (\$/quarter, pc)	Total cons. (\$/quarter, pc)	Food share (%)
<i>Northeast</i>						
	Low (Q1)	112	173	\$965	\$1,663	19.4
	Q2	112	173	\$1,255	\$2,083	17.4
	Q3 (Mid)	112	172	\$1,606	\$2,524	15.9
	Q4	112	172	\$1,884	\$2,848	14.1
	High (Q5)	112	172	\$2,742	\$4,109	12.1
<i>Midwest</i>						
	Low (Q1)	112	216	\$1,051	\$1,657	18.0
	Q2	112	216	\$1,334	\$2,038	16.2
	Q3 (Mid)	112	215	\$1,604	\$2,365	14.8
	Q4	112	216	\$1,827	\$2,625	13.7
	High (Q5)	112	215	\$2,502	\$3,537	11.9
<i>South</i>						
	Low (Q1)	112	329	\$936	\$1,512	20.1
	Q2	112	329	\$1,195	\$1,821	18.1
	Q3 (Mid)	112	329	\$1,460	\$2,166	16.4
	Q4	112	328	\$1,749	\$2,529	14.9
	High (Q5)	112	328	\$2,508	\$3,577	12.4
<i>West</i>						
	Low (Q1)	112	234	\$1,072	\$1,790	18.4
	Q2	112	234	\$1,307	\$2,070	16.9
	Q3 (Mid)	112	234	\$1,602	\$2,470	15.3
	Q4	112	233	\$1,966	\$2,977	13.9
	High (Q5)	112	234	\$2,860	\$4,261	11.9

Notes: Averages over 1992Q1–2019Q4 of quarterly cell statistics; quarterly cell means use CEX survey weights (FINLWT21). Consumption columns report the exponential of the average quarterly log; observations and food share are simple averages. Income quintiles (Q1–Q5) are computed within each calendar year $\times$ Census region using before-tax income: FINCBTXM (BLS imputed) from 2004 and the reported FINCBTAX for 1992–2003, excluding incomplete income reporters in the latter period. Consumption variables are expressed in nominal dollars per capita. “Non-housing consumption” excludes HUSCQ. Sample: CEX Interview Survey, 4 Census regions, 5 income quintiles, 112 quarters.

Table A3: Descriptive Statistics: Swanson (2021) Monetary Policy Shocks

Shock factor	N	Mean	SD	Min	Median	Max
FFR (Fed Funds Rate)	110	0.053	1.033	-5.576	0.255	2.117
FG (Forward Guidance)	110	0.003	1.291	-3.349	0.046	3.751
LSAP (Asset Purchases)	110	0.015	0.913	-5.298	-0.018	2.312

Notes: Quarterly aggregates (sum within quarter) of FOMC-meeting-level Swanson (2021) shocks, 1992Q1–2019Q2.

B Estimation and Inference Details

This appendix reports the details of the estimation and inference in Section 3, including the selection of lag orders, the number of basis functions for the smoothed local projections and the construction of the confidence bands, and the size simulations and bootstrap settings for the Wald tests.

B.1 Lag Selection

The parameter settings are determined by information criteria and the relevant literature (Coibion et al., 2017; Cloyne et al., 2020). The maximum horizon is $H = 12$ (three years), with a Newey–West bandwidth of $h + 2$. Among the lag orders, p_y and p_x are determined by a grid search over 1–8 using four information criteria: AIC, BIC, HQIC, and AICc. All four criteria agree on $p_x = 1$; the four criteria for p_y range from 3 to 6, and I use the BIC result of $p_y = 3$, the most parsimonious choice. p_s is not determined by information criteria, since these measure the trade-off between goodness of fit and parameter parsimony, while the role of p_s is to control for serial correlation in the shock series itself to ensure identification; some adjustment based on the properties of the variable itself and its economic meaning is therefore needed. In the grid, the BIC value decreases monotonically in p_s with no interior optimum, which is consistent with this distinction. I therefore rely on two diagnostics instead. The Ljung–Box white noise test (Ljung and Box, 1978) on the three shock series shows that all three are close to white noise, with only the LSAP series exhibiting a significant autocorrelation at lag 1. A sensitivity analysis on the IRF point estimates shows that the standardized shift in the point estimate when p_s increases from 1 to 2 is only 0.03 standard errors, a shift that increases gradually thereafter and becomes substantial beyond lag 5. I therefore set $p_s = 2$, since this controls for the only significant lag-1 autocorrelation, and controlling for one additional lag has a negligible effect on the estimation results. The final parameters are $(p_s, p_y, p_x) = (2, 3, 1)$. This set of lag orders, together with the 12-horizon lead, determines the anchors available for estimation. Each cell’s series spans 1992Q1–2019Q4, a total of 112 quarters. This is two quarters longer than the 110-quarter period stated in Section 2, because the shocks and control variables end in 2019Q2, whereas the consumption data extend through 2019Q4. The 12-horizon lead from the final anchor, 2016Q3, therefore extends to 2019Q3 in the consumption data. The earliest 3 quarters are reserved for the lag terms and the final 13 quarters for the lead, leaving anchors from 1992Q4 to 2016Q3, or 96 in total. The same set of anchors is used at all horizons, and each regression estimates 13 parameters.

B.2 Smooth LP: Basis Selection and Confidence Bands

The candidate set for K is $\{3, 4, 5\}$, with the upper bound set at $5 = \lfloor H/2 \rfloor - 1$ to prevent overfitting when K is too large. K is selected independently for each cell via GCV, and in practice, $K = 5$, the upper bound, is selected for all region-by-income-group-by-consumption-variable combinations. To check whether this upper bound is too restrictive, I expand the candidate set to $\{2, \dots, 10\}$ and rerun GCV; in the cells examined, the new upper bound of $K = 10$ is selected in every case, rather than the selection converging to some interior value. GCV repeatedly selecting the boundary value after the candidate set is expanded indicates a tendency to overfit, indicating that the upper bound $K_{\max} = \lfloor H/2 \rfloor - 1$ is itself a necessary regularization constraint. I therefore treat $K = 5$ as a fixed regularization choice, rather than claiming it to be an unconstrained optimum in the GCV sense.

The 68% and 90% confidence bands for $Q1$ and $Q5$ in the figures do not come directly from the standard errors of the smoothed specification; instead, they are constructed as follows. Each bootstrap resample path from the standard local projections is smoothed using the same B-spline projection matrix $P = B(B^\top B)^{-1}B^\top$ applied to the point estimates, and the standard deviation of these smoothed paths at each horizon is taken as the width of the confidence band, centered on the point estimate from the smoothed local projections. The confidence bands therefore have the same degree of smoothness as the point estimate curve; their width comes entirely from the bootstrap, while the point estimate itself still comes from the smoothed local projections.

B.3 Size of the Wald Test and the Bootstrap

The size of the joint test is assessed with a Monte Carlo experiment calibrated to the data. For a given cell, I first estimate Equation (1) for $Q1$ and $Q5$ at every horizon with the baseline lag orders, and then set the coefficient on the contemporaneous shock to zero while keeping all other coefficients—own lags, macroeconomic factors, and seasonal dummies—at their estimated values. Each synthetic dataset combines these null coefficients with residuals resampled by the moving block bootstrap ($L = 12$), using one set of block indices for both quintiles and all 12 horizons. The synthetic data therefore share the dynamics and the cross-quintile and cross-horizon dependence of the actual data, while the null hypothesis holds exactly. On each synthetic dataset I compute the Newey–West test of Equation (3) and the bootstrap test with 200 replications, and record whether each rejects. The experiment uses 500 synthetic datasets for each of two cells: non-housing consumption in the Northeast under the LSAP shock, which carries the main regional result, and the food expenditure ratio among mortgagor households under the FFR shock.

Table B1 reports the empirical rejection rates. For the $Q1$ – $Q5$ difference, the Newey–West test rejects the true null in between 35.0 and 44.8 percent of the datasets at the nominal 10% level, between 28.2 and 35.8 percent at 5%, and between 15.8 and 22.4 percent at 1%, that is, 3.5 to 22

times the nominal rate, with the distortion largest at the strictest level. The bootstrap test rejects in 8.4–11.4, 4.4–5.2, and 0.6–1.0 percent of the datasets at the same three levels, within simulation error of the nominal rates. The tests on a single quintile show the same contrast; the bootstrap rejection rate farthest from the nominal level is 12.4 percent, at the 10% level, for $Q5$ among mortgagors.

Table B1: Empirical Rejection Rates under a True Null (%)

Nominal level	Newey–West χ^2			Bootstrap		
	10%	5%	1%	10%	5%	1%
<i>Northeast, LSAP, non-housing consumption</i>						
$Q1$ – $Q5$ difference	35.0	28.2	15.8	11.4	5.2	1.0
$Q1$ alone	46.8	40.4	29.8	10.2	4.4	1.6
$Q5$ alone	47.0	38.8	25.8	10.2	5.0	0.8
<i>Mortgagors, FFR, food expenditure ratio</i>						
$Q1$ – $Q5$ difference	44.8	35.8	22.4	8.4	4.4	0.6
$Q1$ alone	56.2	47.4	32.4	10.2	4.6	0.6
$Q5$ alone	55.6	49.0	37.8	12.4	7.2	1.8

Notes: each entry is the share of 500 synthetic datasets, generated with the shock coefficient set to zero at every horizon, in which the test rejects at the stated nominal level. The bootstrap test uses 200 replications within each synthetic dataset and block length $L = 12$. With 500 datasets the simulation standard error of a rejection rate is about 1.3 percentage points at the 10% level, 1.0 at 5%, and 0.4 at 1%.

In the bootstrap used for inference, the same set of block indices is applied across all 12 horizons, so that each resample corresponds to the same set of anchors at every horizon. The conclusions are insensitive to the choice of block length: 82 of 84 cells yield the same significance judgments across $L \in \{4, 8, 12, 16\}$.

C Standard LP and Smooth LP Comparison

This appendix presents the standard and smoothed local projection estimates side by side, covering the results reported in Sections 4 and 5. The two specifications differ only in that the smoothed specification constrains the sequence $\{\hat{\beta}_g^h\}_{h=1}^H$ in Equation (1) to a B-spline basis; they are otherwise identical. The main-text figures for the regional estimation use the smoothed specification, so the corresponding standard local projection estimates are given here; the main-text figures for the tenure estimation use the standard specification, so the corresponding smoothed estimates are given here. Since all Wald tests and significance annotations reported in the main text are already based on the standard errors from the standard local projections (Section 3.2), the comparisons below concern only the shape of the plotted paths and do not affect any of the reported significance conclusions.

C.1 LSAP Shock — Non-Housing Consumption (cf. Figure 2)

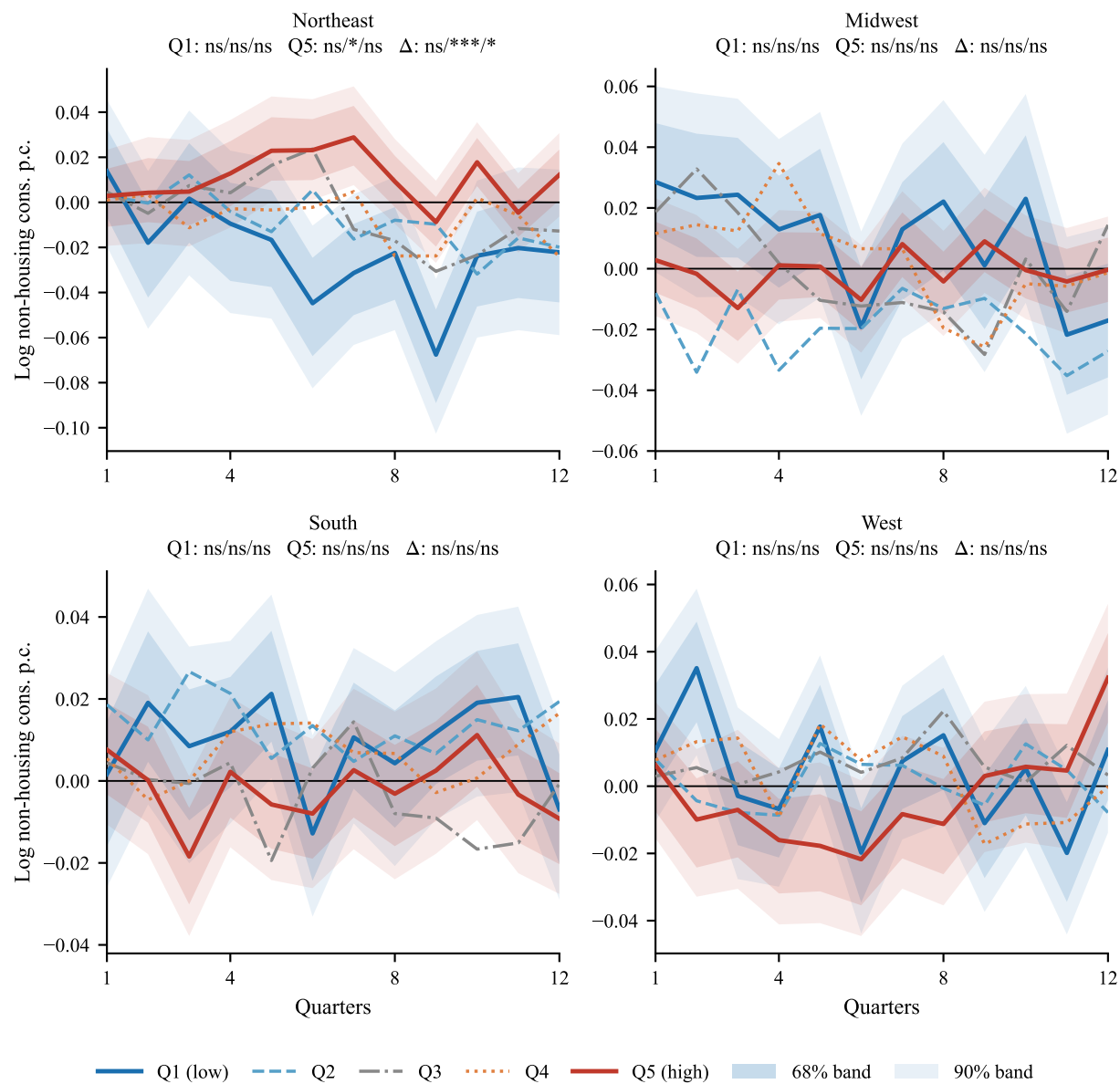


Figure C1: Standard LP IRF: Log Non-Housing Consumption — LSAP Shock. Compare with Figure 2 (Smooth LP).

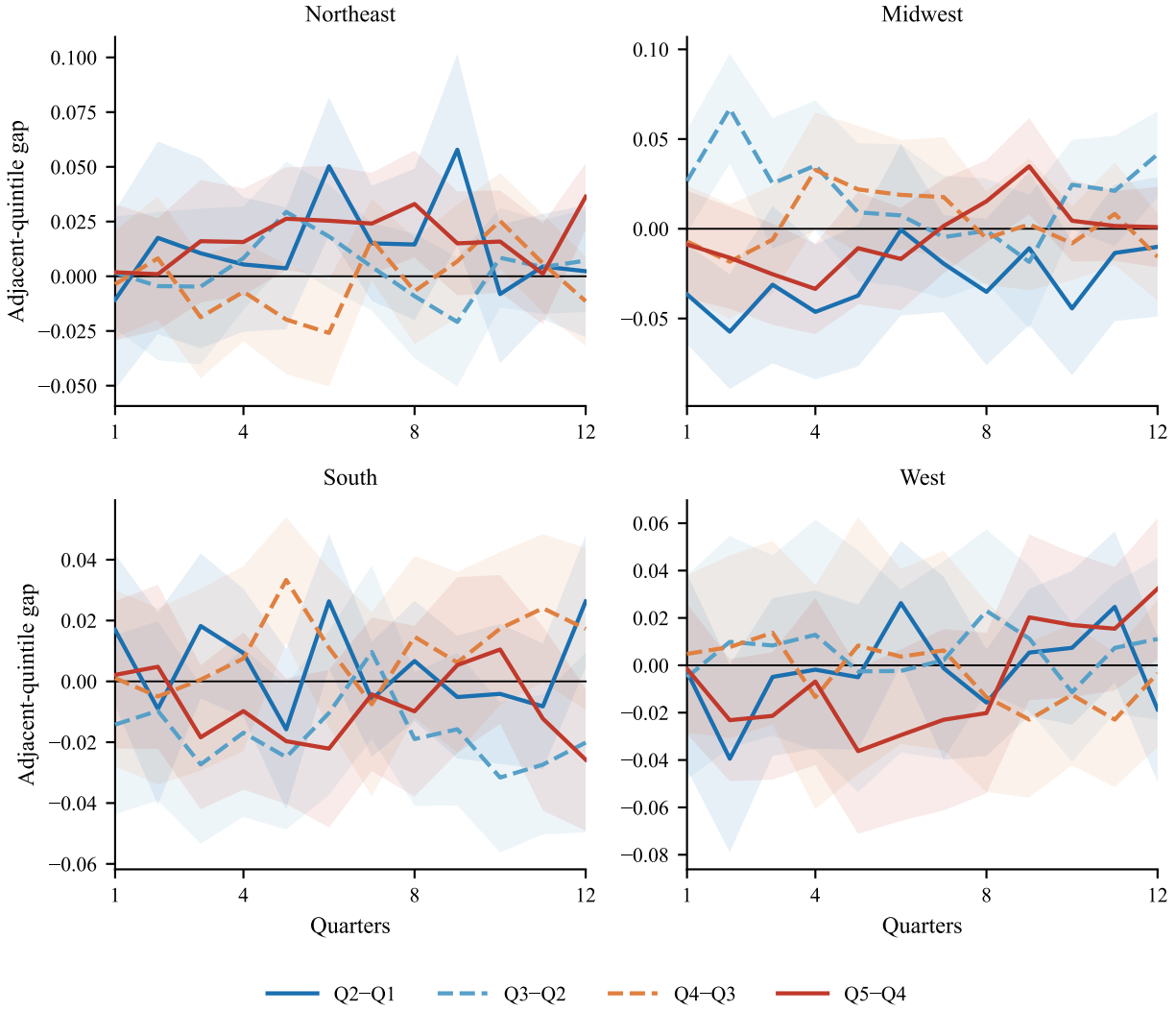


Figure C2: Standard LP Adjacent Gap: Log Non-Housing Consumption — LSAP Shock. Compare with Figure D4 (Smooth LP, this appendix).

C.2 FG Shock — Non-Housing Consumption (cf. Figure D5)

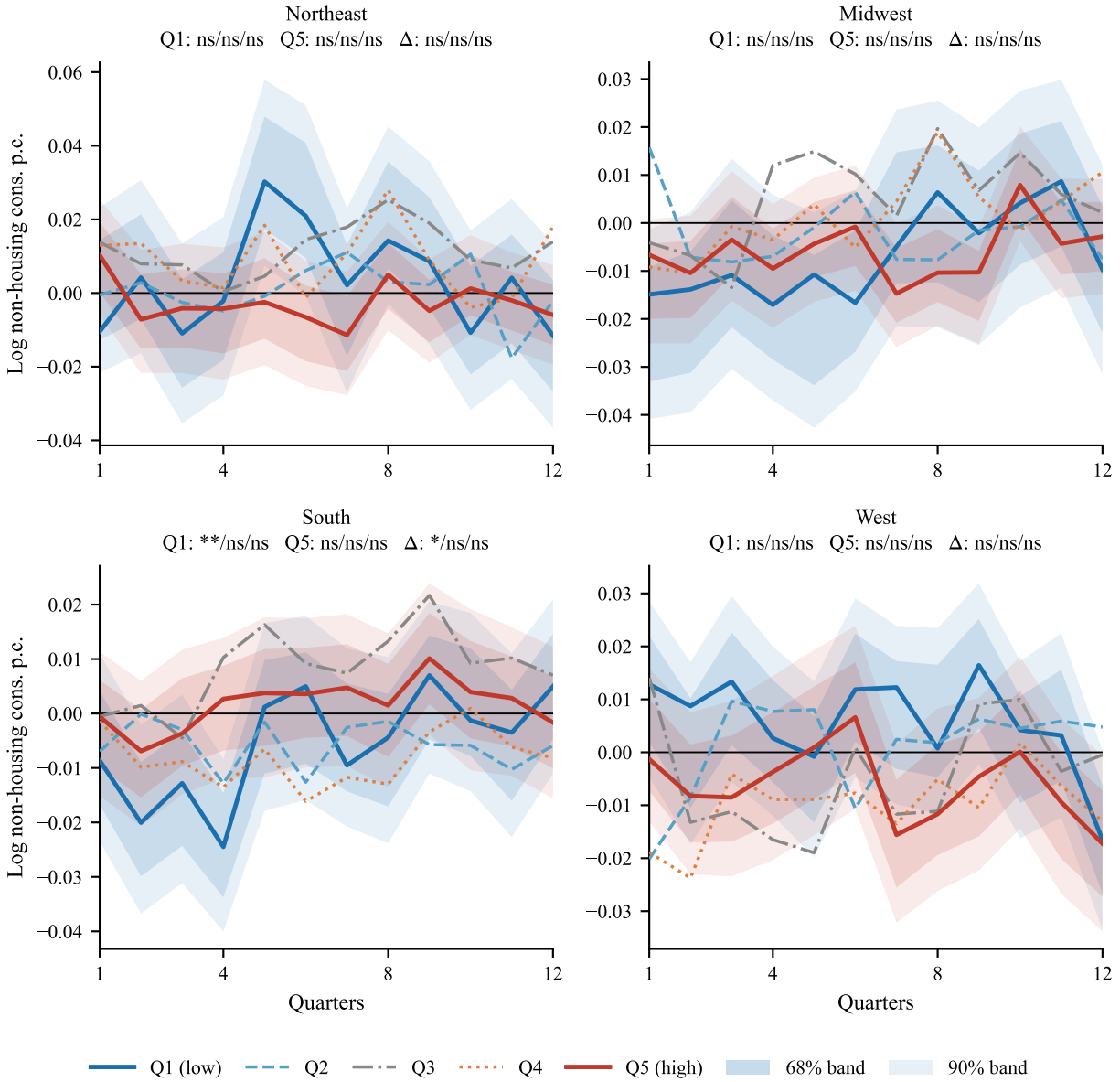


Figure C3: Standard LP IRF: Log Non-Housing Consumption — FG Shock. Compare with Figure D5 (Smooth LP).

C.3 FFR Shock — Non-Housing Consumption

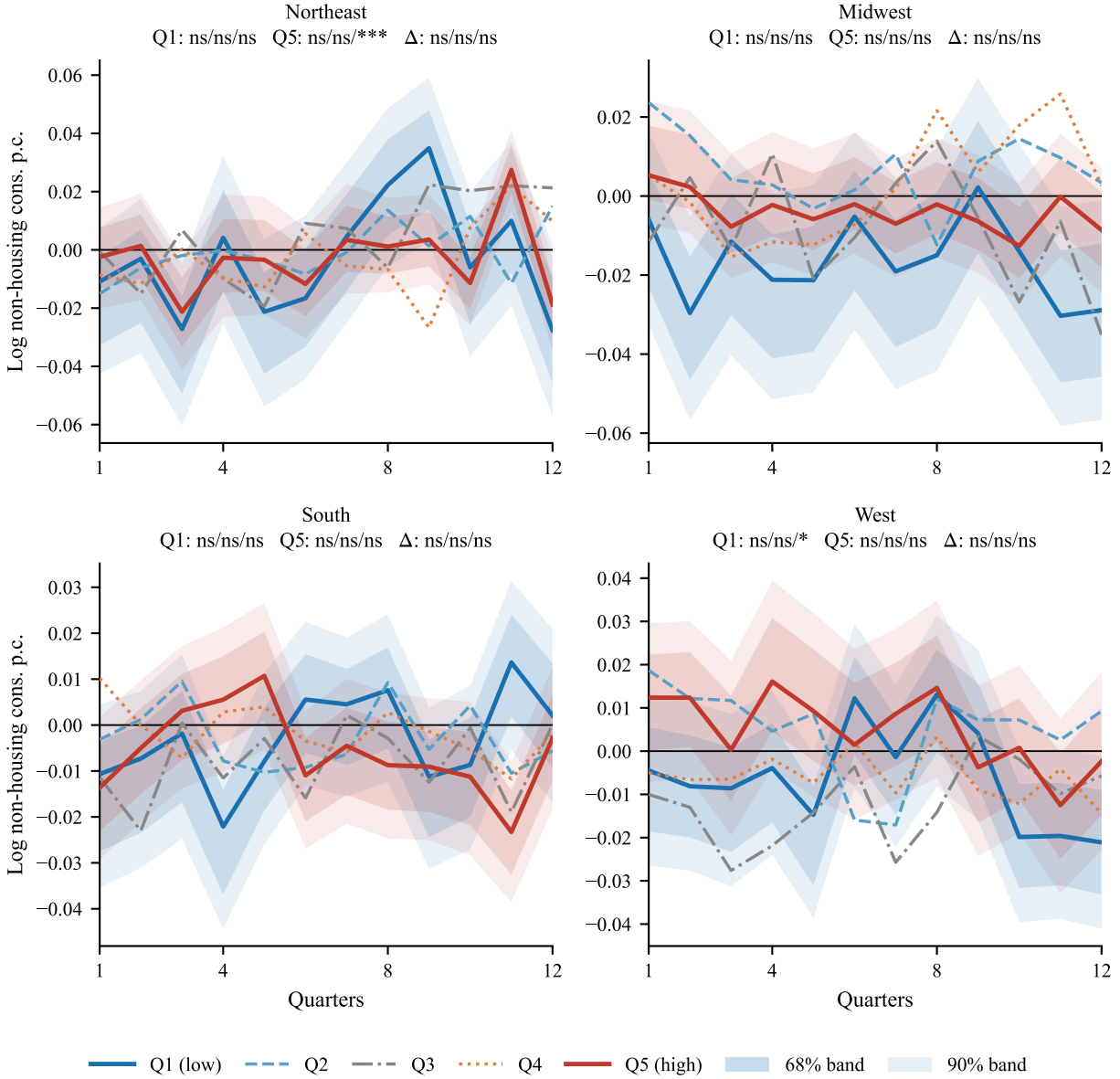


Figure C4: Standard LP IRF: Log Non-Housing Consumption — FFR Shock

C.4 Tenure Estimation — LSAP Shock (cf. Figure E1)

The tenure decomposition in both Figure 4 in the main text and Figure E1 in the appendix uses standard local projections rather than smoothed local projections. Figure C5 presents the smoothed version of the same estimates for comparison: under the tenure grouping, the sample size in each cell is smaller than under the regional grouping, so the signal is correspondingly weaker, and the B-spline parameterization produces visible magnitude shifts at several horizons, most notably for the outright owner group. The overall shape of the paths is the same under both specifications, and the relative position of $Q1$ and $Q5$ is unchanged within each group, but because the shifts

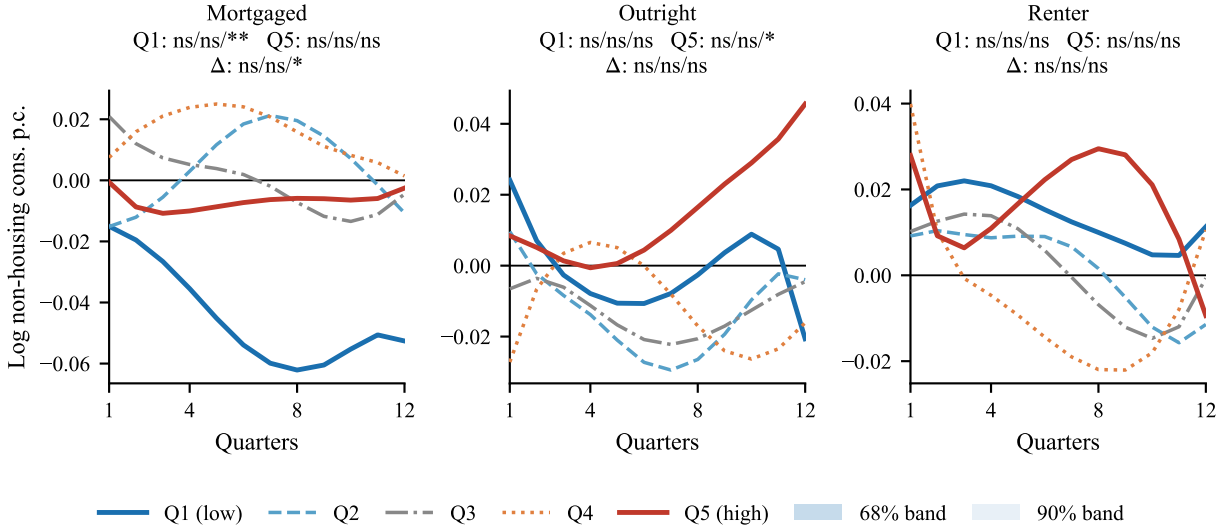


Figure C5: Smooth LP IRF: Log Non-Housing Consumption per Capita — LSAP Shock, by Housing Tenure $\times$ Income Quintile. Compare with Figure E1 (Standard LP).

Notes: The direction of this comparison is the reverse of the preceding subsections: Figure E1 in Appendix E reports the Standard LP estimates for the tenure estimation, and the Smooth LP counterpart is shown here. Cell sizes are smaller under the tenure grouping than under the regional grouping, so the signal is weaker and the B-spline parameterization shifts the path visibly at several horizons. See the notes to Figure 2 for the confidence bands, significance annotations, and sample period.

introduced by the smoothed specification are not negligible here, the main text uses the standard local projections for the tenure decomposition figures.

C.5 Wald Significance — Standard LP (All Outcomes)

Individual-quintile Wald significance results are reported separately by shock in Tables C1–C3.

Table C1: LP Wald Test ($H_0: \beta^h = 0, h = 1, \dots, 12$) — Regional Estimation, FFR Shock: Region $\times$ Income Quintile

Outcome	Quintile	Northeast	Midwest	South	West
Non-Housing Cons.	Q1	***/(0.28)	***/(0.12)	(0.16)/(0.30)	***/(0.12)
	Q2	(0.69)	(0.36)	(0.42)	(0.13)
	Q3	***	***	(0.13)	*
	Q4	***	(0.11)	(0.72)	(0.83)
	Q5	***/(0.25)	(0.45)/(0.83)	(0.16)/(0.21)	(0.74)/(0.77)
Total Consumption	Q1	(0.12)/(0.49)	*/(0.31)	(0.26)/(0.40)	*/(0.40)
	Q2	(0.80)	(0.59)	(0.52)	(0.26)
	Q3	***	*	(0.62)	(0.22)
	Q4	(0.53)	**	(0.78)	(0.99)
	Q5	***/(0.25)	(0.80)/(0.90)	(0.11)/(0.17)	(0.76)/(0.67)
Food Share	Q1	(0.28)/(0.27)	***/**	(0.25)/(0.38)	*/(0.19)
	Q2	**	***	(0.49)	***
	Q3	(0.51)	***	***	***
	Q4	(0.12)	***	***	(0.17)
	Q5	***/*	*/(0.24)	***/*	*/(0.24)
Housing Expenditure	Q1	(0.27)/(0.29)	(0.21)/(0.66)	**/*	(0.96)/(0.94)
	Q2	***	*	***	(0.71)
	Q3	***	***	(0.20)	(0.16)
	Q4	(0.62)	***	*	(0.90)
	Q5	(0.15)/(0.64)	(0.22)/(0.40)	*/(0.29)	(0.30)/(0.46)

Notes: cells show HAC significance for Q1–Q5 (Bootstrap significance additionally shown for Q1/Q5, as HAC/Bootstrap). *p<0.10, **p<0.05, ***p<0.01; parentheses show exact p -value. HAC: Newey–West SE ($bw=h+2$), diagonal $\chi^2(12)$ joint test, $(p_s, p_y, p_x) = (2, 3, 1)$. Bootstrap: residual moving block bootstrap, $B = 5000, L = 12$; not computed for Q2–Q4. Horizons $h = 1, \dots, 12$.

Table C2: LP Wald Test ($H_0: \beta^h = 0, h = 1, \dots, 12$) — Regional Estimation, FG Shock: Region $\times$ Income Quintile

Outcome	Quintile	Northeast	Midwest	South	West
Non-Housing Cons.	Q1	(0.54)/(0.51)	(0.51)/(0.46)	*/(0.34)	*/(0.45)
	Q2	(0.77)	(0.89)	(0.60)	(0.75)
	Q3	***	**	***	*
	Q4	***	(0.44)	***	**
	Q5	(0.69)/(0.89)	*/(0.37)	(0.83)/(0.96)	(0.20)/(0.53)
Total Consumption	Q1	(0.86)/(0.91)	(0.68)/(0.62)	*/(0.36)	(0.49)/(0.90)
	Q2	(0.79)	(0.92)	(0.24)	(0.53)
	Q3	***	(0.31)	***	(0.11)
	Q4	**	(0.58)	(0.13)	*
	Q5	(0.74)/(0.95)	(0.34)/(0.46)	(0.98)/(1.00)	(0.78)/(0.83)
Food Share	Q1	***/*	(0.33)/(0.58)	(0.21)/(0.68)	(0.76)/(0.98)
	Q2	(0.98)	*	(0.19)	***
	Q3	***	(0.34)	(0.83)	(0.54)
	Q4	***	(0.18)	**	(0.37)
	Q5	(0.40)/(0.54)	(0.98)/(0.99)	(0.35)/(0.59)	(0.36)/(0.66)
Housing Expenditure	Q1	(0.46)/(0.77)	(0.86)/(0.85)	(0.16)/(0.26)	(0.36)/(0.53)
	Q2	**	*	(0.11)	***
	Q3	*	(0.55)	(0.29)	(0.67)
	Q4	(0.42)	(0.44)	**	(0.22)
	Q5	(0.45)/(0.60)	(0.63)/(0.58)	(0.78)/(0.86)	(0.13)/(0.47)

Notes: cells show HAC significance for Q1–Q5 (Bootstrap significance additionally shown for Q1/Q5, as HAC/Bootstrap). *p<0.10, **p<0.05, ***p<0.01; parentheses show exact p -value. HAC: Newey–West SE (bw= $h + 2$), diagonal $\chi^2(12)$ joint test, $(p_s, p_y, p_x) = (2, 3, 1)$. Bootstrap: residual moving block bootstrap, $B = 5000, L = 12$; not computed for Q2–Q4. Horizons $h = 1, \dots, 12$.

Table C3: LP Wald Test ($H_0: \beta^h = 0, h = 1, \dots, 12$) — Regional Estimation, LSAP Shock: Region $\times$ Income Quintile

Outcome	Quintile	Northeast	Midwest	South	West
Non-Housing Cons.	Q1	***/(0.12)	(0.15)/(0.33)	*/(0.51)	(0.22)/(0.35)
	Q2	**	***	***	(0.98)
	Q3	***	(0.12)	(0.68)	(1.00)
	Q4	***	**	(0.25)	(0.63)
	Q5	**/(0.16)	(0.99)/(0.97)	(0.80)/(0.69)	(0.43)/(0.43)
Total Consumption	Q1	***/**	**/(0.20)	(0.65)/(0.85)	(0.27)/(0.41)
	Q2	**	***	***	(0.97)
	Q3	***	**	(0.92)	(1.00)
	Q4	**	***	(0.79)	(0.65)
	Q5	**/(0.37)	(0.99)/(0.94)	(0.62)/(0.49)	(0.20)/(0.35)
Food Share	Q1	(0.11)/(0.39)	(0.49)/(0.38)	(0.11)/(0.47)	(0.19)/(0.40)
	Q2	***	**	***	(0.46)
	Q3	(0.90)	***	**	(0.72)
	Q4	**	***	***	*
	Q5	***/*	(0.21)/(0.42)	(0.21)/(0.53)	(0.15)/(0.40)
Housing Expenditure	Q1	***/(0.13)	*/(0.25)	(0.27)/(0.56)	(0.77)/(0.77)
	Q2	***	***	(0.78)	(0.25)
	Q3	***	(0.25)	(0.84)	(0.82)
	Q4	**	***	(0.74)	(0.27)
	Q5	(0.22)/(0.50)	(0.86)/(0.69)	**/(0.11)	*/(0.37)

Notes: cells show HAC significance for Q1–Q5 (Bootstrap significance additionally shown for Q1/Q5, as HAC/Bootstrap). *p<0.10, **p<0.05, ***p<0.01; parentheses show exact p -value. HAC: Newey–West SE ($bw=h+2$), diagonal $\chi^2(12)$ joint test, $(p_s, p_y, p_x) = (2, 3, 1)$. Bootstrap: residual moving block bootstrap, $B = 5000, L = 12$; not computed for Q2–Q4. Horizons $h = 1, \dots, 12$.

D Additional Outcome Variables and Regional Detail

This appendix reports the smoothed local projections results for the regional estimation that are not shown in the main text. Section 4 presents only the fan chart for non-housing consumption under the LSAP shock (Figure 2). The corresponding figure for the FG shock, the adjacent quintile gap charts for both, and the fan chart for the food expenditure ratio under the LSAP shock are all reported in this appendix, along with supplementary material along two dimensions.

First, total consumption and housing expenditure are reported for shocks where the main text discusses only non-housing consumption. Second, the FFR shock and the food expenditure ratio for FG and FFR are reported here in full, since Section 4.3 discusses FFR only briefly and refers to this appendix for the underlying figures. All figures use the same Smooth LP specification, confidence bands, and significance annotations as the main-text figures; see the notes to Figure 2 for details. Adjacent gap figures are provided alongside the corresponding fan charts throughout this appendix, including the non-housing consumption gap charts for LSAP and FG referenced in Section 4 and the food expenditure ratio gap charts for LSAP, FG, and FFR. The full set of Q1–Q5 joint test results across all four outcome variables, shocks, and regions—extending the three-outcome summary reported in the main text as Table 1—is reported in Table D2 below.

D.1 Regional Dynamics: LSAP Non-Housing Consumption

The magnitude and shape of the effect differ considerably across regions. The Northeast shows the largest full-horizon effect among the four regions, with the path peaking in the second year and narrowing thereafter, crossing the zero line only once over the full horizon; the joint test is significant in the second year and remains significant in the third. The full-horizon means for the remaining three regions are all negative, at approximately one percentage point in absolute value, in the opposite direction to the Northeast, and none is statistically significant. The differences for the Midwest and West move from negative to positive, so their full-horizon means do not summarize their paths; the difference for the South is negative in most horizons, and its full-horizon mean is close to the means of the three windows. Region-specific values are reported in Table D1.

Table D1: LSAP: The $Q1$ – $Q5$ Gap in Non-Housing Consumption by Region and Horizon Window

Region	Horizon	$\bar{\delta}_w^a$	HAC / Bootstrap ^b	Path shape ^c
Northeast	1–4	+0.92 pp	(0.59)/(0.52)	One sign change
	5–8	+4.98 pp	***/**	
	9–12	+3.76 pp	***/*	
Midwest	1–4	–2.50 pp	(0.14)/(0.26)	Unstable (sign changes 5×)
	5–8	–0.98 pp	(0.72)/(0.66)	
	9–12	+0.47 pp	(0.65)/(0.54)	
South	1–4	–1.24 pp	(0.39)/(0.37)	Unstable (sign changes 3×)
	5–8	–0.94 pp	(0.46)/(0.49)	
	9–12	–1.08 pp	(0.70)/(0.71)	
West	1–4	–1.57 pp	(0.28)/(0.31)	One sign change
	5–8	–1.99 pp	(0.17)/(0.23)	
	9–12	+1.51 pp	(0.54)/(0.58)	

^a $\bar{\delta}_w \equiv \text{mean of } \beta_{Q5}^h - \beta_{Q1}^h$ over the horizons of the stated window, in percentage points per unit contractionary LSAP shock, Standard LP with $(p_s, p_y, p_x) = (2, 3, 1)$. A positive value means $Q1$ loses more, i.e. the progressive case.

^b Joint test that $\delta^h = 0$ for every h in the window. HAC: independence-approximation SE with a diagonal χ^2 . Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show the exact p -value.

^c Number of times the 12-horizon path of δ^h crosses zero. A path that changes sign repeatedly is not summarised by any single average, which is why the windows are reported separately rather than pooled.

D.2 Total Consumption (including housing costs)

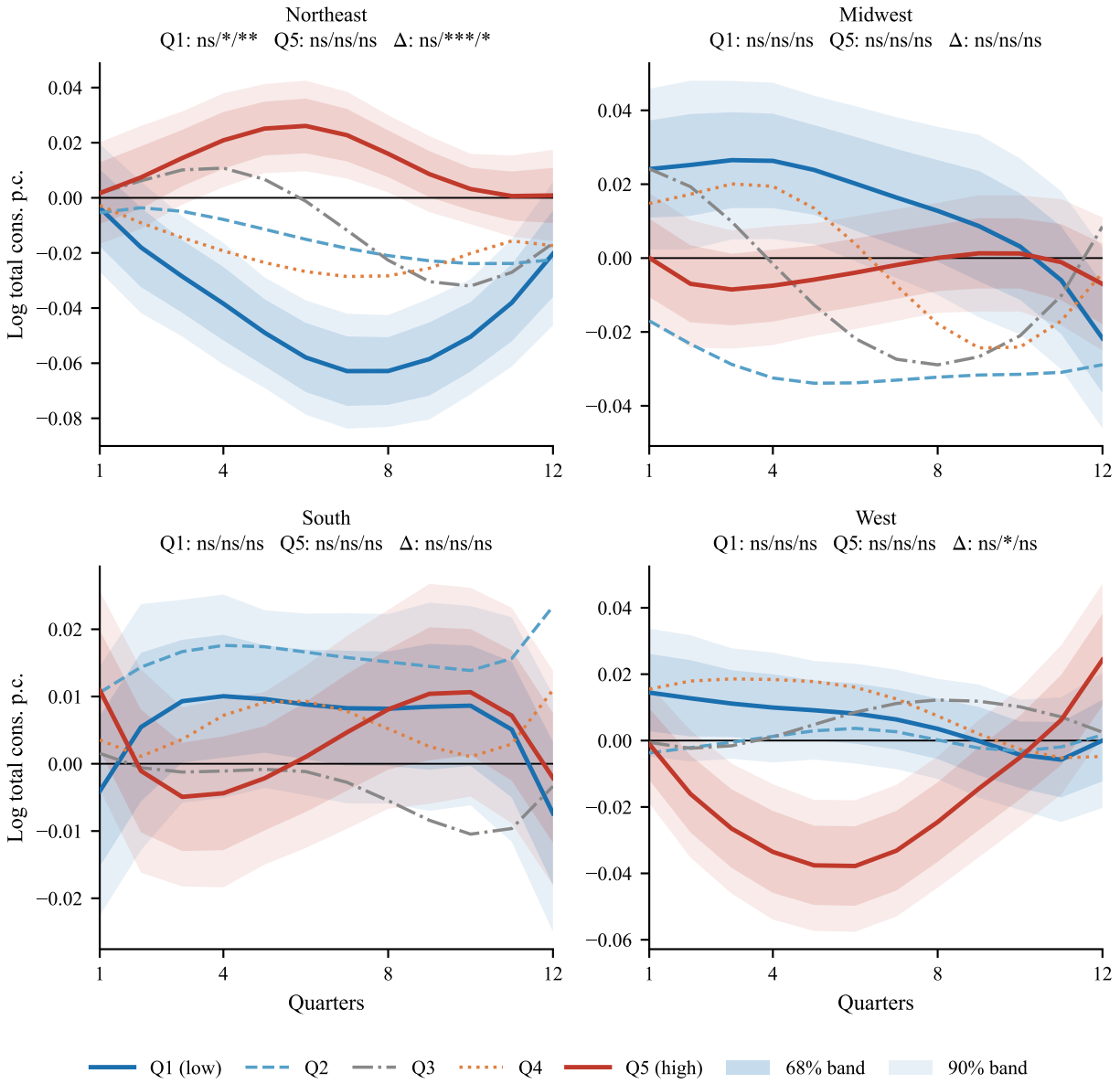


Figure D1: Smooth LP IRF: Log Total Consumption per Capita — LSAP Shock

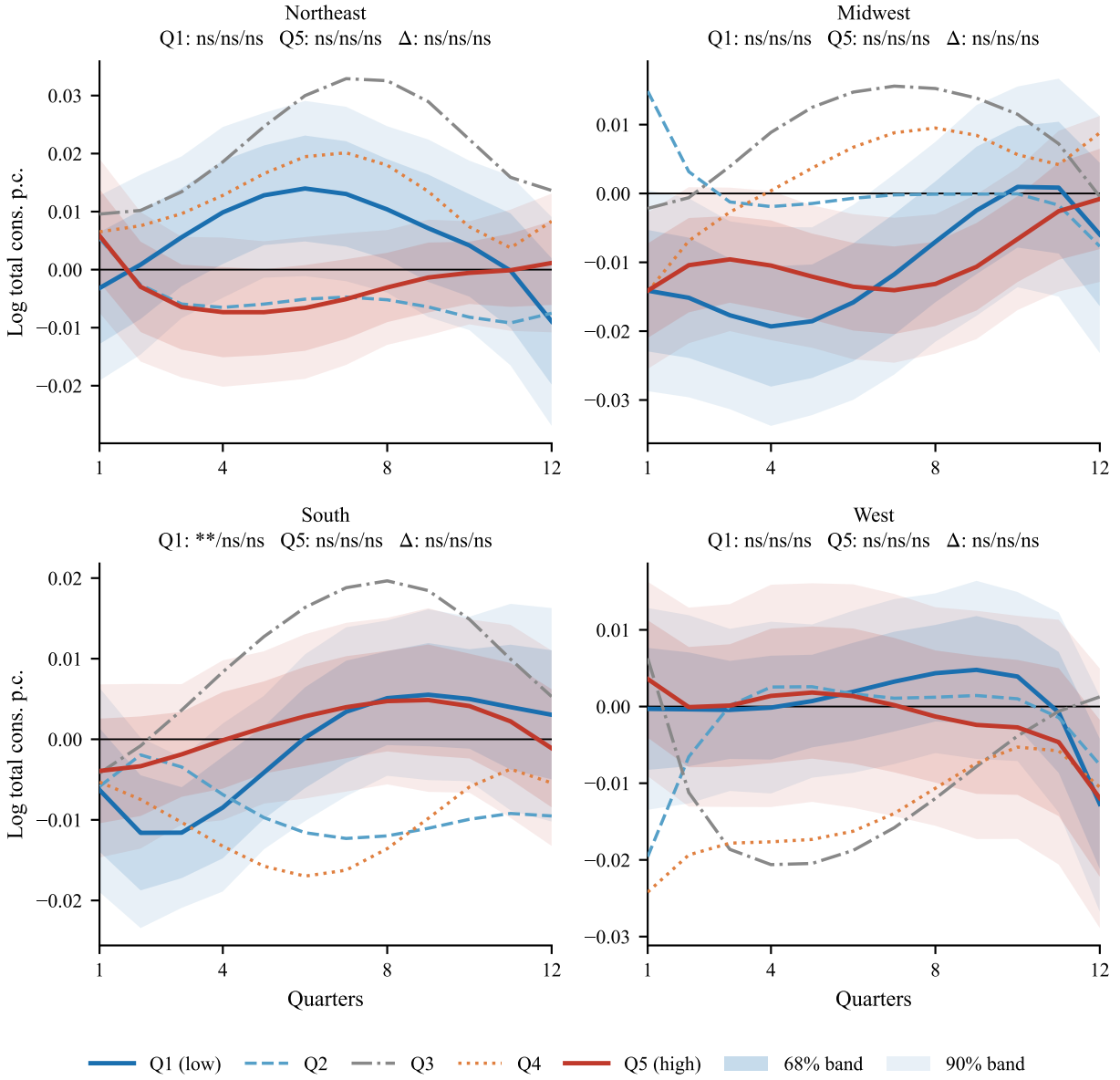


Figure D2: Smooth LP IRF: Log Total Consumption per Capita — FG Shock

D.3 Housing Expenditure

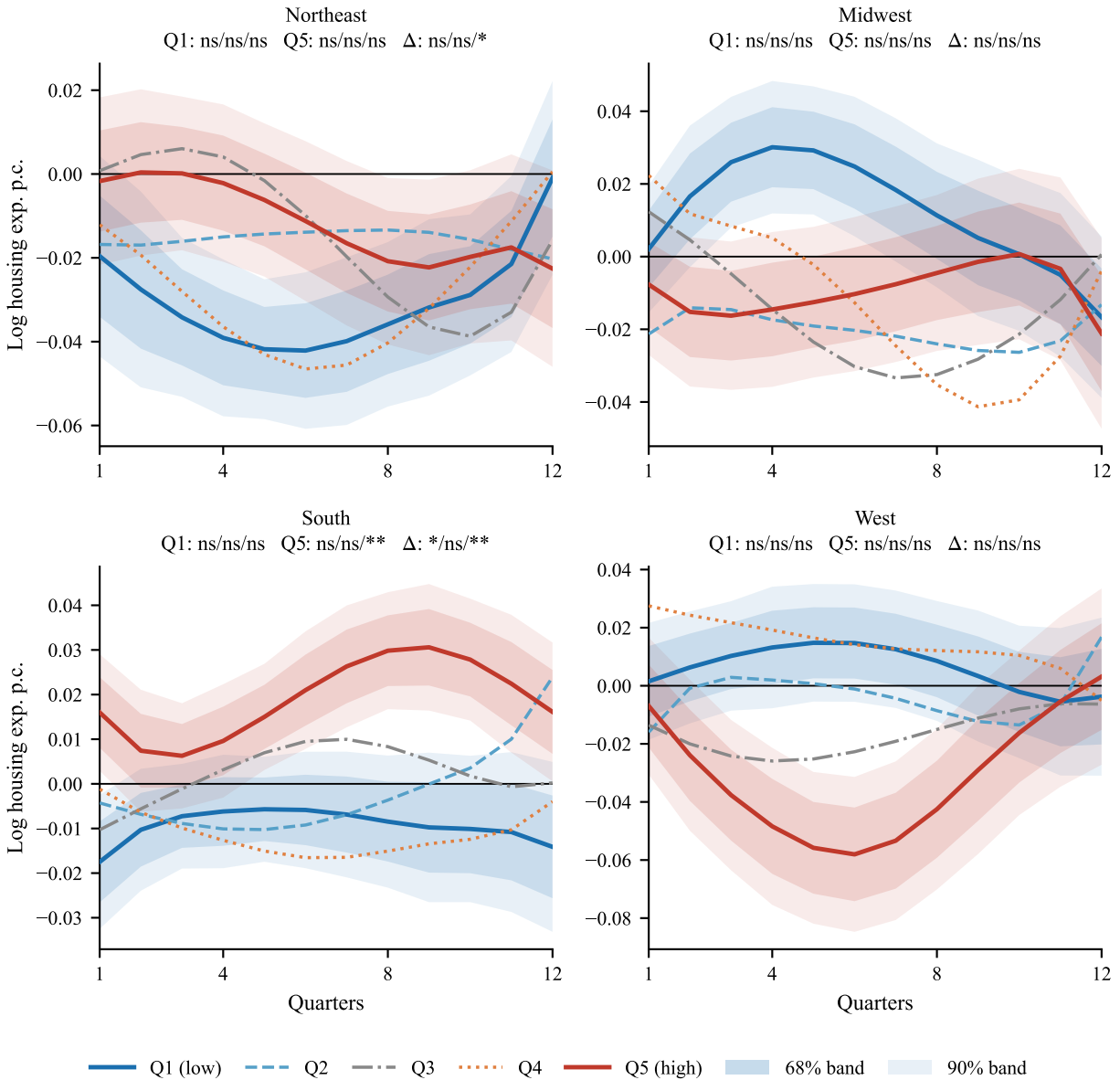


Figure D3: Smooth LP IRF: Log Housing Expenditure per Capita — LSAP Shock

D.4 Adjacent Gap Figures — All Outcomes and Shocks

The adjacent gap chart plots the difference in impulse responses between each pair of neighboring income quintiles, indicating whether consumption inequality is concentrated among lower- or

higher-income groups. It is defined as follows:

$$G_q^h = \hat{\beta}_{g,q}^h - \hat{\beta}_{g,q-1}^h, \quad q \in \{2, 3, 4, 5\} \quad (\text{D.1})$$

where $\hat{\beta}_{g,q}^h$ denotes the estimated impulse response (equation (1)) of income quintile q within cell-group g at horizon h , and $q \in \{2, 3, 4, 5\}$ indexes adjacent income quintiles. G_q^h thus measures the marginal difference in the consumption response between quintile q and the next-lower quintile $q - 1$ at horizon h .

This measure has direct implications for policy design. Depending on the income structure of a given society, inequality may be concentrated at the bottom of the distribution—reflecting an outsized gap between the poorest households and the rest—or between the middle class and the wealthy. These two patterns call for different policy responses. Lower-income households are more sensitive to prices of necessities such as food and to employment conditions in low-wage labor markets, whereas middle-income households are more exposed to mortgage rates, housing markets, taxation, and financial asset prices. Since different monetary policy instruments operate through different transmission channels, they affect income groups with different degrees of sensitivity, generating heterogeneous consumption responses across the distribution.

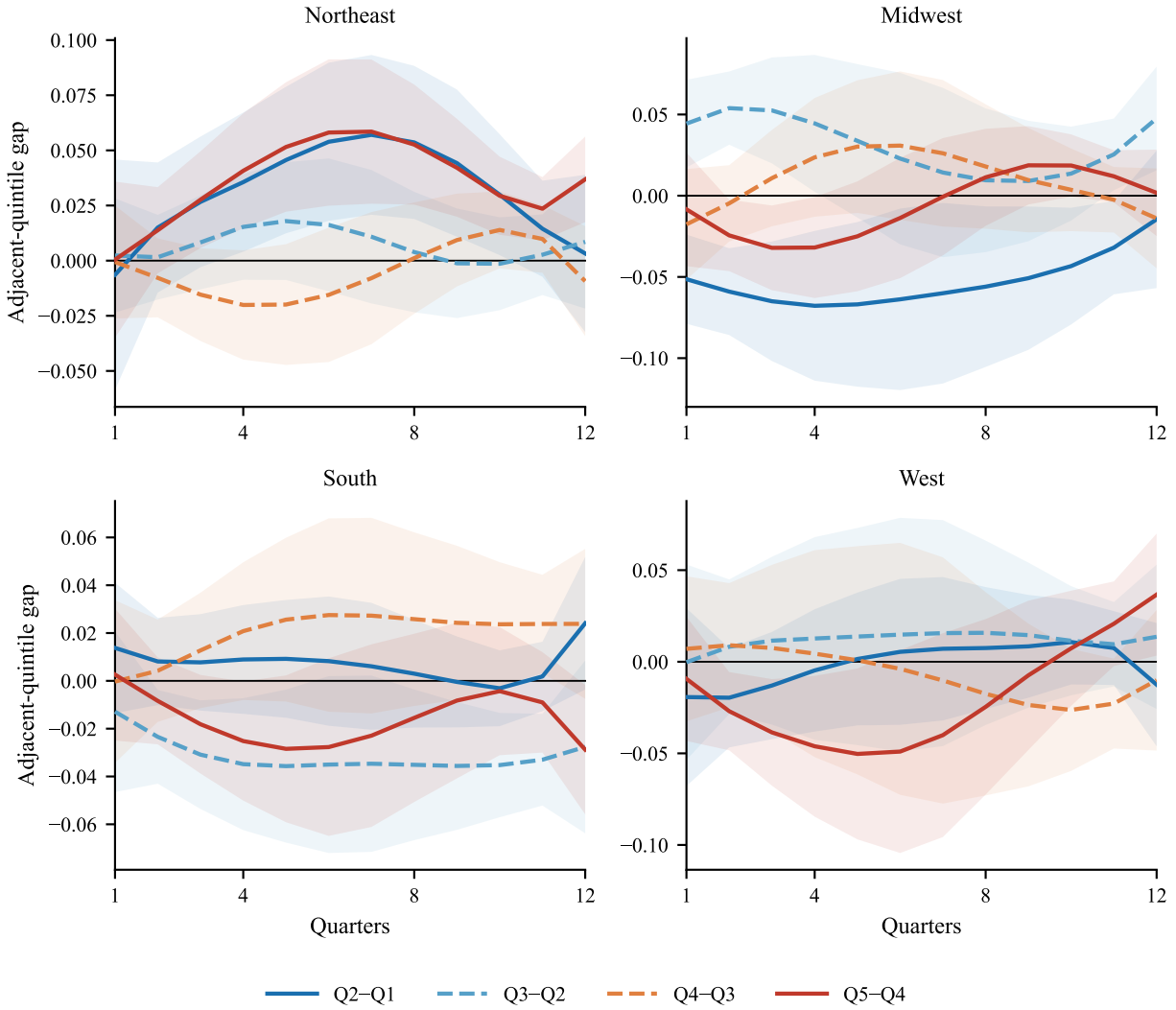


Figure D4: Adjacent Income Gap: Log Non-Housing Consumption — LSAP Shock

Notes: (1) $G_q^h = \hat{\beta}_q^h - \hat{\beta}_{q-1}^h$, the difference in cumulative impulse responses between adjacent income quintiles. (2) Shaded bands: 68% (inner) and 90% (outer) confidence intervals from the residual moving block bootstrap ($B = 5000$, $L = 12$), for $Q1$ and $Q5$ only. (3) Contractionary LSAP shock direction; see Figure 2 notes for general specification.

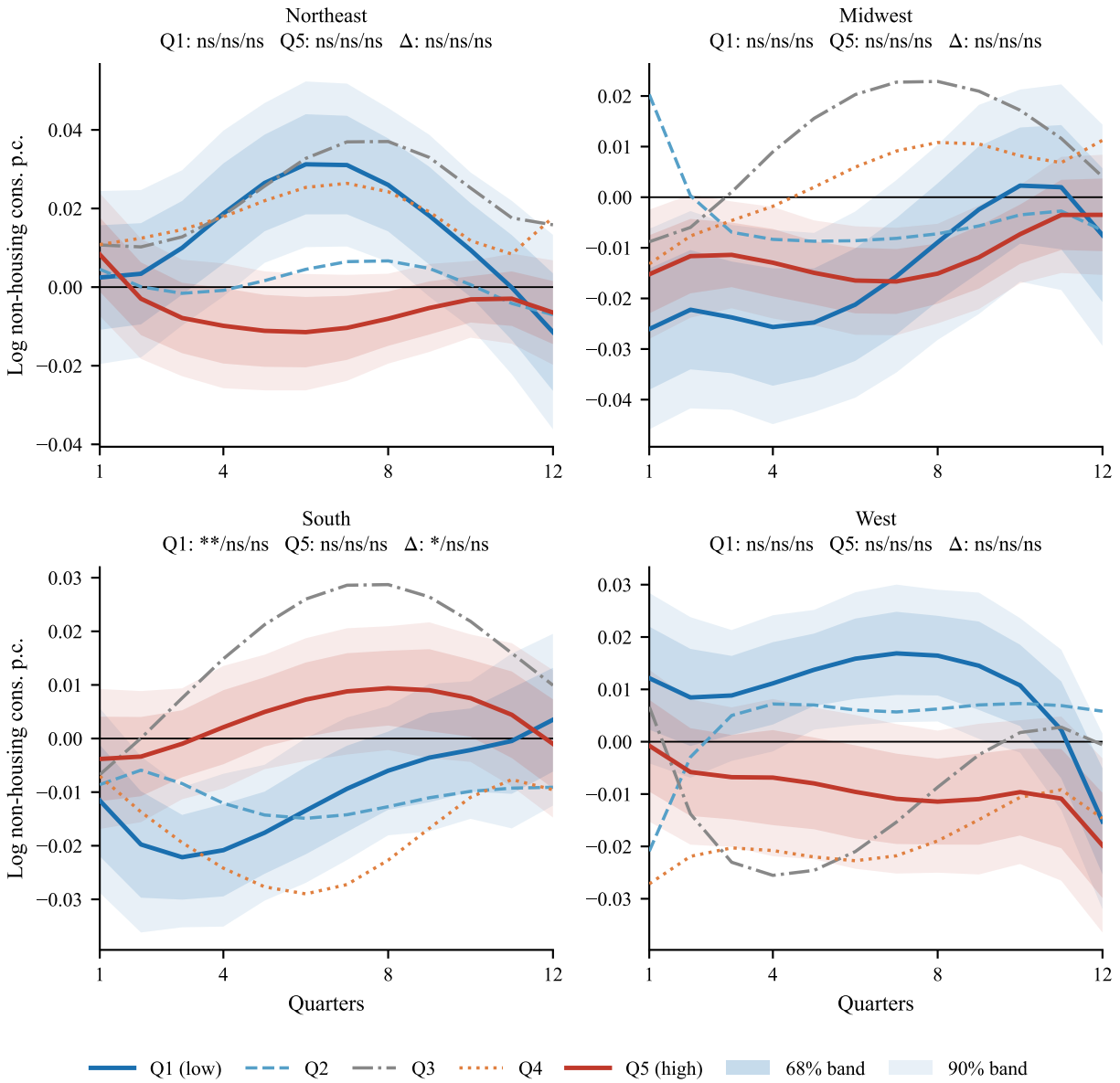


Figure D5: IRF: Log Non-Housing Consumption per Capita — FG Shock

Notes: Smooth LP impulse responses by Census region and income quintile, as in Figure 2 but with FG as the shock variable. See the notes to Figure 2 for the confidence bands, significance annotations, and sample period.

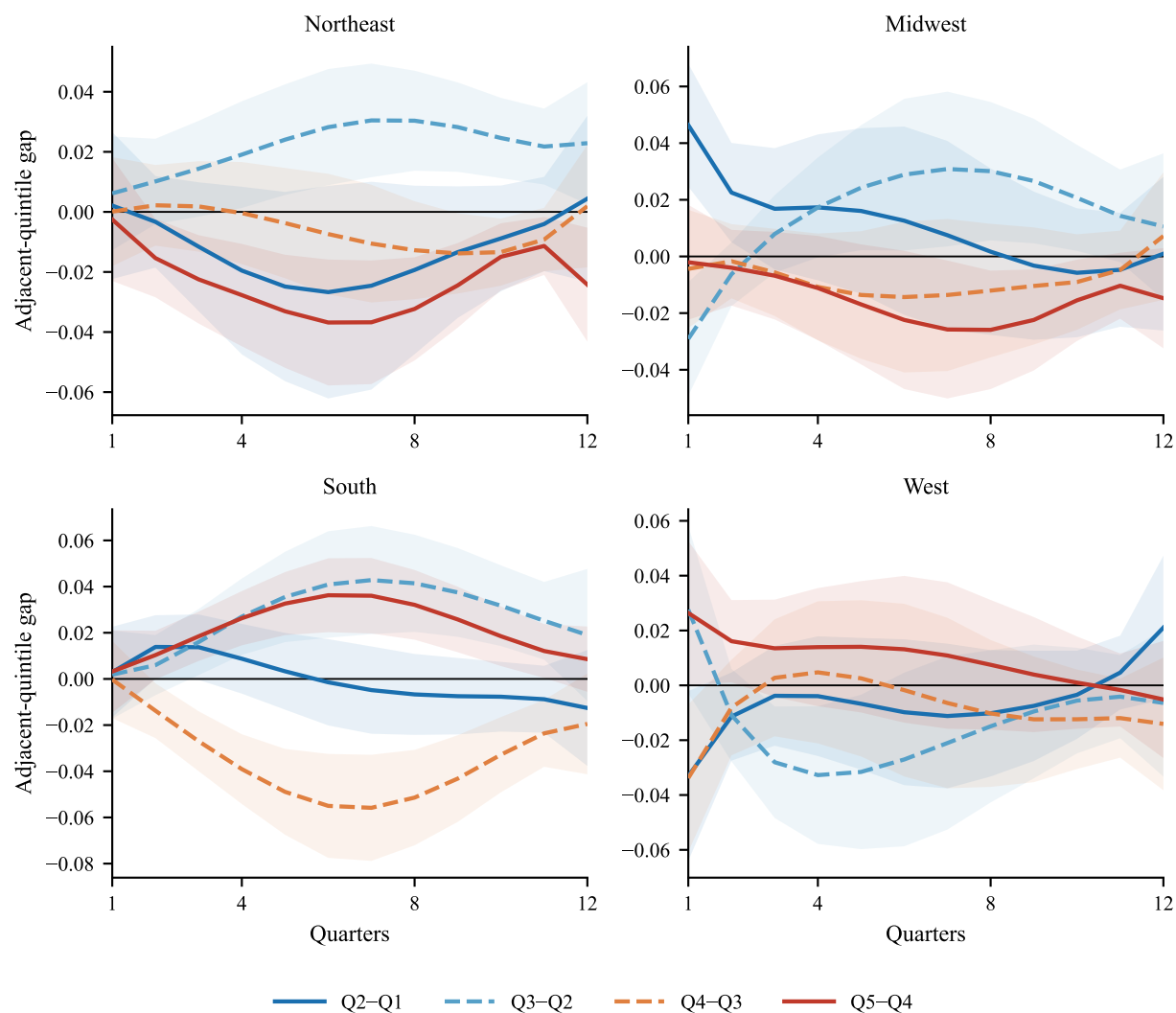


Figure D6: Adjacent Income Gap: Log Non-Housing Consumption — FG Shock
Notes: Adjacent income gap; see Figure D4 notes for definition. FG shock; see Figure 2 notes for general specification.

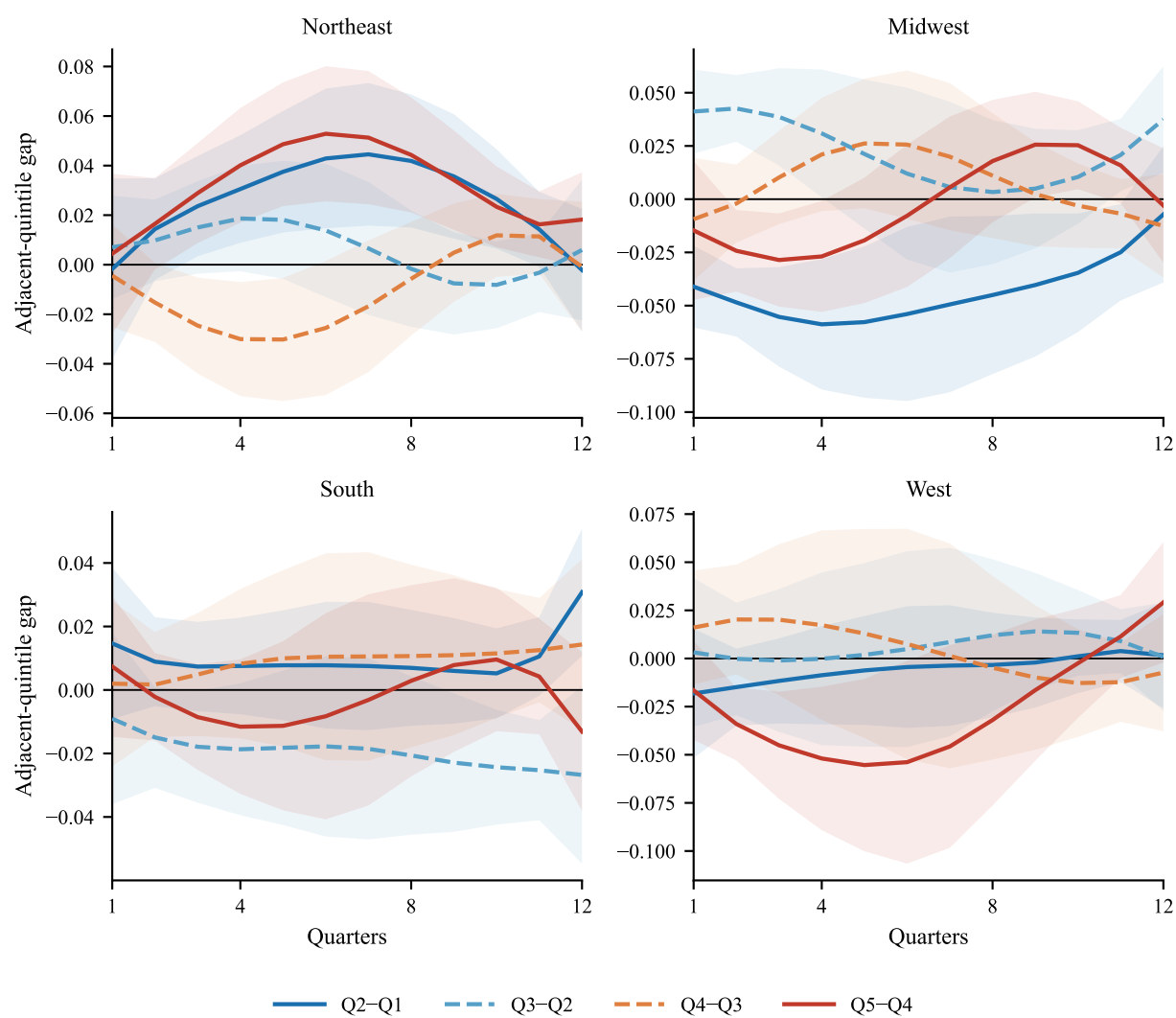


Figure D7: Adjacent Income Gap: Total Consumption — LSAP Shock

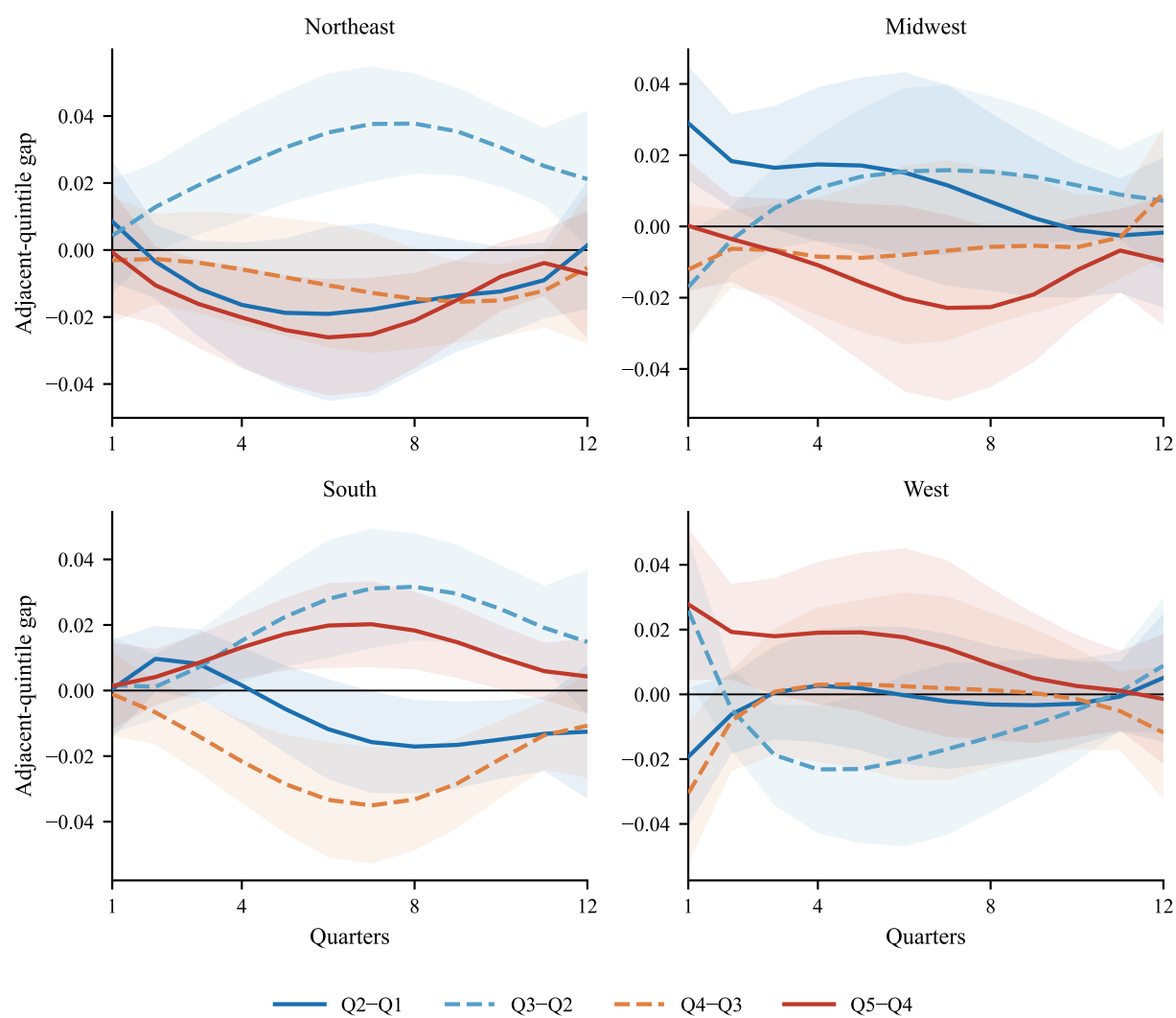


Figure D8: Adjacent Income Gap: Total Consumption — FG Shock

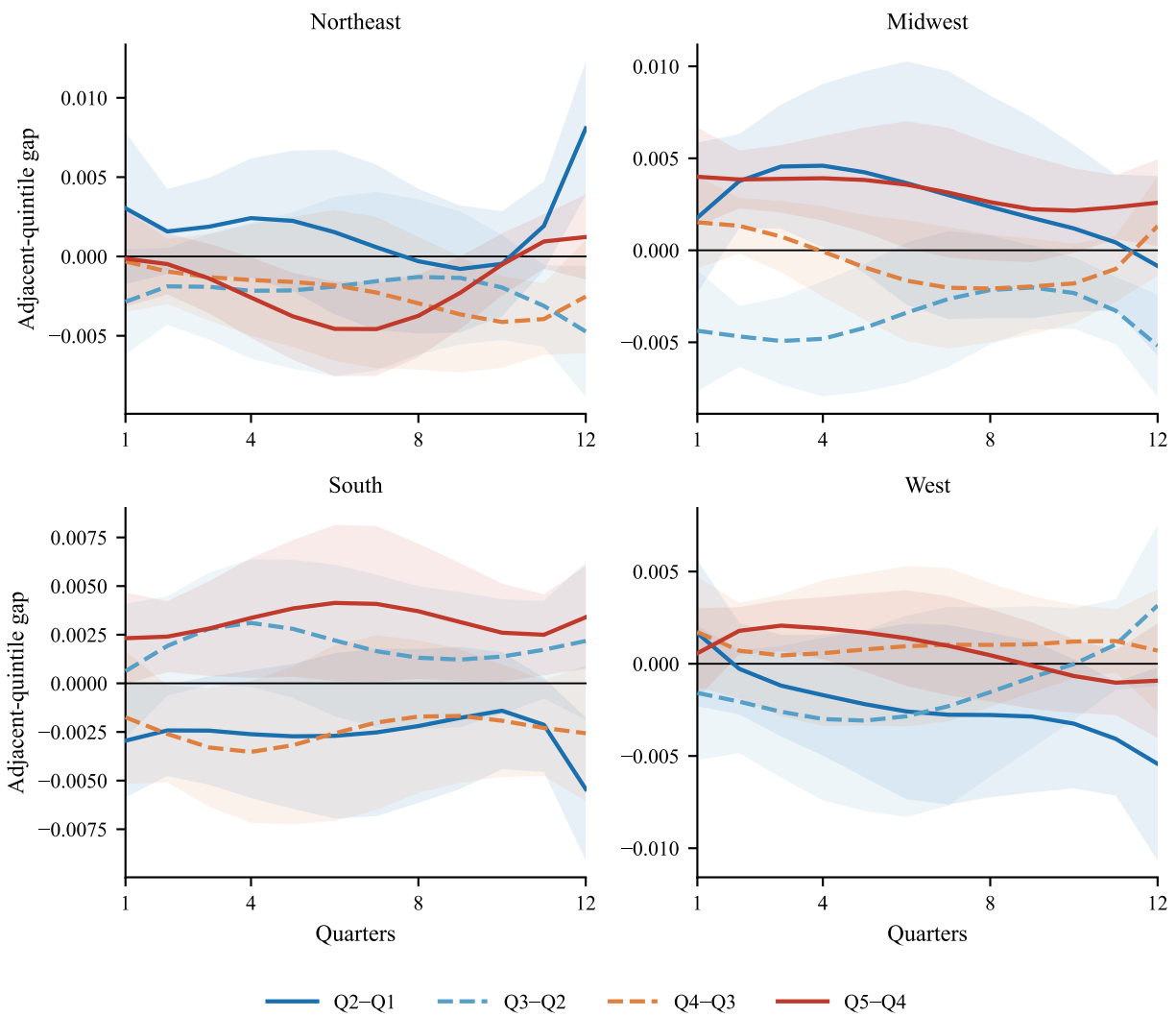


Figure D9: Adjacent Income Gap: Food Share — LSAP Shock

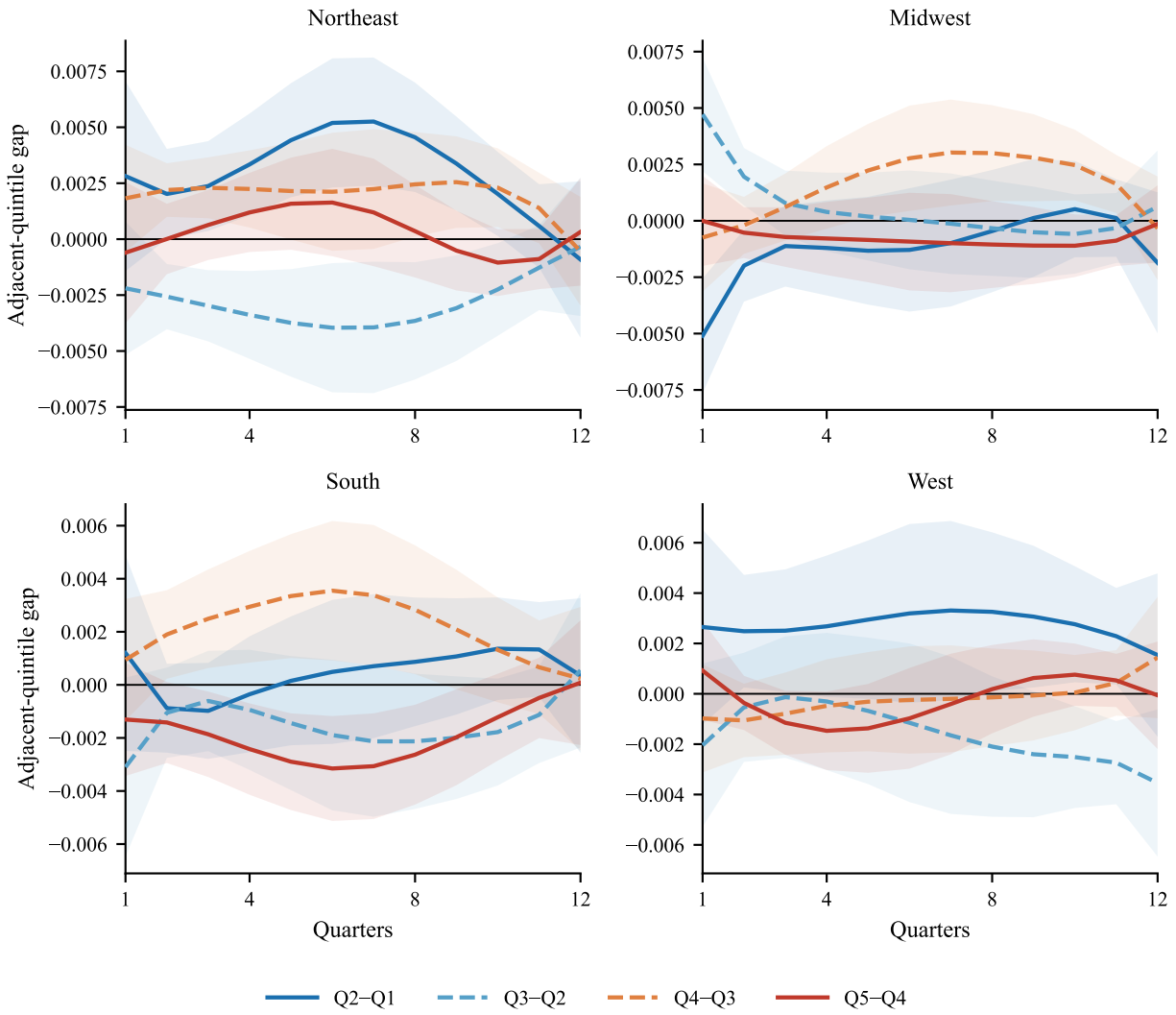


Figure D10: Adjacent Income Gap: Food Share — FG Shock

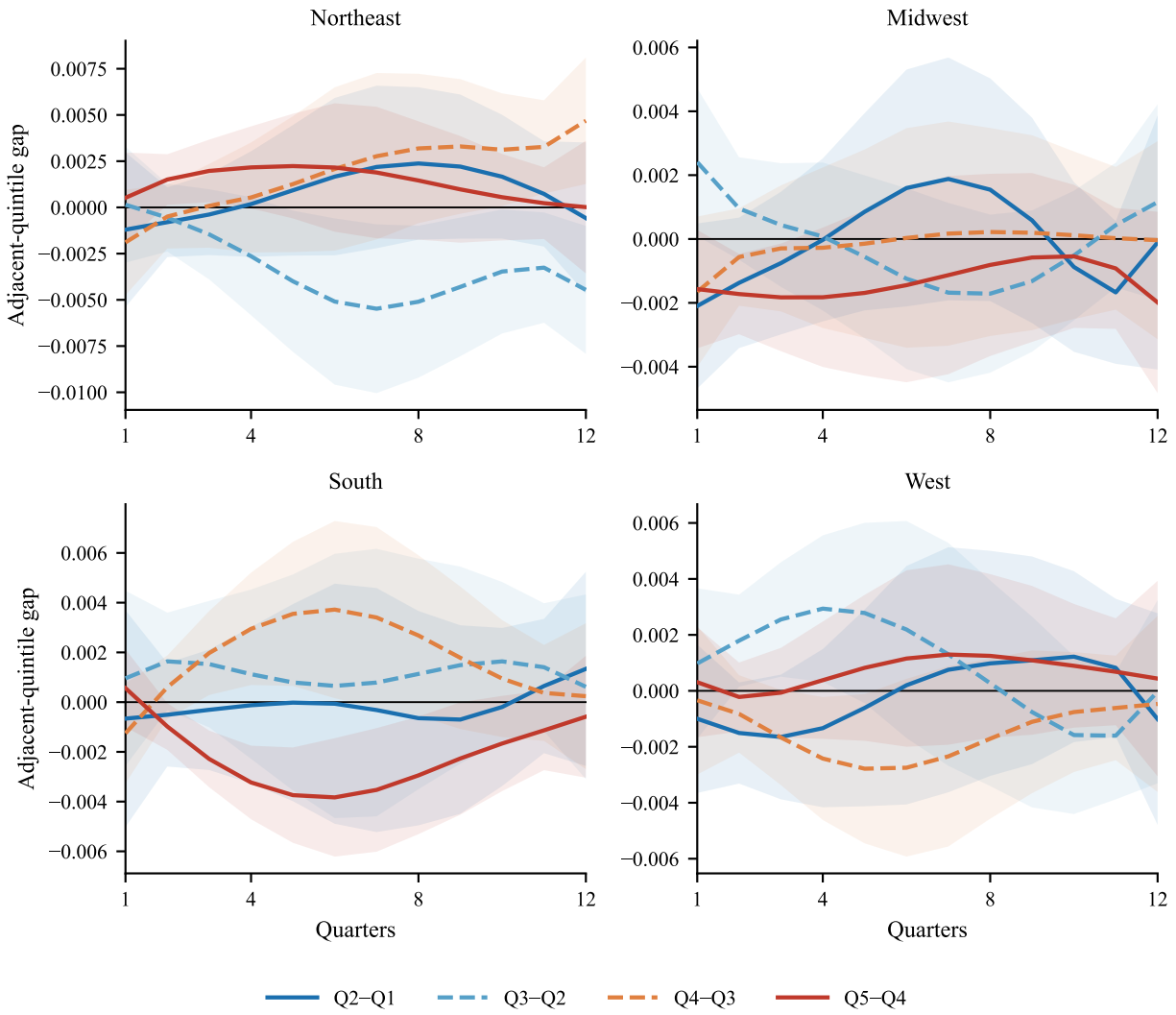


Figure D11: Adjacent Income Gap: Food Share — FFR Shock

D.5 FFR Shock — Non-Housing Consumption

Figures D12 and D13 report the Smooth LP fan and adjacent-gap charts for the FFR shock referenced in Section 4.3; the corresponding Standard LP version is shown in Appendix C.

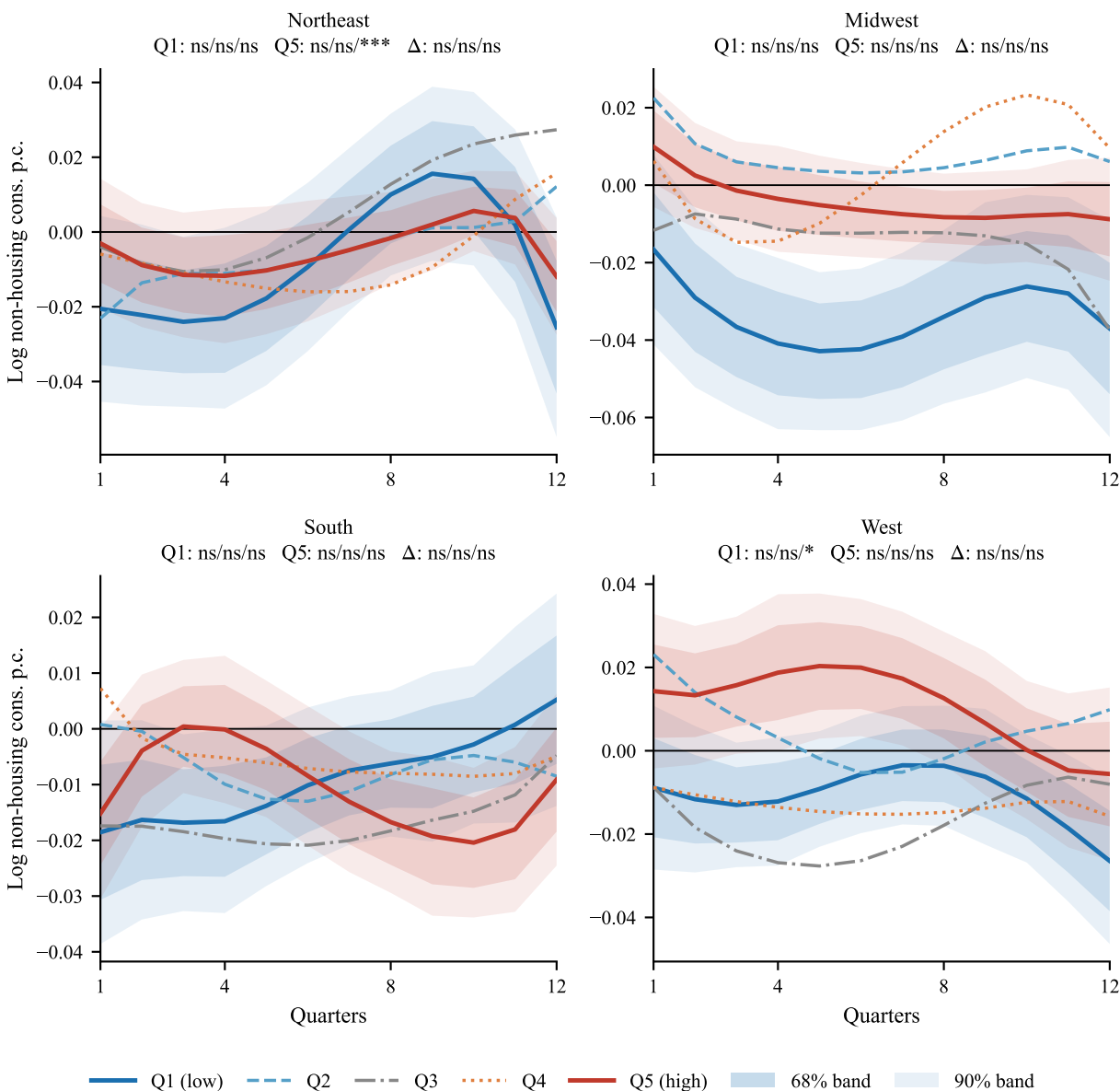


Figure D12: Smooth LP IRF: Log Non-Housing Consumption — FFR Shock

The distributional effects of FFR shocks show complex regional heterogeneity, with the relative position of $Q1$ and $Q5$ differing from the LSAP results. The fan charts in Figure D12 (smoothed local projections, used for display only and not for inference) show that non-housing consumption for $Q1$ declines in all four regions following a contractionary shock. Only in the South is the decline for $Q1$ slightly smaller than that for $Q5$. In the Midwest and West, the decline for $Q1$ is

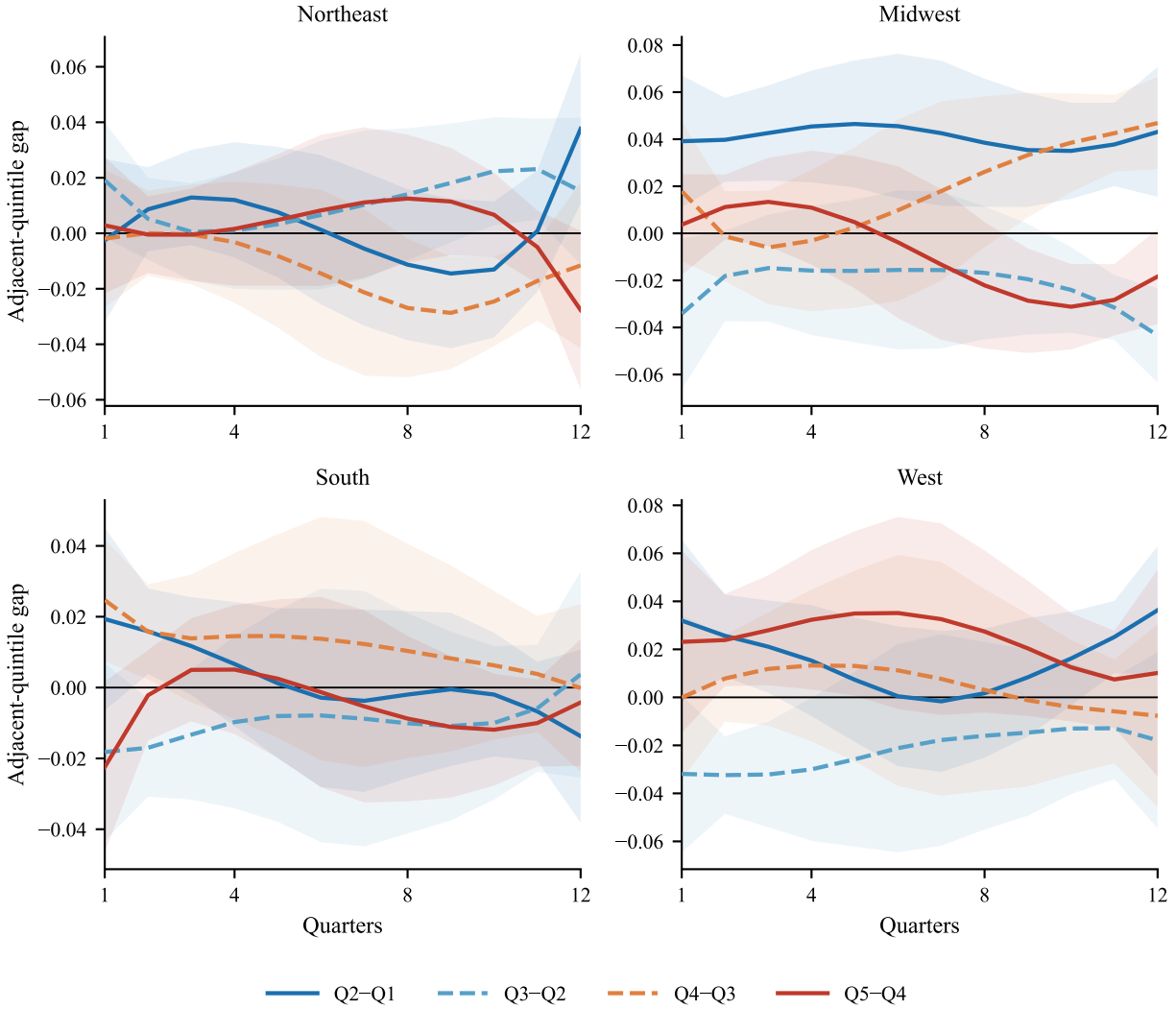


Figure D13: Adjacent Income Gap: Non-Housing Consumption — FFR Shock

larger than that for $Q5$, consistent with the direction observed under LSAP, with the difference most pronounced in the Midwest; in the Northeast, the two responses are nearly identical. The specific point estimates for the South are provided in Table 2.

The rise in total nominal consumption for $Q1$ in some regions requires interpretation alongside other consumption indicators. Under the FFR shock, the $Q1$ – $Q5$ gap in the food expenditure ratio fails the joint test in all four regions, so this measure cannot currently provide robust supplementary evidence on whether the rise in nominal consumption in the South (or any region) reflects a genuine improvement in welfare. At the same time, housing consumption for $Q1$ is positive in the Northeast, Midwest, and South, but negative in the West. This is consistent with rents rising with interest rates following a contractionary shock, while monthly payments on existing fixed-rate loans do not respond to short-term rates. This is an inherent limitation of nominal consumption measures, and

is the reason the main text uses non-housing consumption as the primary outcome variable.

The joint test for non-housing consumption does not reach conventional significance in any of the four regions, including the South. In terms of point estimates, the $Q1$ – $Q5$ difference in the Midwest and West runs in the same direction as under LSAP, the Northeast is close to zero, and only the South runs in the opposite direction, with a small magnitude. This does not rule out the possibility that FFR shocks have distributional effects in more regions; the current sample may simply lack the statistical power to detect them, particularly during the 2009–2015 ZLB period when the FFR was near zero and the shock series lacked effective variation.

D.6 Food Share — All Three Shocks

The food expenditure ratio provides a complementary perspective based on consumption composition. Its primary function is to provide cross-validation for the findings from other measures, such as non-housing consumption and total consumption, particularly when a result appears to conflict with the main findings or requires additional explanation. It does not, in itself, constitute independent directional evidence. Engel’s Law posits that as income levels rise, the proportion of household budgets allocated to food expenditures declines. Consequently, fluctuations in the food expenditure ratio in response to monetary policy shifts can be regarded as an indirect measure of alterations in expenditure composition. In contrast to nominal consumption levels, the food expenditure ratio is not directly influenced by aggregate price-level movements. This measure also has limitations. Griffith et al. (2016) find that during the Great Recession, households reduced food spending primarily by switching to cheaper items within food categories, purchasing more items on promotion, and changing retail outlets, while largely maintaining their nutritional intake. Changes in the food expenditure ratio therefore reflect both a budget-tightening component and a within-basket substitution component, with the magnitude of the latter not being fully independent of the unit food prices households actually pay. For the results in this paper, however, this channel biases the estimates in a conservative direction: downgrading behavior reduces food spending and hence lowers the ratio, implying that the rise in the food expenditure ratio for $Q1$ under contractionary shocks represents a net increase that remains after accounting for this effect.

The robust result for the Northeast is discussed in the main text (see Section 4.1); this section reports the evidence for the remaining three regions. The West ($p = 0.482$) runs in the same direction as the Northeast. The Midwest ($p = 0.268$) runs in the opposite direction, with $Q5$ rising more than $Q1$, and its direction changes repeatedly across the 12 horizons. The $Q1$ – $Q5$ gap in the South ($p = 0.723$) is close to zero. None of the joint tests in the three regions pass, and no individual quintile test reaches conventional significance after bootstrap correction.

Under the FFR shock, the joint $Q1$ – $Q5$ test for the food expenditure ratio is insignificant in all

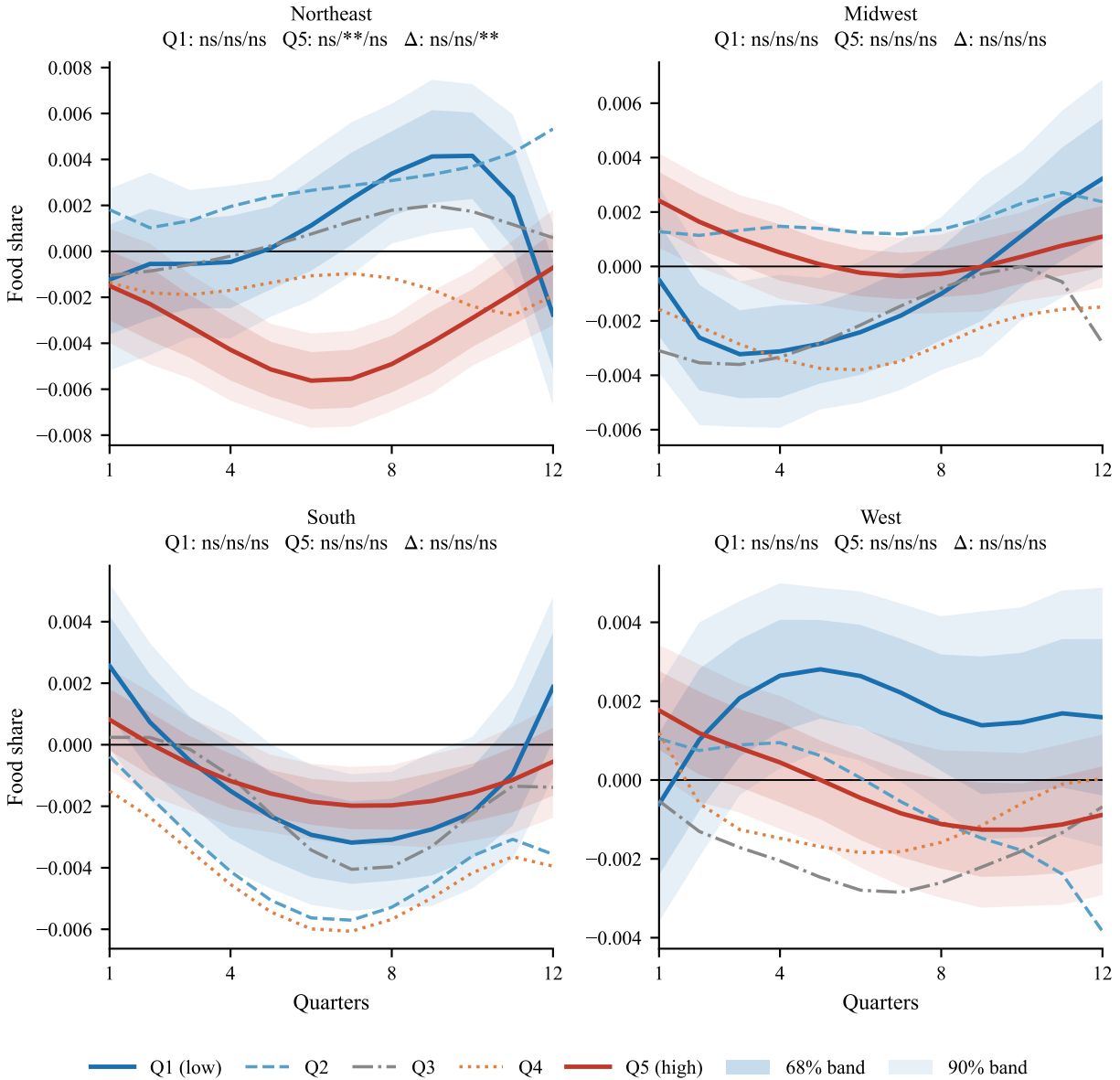


Figure D14: IRF: Food Share — LSAP Shock

Notes: (1) Food share is food expenditure divided by total consumption expenditure. Unlike non-housing consumption, a progressive effect here implies *Q1 above Q5*: tightening worsens *Q1*'s consumption composition more (food share rises more). (2) See Figure 2 notes for general specification.

four regions. Under the FG shock, the Northeast is significant only in the second-year window and insignificant over the full horizon, whereas the Midwest is insignificant in every window.

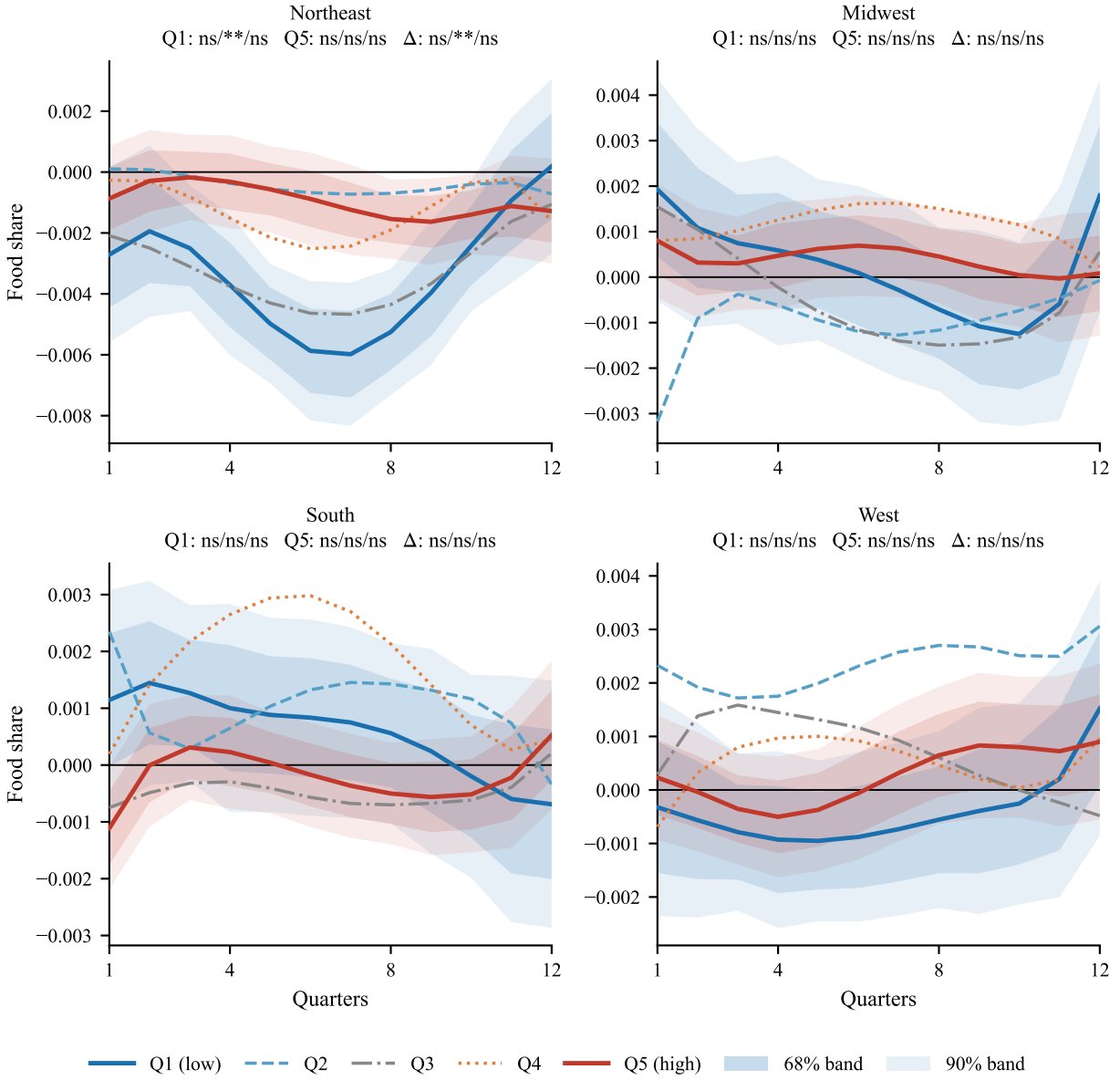


Figure D15: Smooth LP IRF: Food Share — FG Shock

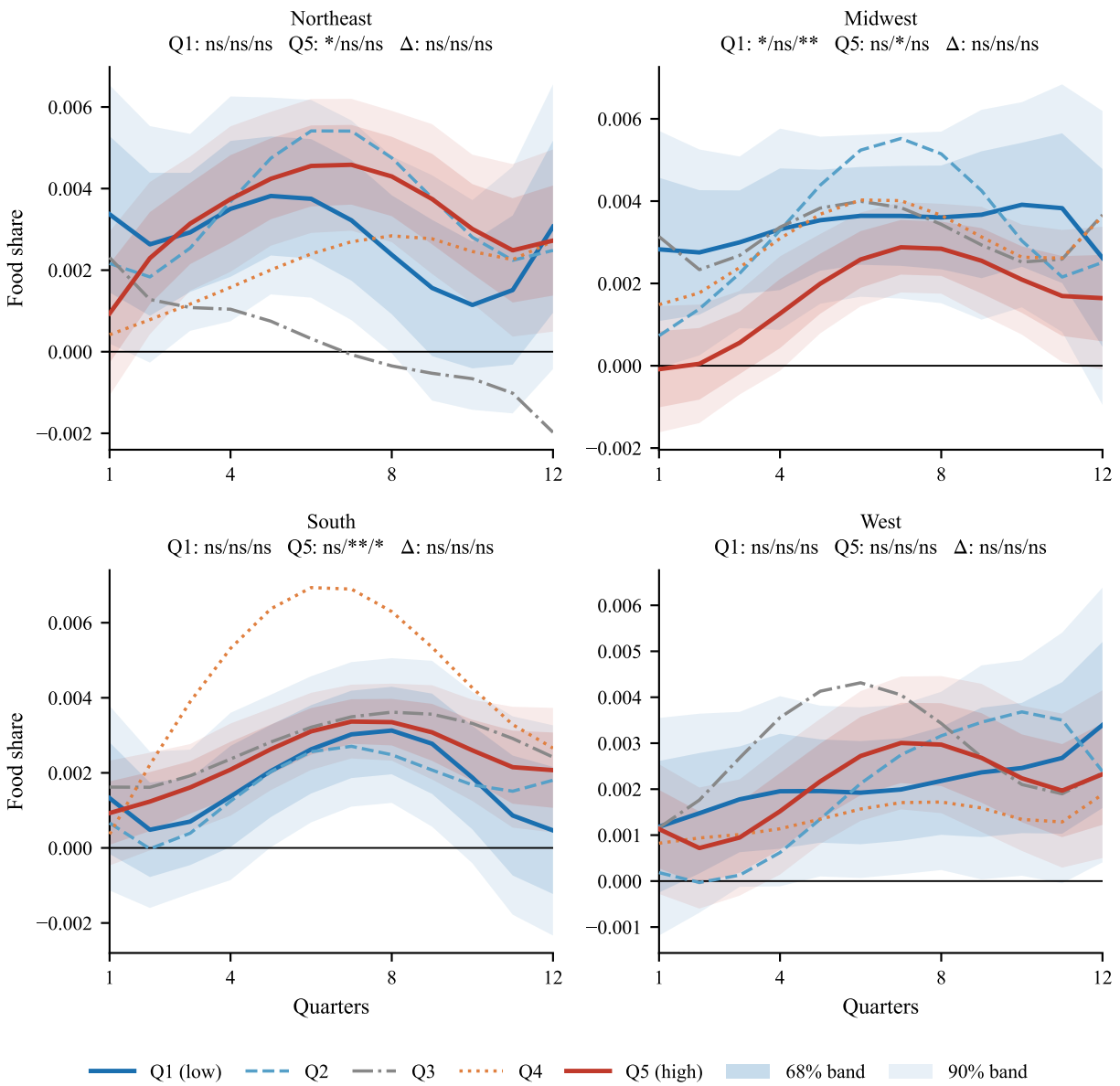


Figure D16: Smooth LP IRF: Food Share — FFR Shock

D.7 Q1–Q5 Joint Test: All Outcomes

Table D4 directly compares the $Q1$ – $Q5$ differences between pairs of instruments within the same region and consumption measure, covering all three instrument pairs; the final column pools the four regions into a single test.

Table D2: Joint Test for Distributional Effect by Horizon Window: $H_0 : \beta_{Q1}^h = \beta_{Q5}^h$ for all h in the Window (Full Four-Outcome Tabulation, HAC vs. Bootstrap)

Outcome	Shock	Horizon	NE	MW	S	W
Non-Housing Cons.	FFR	1–4	(0.75)/(0.85)↑	*/(0.15)↑	(0.15)/(0.13)↑	(0.31)/(0.29)↑
		5–8	(0.27)/(0.63)↑	(0.38)/(0.58)↑	(0.29)/(0.25)↓	(0.40)/(0.60)↑
		9–12	(0.37)/(0.51)↓	(0.25)/(0.33)↑	*/(0.11)↓	(0.11)/(0.38)↑
	FG	1–4	(0.82)/(0.86)↑	(0.88)/(0.91)↑	*/*/↑	(0.55)/(0.55)↓
		5–8	*/(0.14)↓	(0.44)/(0.54)↓	(0.83)/(0.91)↑	(0.23)/(0.31)↓
		9–12	(0.76)/(0.82)↓	(0.81)/(0.80)↓	(0.84)/(0.91)↑	(0.31)/(0.52)↓
	LSAP	1–4	(0.59)/(0.52)↑	(0.14)/(0.26)↓	(0.39)/(0.37)↓	(0.28)/(0.31)↓
		5–8	***/*↑	(0.72)/(0.66)↓	(0.46)/(0.49)↓	(0.17)/(0.23)↓
		9–12	***/*↑	(0.65)/(0.54)↑	(0.70)/(0.71)↓	(0.54)/(0.58)↑
Total Consumption	FFR	1–4	(0.65)/(0.73)↓	*/(0.14)↑	(0.44)/(0.35)↑	(0.26)/(0.30)↑
		5–8	(0.71)/(0.80)↓	(0.54)/(0.69)↑	(0.57)/(0.46)↓	(0.31)/(0.44)↑
		9–12	(0.22)/(0.24)↓	(0.76)/(0.82)↑	***/*↓	(0.29)/(0.40)↑
	FG	1–4	(0.63)/(0.68)↑	(0.94)/(0.96)↑	(0.10)/(0.13)↑	(0.99)/(0.99)↓
		5–8	(0.31)/(0.39)↓	(0.63)/(0.63)↓	(0.81)/(0.91)↓	(0.81)/(0.75)↓
		9–12	(0.79)/(0.82)↑	(0.85)/(0.80)↓	(0.97)/(0.99)↓	(0.85)/(0.88)↓
	LSAP	1–4	(0.34)/(0.34)↑	*/(0.16)↓	(0.43)/(0.37)↓	(0.15)/(0.32)↓
		5–8	***/*↑	(0.64)/(0.59)↓	(0.68)/(0.66)↓	*//*↓
		9–12	***/*↑	(0.70)/(0.55)↑	(0.92)/(0.87)↑	(0.47)/(0.53)↑
Food Share	FFR	1–4	(0.77)/(0.76)↓	*/(0.17)↓	(0.67)/(0.68)↑	(0.47)/(0.57)↓
		5–8	(1.00)/(1.00)↑	(0.92)/(0.93)↓	(0.88)/(0.84)↑	(0.58)/(0.47)↑
		9–12	(0.70)/(0.60)↑	(0.10)/(0.12)↓	(0.34)/(0.22)↑	(0.82)/(0.76)↓
	FG	1–4	(0.46)/(0.41)↑	(0.59)/(0.66)↓	(0.45)/(0.35)↓	(0.97)/(0.96)↑
		5–8	***/*↑	(0.75)/(0.73)↑	(0.58)/(0.65)↓	(0.96)/(0.96)↑
		9–12	(0.89)/(0.93)↓	(0.36)/(0.43)↑	(0.91)/(0.91)↑	(0.82)/(0.84)↑
	LSAP	1–4	(0.76)/(0.78)↓	*/(0.11)↑	(0.65)/(0.71)↓	(0.58)/(0.62)↓
		5–8	*/(0.12)↓	(0.67)/(0.57)↑	(0.82)/(0.79)↑	(0.19)/(0.32)↓
		9–12	***/*↓	(0.85)/(0.70)↓	(0.34)/(0.40)↓	(0.47)/(0.43)↓
Housing Expenditure	FFR	1–4	***/*↓	(0.37)/(0.50)↑	(0.39)/(0.64)↑	(0.34)/(0.43)↑
		5–8	(0.78)/(0.66)↓	(0.15)/(0.35)↓	*/(0.15)↓	(0.32)/(0.40)↑
		9–12	(0.19)/(0.22)↓	(0.17)/(0.32)↓	***/*↓	(0.38)/(0.35)↑
	FG	1–4	(0.62)/(0.65)↑	(0.21)/(0.37)↓	(0.96)/(0.96)↑	*/(0.17)↑
		5–8	*/(0.14)↑	(0.96)/(0.93)↓	***/*↓	*/(0.15)↑
		9–12	(0.31)/(0.28)↑	(0.99)/(0.98)↑	(0.20)/(0.37)↓	(0.78)/(0.84)↑
	LSAP	1–4	(0.45)/(0.48)↑	*/(0.12)↓	*//*↑	(0.18)/(0.40)↓
		5–8	*/(0.31)↑	(0.50)/(0.46)↓	(0.44)/(0.36)↑	*/(0.10)↓
		9–12	***/*↓	(0.93)/(0.83)↑	***/*↑	(0.85)/(0.88)↑

Notes: each cell shows HAC/Bootstrap significance for the joint $Q1$ – $Q5$ difference test restricted to the stated window ($H_0 : \delta^h = 0$ for all h in that window), followed by the direction of δ^h averaged over that window, so the arrow may differ across windows of the same cell. The test is reported by horizon window rather than as a single test over $h = 1, \dots, 12$ because the latter averages over the timing of the response: a distributional effect confined to one year is diluted by the eight horizons in which nothing happens.

↑ denotes $\delta > 0$, i.e. $Q1$'s cumulative response lies below $Q5$'s. All three shock series are oriented so that a positive value is a monetary tightening, so the arrow has the same meaning for every instrument and no instrument-specific sign adjustment is applied. For the three consumption measures ↑ is the progressive case ($Q1$ loses more under a contraction); for the food expenditure share the reading is reversed, since a smaller rise in $Q1$'s food share indicates $Q1$ is relatively better off.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show the exact p -value. HAC: independence-approximation SE, diagonal χ^2 with degrees of freedom equal to the number of horizons in the window. Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$; the draws are shared across windows, so a window statistic is a subset of the same squared terms. $(p_s, p_y, p_x) = (2, 3, 1)$.

Table D3: LSAP versus FFR within the Same Region, Outcome and Horizon Window

Region	Outcome	Horizon	$\bar{\delta}_{\text{LSAP}}^a$	$\bar{\delta}_{\text{FFR}}$	Difference	Boot.
South	Housing expenditure	1–4	+1.45	+0.10	+1.35	(0.54)
		5–8	+0.92	−0.86	+1.78	(0.20)
		9–12	+2.52	−2.23	+4.76	***
South	Non-housing cons.	1–4	−1.12	+1.13	−2.25	(0.19)
		5–8	−0.99	−0.52	— ^b	(0.28)
		9–12	−1.07	−0.99	— ^b	(0.95)
Northeast	Housing expenditure	1–4	+1.73	−2.70	+4.43	**
		5–8	+1.44	−0.69	+2.13	(0.33)
		9–12	−0.34	−0.52	— ^b	(0.34)
Northeast	Non-housing cons.	1–4	+1.03	+0.76	— ^b	(0.69)
		5–8	+5.02	+0.24	+4.78	*
		9–12	+3.83	−0.08	+3.91	(0.20)

^a $\bar{\delta}_{\text{LSAP}}$ and $\bar{\delta}_{\text{FFR}}$ are window averages of the gap defined in Table 2, note a; the difference is LSAP minus FFR. The table holds region, outcome, window, sample, specification and inference method fixed and varies only the instrument, and is computed on the common 96-anchor design so that the two instruments use identical observations at every horizon; this is why $\bar{\delta}_{\text{LSAP}}$ differs slightly from the corresponding entry in Table 2. The shock factors are orthogonal by construction (Swanson, 2021). The test is $H_0 : \delta_{\text{LSAP}}^h = \delta_{\text{FFR}}^h$ for all h in the window, taken directly rather than inferred from the two separate tests against zero, which would not license that conclusion. 3 of the 4 contrasts reject at the 10% level in at least one window: South housing expenditure (9–12, $p = 0.002$), Northeast housing expenditure (1–4, $p = 0.036$), Northeast non-housing consumption (5–8, $p = 0.071$). The remaining contrast rejects in no window: South non-housing consumption. The instrument comparison in Table 2 is confounded, since LSAP’s significant cells fall in the Northeast and FFR’s in the South; this table removes that confound by holding the region fixed.

^b A window average is reported only when it summarises the path. The joint test rejects $\delta^h = 0$ at every h in the window, which a path that changes sign within the window can do while averaging to almost nothing; the average would then be read as an effect size it is not. Where the mean absolute deviation exceeds twice the absolute mean, the average is suppressed. 4 entries are suppressed on this rule.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show the exact p -value. Inference is the bootstrap throughout.

Table D4: Instrument Contrasts of the $Q1$ – $Q5$ Gap by Horizon Window: $H_0 : \delta_A^h = \delta_B^h$ for all h in the Window

Outcome	Contrast	NE	MW	S	W	All regions ^a
Non-Housing Cons.	LSAP – FFR	ns/*/ns↑	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns
	LSAP – FG	ns/***/ns↑	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↑	ns/**/ns
	FG – FFR	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↑	*/ns/ns↓	ns/ns/ns
Total Consumption	LSAP – FFR	ns/**/ns↑	*/ns/ns↓	ns/ns/ns↑	ns/*/ns↓	ns/*/ns
	LSAP – FG	ns/***/ns↑	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↓	ns/*/ns
	FG – FFR	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns↑	ns/ns/ns↓	ns/ns/ns
Food Share	LSAP – FFR	ns/ns/ns↓	*/ns/ns↑	ns/ns/ns↓	ns/ns/ns↓	ns/ns/ns
	LSAP – FG	ns/*/ns↓	ns/ns/ns↑	ns/ns/ns↑	ns/ns/ns↓	ns/ns/ns
	FG – FFR	ns/ns/ns↑	ns/ns/ns↑	ns/ns/ns↓	ns/ns/ns↑	ns/ns/ns
Housing Exp.	LSAP – FFR	**/ns/ns↑	ns/ns/ns↓	ns/ns/**↑	ns/*/ns↓	ns/ns/ns
	LSAP – FG	ns/ns/**↑	ns/ns/ns↓	ns/ns/**↑	ns/*/ns↓	ns/ns/ns
	FG – FFR	**/ns/*↑	ns/ns/ns↓	ns/ns/ns↑	ns/ns/ns↑	ns/ns/ns

^a Joint test that the two instruments' gap paths coincide in all four regions at once ($H_0 : \delta_{A,r}^h = \delta_{B,r}^h$ for every region r and every h in the window), answering whether the two instruments differ on that outcome without selecting a region. It imposes four times as many restrictions as a single-region test and has correspondingly lower power.

Notes: $\delta_s^h \equiv \beta_{Q5,s}^h - \beta_{Q1,s}^h$ is the $Q1$ – $Q5$ gap under instrument s , per unit contractionary shock, and each row contrasts instrument A with instrument B as labelled ($A - B$). Each cell reports the bootstrap significance of the joint test for the three horizon windows in order, $h = 1$ – 4 / 5 – 8 / 9 – 12 , followed by the direction of $\delta_A^h - \delta_B^h$ averaged over $h = 1, \dots, 12$: ↑ means that $Q1$ loses more relative to $Q5$ under instrument A than under instrument B . Region, outcome, sample, specification and anchor design are held fixed, and only the instrument varies; the test is taken directly rather than inferred from the two separate tests against zero. The LSAP – FFR cells for Northeast and South non-housing consumption and housing expenditure reproduce Table D3. Rescaling each shock to one standard deviation (FFR 1.03, FG 1.29, LSAP 0.91) instead of one unit leaves the pattern unchanged; for example, Northeast non-housing consumption in $h = 5$ – 8 gives $p = 0.071$ (LSAP – FFR) and $p = 0.001$ (LSAP – FG) per unit shock, and $p = 0.093$ and $p = 0.002$ per standard deviation.

ns denotes $p \geq 0.10$; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Residual moving block bootstrap ($B = 5000$, $L = 12$) with one block index per draw shared by all 24 regressions of an outcome ($4 \text{ regions} \times Q1/Q5 \times 3 \text{ instruments}$). $(p_s, p_y, p_x) = (2, 3, 1)$.

E Full Results: Housing Tenure $\times$ Income Quintile (Tenure Estimation)

This appendix reports the full set of results for the tenure-by-income estimation underlying the mechanism analysis in Section 5. Figure 4 in the main text presents the $Q1$ – $Q5$ gap paths within each tenure group under LSAP using the standard local projections, as the smoothed local projections are not reliable for this estimation (see Appendix C). This appendix provides the corresponding fan charts, with LSAP estimated using the standard local projections and FG and FFR using the smoothed local projections, together with housing consumption and the food expenditure ratio under LSAP and the corresponding adjacent-income-gap charts. All Wald tests and joint significance results reported in this appendix are based on the standard local projections estimator and follow the inference conventions described in Section 3.3, with bootstrap-corrected p -values as the basis for judgment.

E.1 Tenure Composition by Income Quintile

Table E1 reports the share of households in each income quintile falling into each of the three tenure categories, referenced in Section 5.2. Renters account for a substantially larger share of $Q1$ than of $Q5$ (60.4% vs. 11.3%), while mortgaged owners show the opposite pattern, rising from 11.5% of $Q1$ to 70.2% of $Q5$. Outright owners are most prevalent among $Q1$ – $Q2$ households (28–33%) and least prevalent in $Q5$ (18.4%), a pattern consistent with this group disproportionately comprising older, retired homeowners whose current income is low relative to their accumulated housing wealth. This systematic association between tenure type and income is the reason the tenure decomposition in Section 5 must compare across tenure groups within a fixed income quintile, rather than across quintiles directly.

Table E2 reports the tenure composition of each region and the share of housing in local household budgets. This table supports two claims in Section 5.3. First, the tenure composition is very similar across the four regions: the gap in the mortgagor share between the highest and lowest income quintiles is around 60 percentage points in each region, and the Northeast does not have the largest gap. Thus, differences across regions cannot be attributed to differences in tenure composition. Second, the Northeast has the highest share of housing in total expenditure among the four regions, particularly in the lowest income quintile.

Table E1: Housing Tenure Composition by Income Quintile

Tenure Type	Q1	Q2	Q3	Q4	Q5
Mortgaged	11.5%	21.6%	37.6%	56.2%	70.2%
Outright	28.1%	32.9%	25.5%	20.5%	18.4%
Renter	60.4%	45.5%	36.9%	23.3%	11.3%
<i>Household-level observations (sum over sample period)</i>					
Mortgaged	12,092	22,779	39,825	59,690	75,286
Outright	29,405	34,670	26,952	21,750	19,259
Renter	65,213	49,116	39,725	24,791	11,749

Notes: Each column reports the share of households within that income quintile (survey-weighted, summed over the full 1992Q1–2019Q4 CEX panel) falling into each of the three housing tenure categories; columns sum to 100%. The lower panel reports the corresponding unweighted household-level observation counts. Renters account for a substantially larger share of Q1 than of Q5 (60.4% vs. 11.3%). Mortgaged owners are most heavily concentrated in Q5 (70.2%) and least concentrated in Q1 (11.5%); outright owners are most prevalent in Q1–Q2 (28–33%) and least prevalent in Q5 (18.4%), consistent with this group disproportionately comprising older, retired homeowners with substantial housing wealth but comparatively low current income.

Table E2: Housing Tenure Composition and the Weight of Housing in the Budget, by Region

Region	Tenure composition			Housing share of expenditure
	Mortgaged	Outright	Renter	
<i>Panel A. All households</i>				
Northeast	36.3%	25.6%	38.1%	38.4%
Midwest	41.1%	26.8%	32.2%	34.7%
South	38.8%	27.7%	33.5%	34.9%
West	40.4%	18.9%	40.7%	37.7%
<i>Panel B. Within Q1</i>				
Northeast	7.9%	24.5%	67.7%	43.0%
Midwest	10.1%	27.5%	62.4%	39.4%
South	13.2%	33.2%	53.6%	40.1%
West	13.1%	23.3%	63.6%	42.3%
<i>Panel C. Within Q5</i>				
Northeast	66.7%	20.5%	12.8%	34.4%
Midwest	73.3%	19.9%	6.7%	30.7%
South	69.8%	19.1%	11.1%	31.2%
West	71.0%	13.9%	15.1%	34.1%

Memo: Q5 – Q1 mortgaged-share wedge (Panel C – Panel B)

Northeast +58.8 pp, Midwest +63.2 pp, South +56.6 pp, West +57.9 pp

Notes: Each panel covers one population: all households, those in Q1, and those in Q5. The first three columns report the share of that population falling into each of the three housing tenure categories and sum to 100%; the last column reports the survey-weighted mean of the household-level ratio of housing expenditure to total expenditure for the same population. All figures are survey-weighted and summed over the 1992Q1–2019Q4 CEX panel. Two features are referenced in Section 5.3. First, tenure composition is close to uniform across regions: the Q5 – Q1 mortgaged-share wedge ranges from 56.6 to 63.2 percentage points (South to Midwest), and the Northeast is not the extreme case, so differences in composition cannot account for the regional pattern in Section 4. Second, the Northeast has the largest housing share of total expenditure of the four regions, both overall (38.4%) and within Q1 (43.0%). Sample is the estimation sample of Section 2.

E.2 Detailed Transmission Discussion: Renters and Outright Owners

Among the three tenure groups, mortgagor households are the direct recipients of the cash flow channel. Mortgages in the United States are predominantly long-term and fixed-rate, so the monthly payment on an existing loan does not move with short-term rates, and monetary policy can affect the cash flow of these households only through the prepayment option. Asset purchases compress long-term rates and mortgage-backed securities spreads, which determine whether refinancing is worthwhile, but exercising the option requires paying closing costs and meeting income, equity, and credit requirements. Low-income borrowers have the highest ratio of monthly payments to disposable income and the thinnest liquidity buffers, making these requirements the hardest for them to meet. They are therefore squeezed most when a contractionary shock closes the refinancing channel. In Figure 4, the difference path for the mortgagor group is close to zero in the first year and widens window by window thereafter, consistent with the effects of refinancing taking time to accumulate.

Neither renter households nor outright owner households lie on this channel. The housing cash flow of renters adjusts through rents as short-term rates move, while asset purchases do not act on rents directly. Accordingly, no income gradient appears within the renter group: non-housing consumption, total consumption, and housing consumption are insignificant in every window, and only the food expenditure ratio is marginally significant in the first year. It should be noted, however, that a gradient does appear within the renter group in the third year after excluding extreme shock quarters, so this comparison likewise depends on crisis-period shock variation (Appendix F). Outright owner households have neither monthly mortgage payments nor a refinancing margin. The only significant results for this group are housing consumption and total consumption, which includes housing consumption, reflecting changes in holding costs such as property taxes, insurance, and maintenance with interest rates rather than a response of consumption demand. This group contains a higher share of older households who have paid off their mortgages and rely more on fixed sources of income such as pensions, which is also consistent with the direction of the $Q1$ response for this group under forward guidance being opposite to that of the other two groups.

E.3 Non-Housing Consumption by Tenure Type — All Three Shocks

The three figures below present fan charts of non-housing consumption by tenure group under the three shocks, with the three panels in each figure showing mortgagor, outright owner, and renter households, respectively. The LSAP figure uses the standard local projections for the reason given in Appendix C, whereas the FFR and FG figures use the smoothed local projections.

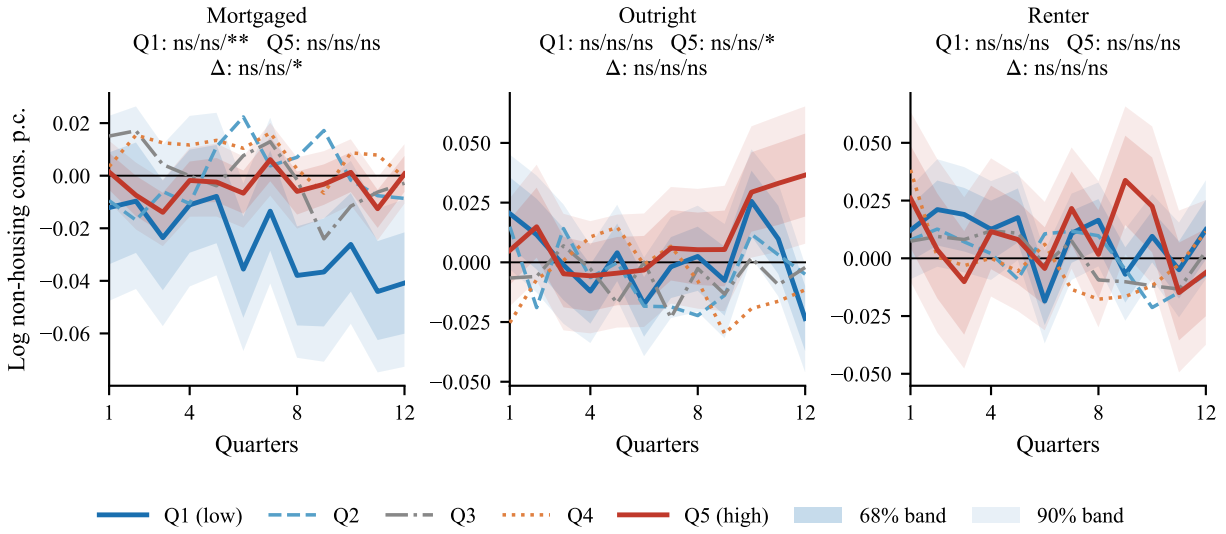


Figure E1: IRF: Log Non-Housing Consumption per Capita — LSAP Shock, by Housing Tenure × Income Quintile

Notes: (1) Tenure estimation (housing tenure × income quintile); estimates are national-level. Three panels correspond to mortgaged owners, outright owners, and renters; lines within each panel are Q1–Q5 income quintiles. (2) This figure uses Standard LP estimates (not Smooth LP) because the B-spline smoothing introduces directional bias for the tenure estimation under LSAP shocks; see Appendix C. (3) Confidence bands and significance annotations come from the residual moving block bootstrap ($B = 5000$, $L = 12$); panel titles report the bootstrap joint tests by horizon window ($h = 1-4 / 5-8 / 9-12$) for $Q1$, $Q5$ and Δ . See Figure 2 notes for the band definitions, significance symbols, and sample period.

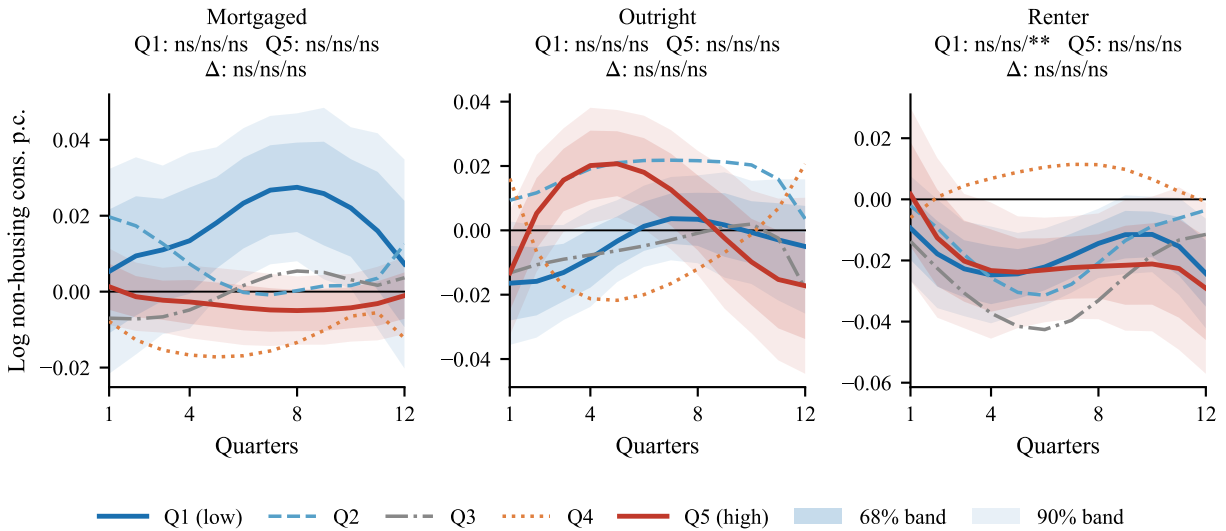


Figure E2: Smooth LP IRF: Log Non-Housing Consumption — FFR Shock, by Tenure

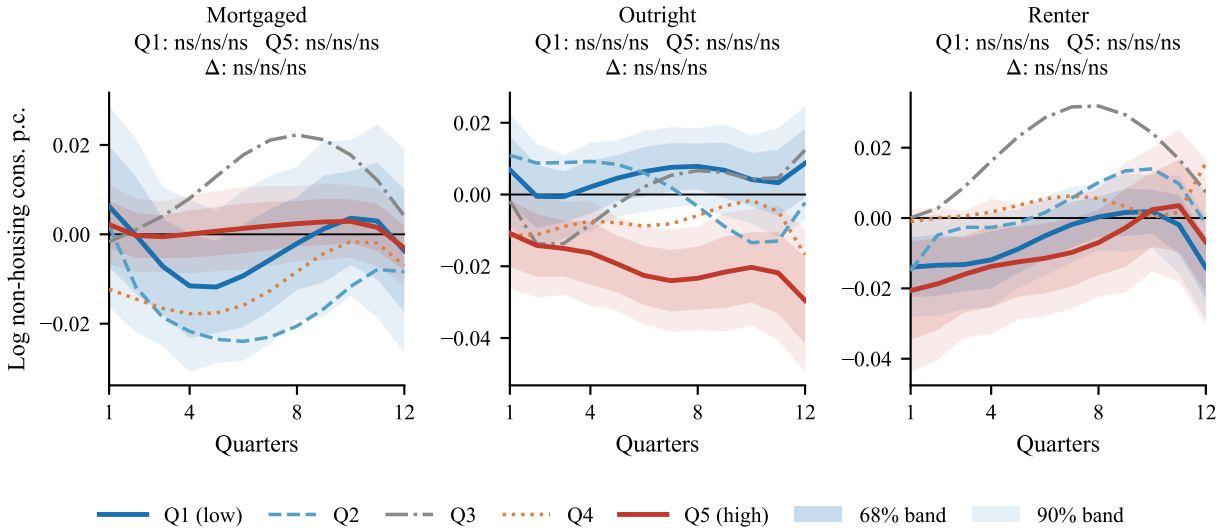


Figure E3: Smooth LP IRF: Log Non-Housing Consumption — FG Shock, by Tenure

E.4 Housing Expenditure by Tenure Type

As in the regional estimation, housing consumption responds directly to interest rates: for renters, this operates mainly through rents, whereas for mortgagors, it operates mainly through refinancing activity rather than changes in existing monthly payments. Figure E4 presents this result for LSAP as background to the non-housing consumption results discussed above.

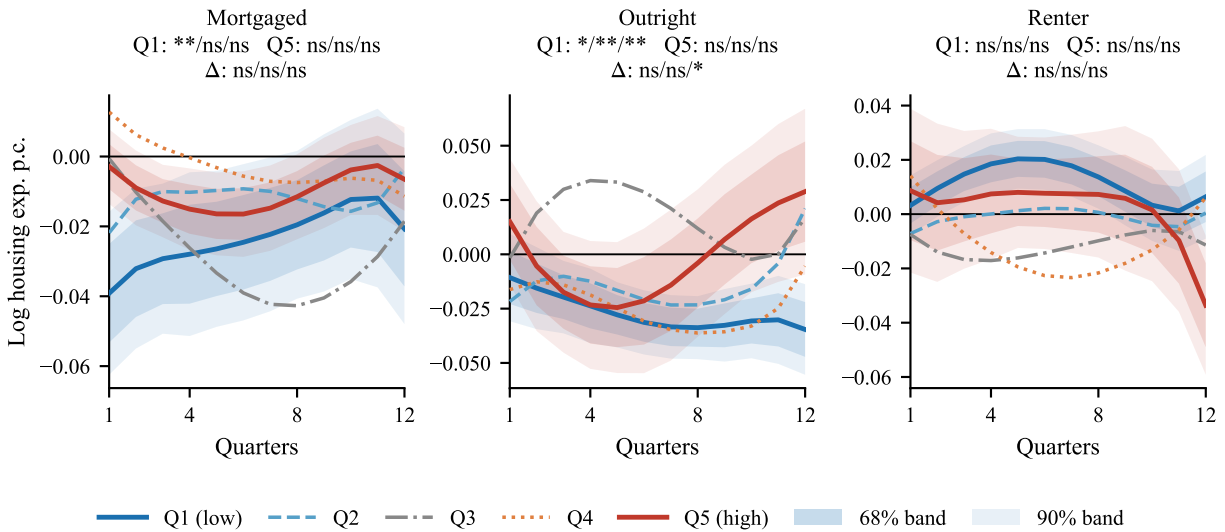


Figure E4: Smooth LP IRF: Log Housing Expenditure — LSAP Shock, by Tenure

E.5 Food Share by Tenure Type

Figure E5 reports the food expenditure ratio response to LSAP by tenure type, complementing the regional food-share evidence discussed in Section 4.1.

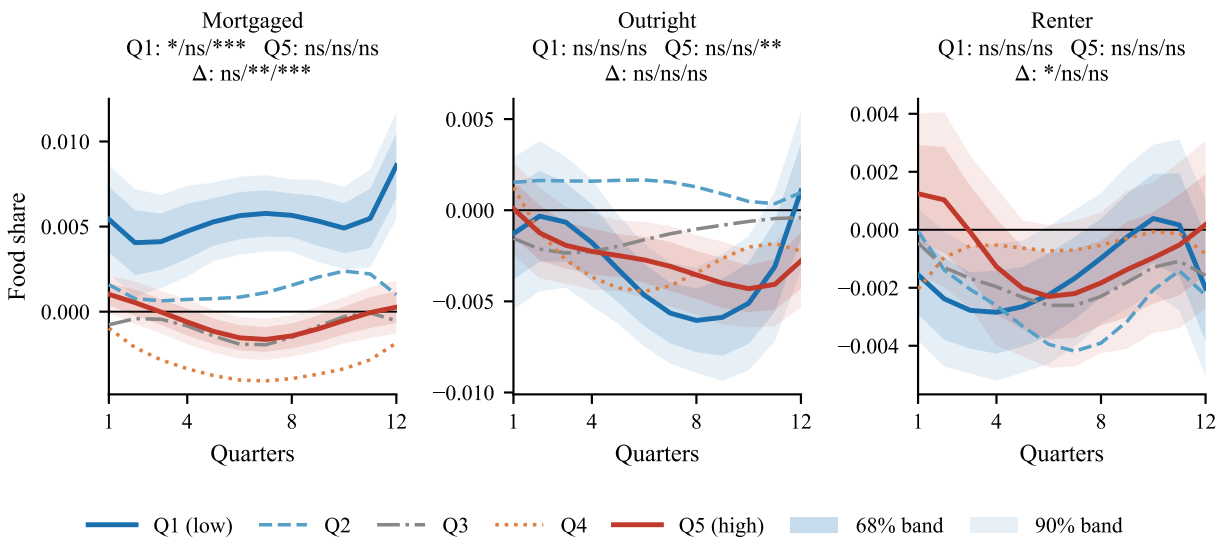


Figure E5: Smooth LP IRF: Food Share — LSAP Shock, by Tenure

E.6 Adjacent Gap Figures — Tenure Estimation

This subsection reports adjacent income gap figures for the tenure estimation, paralleling the fan charts above. For non-housing consumption under LSAP, the main text uses Standard LP for display (Figure 4) because the Smooth LP fit is unreliable for this shock in the tenure estimation (Appendix C); the corresponding gap figure below uses the same Standard LP specification for consistency. The remaining gap figures use Smooth LP, matching the fan charts reported earlier in this appendix.

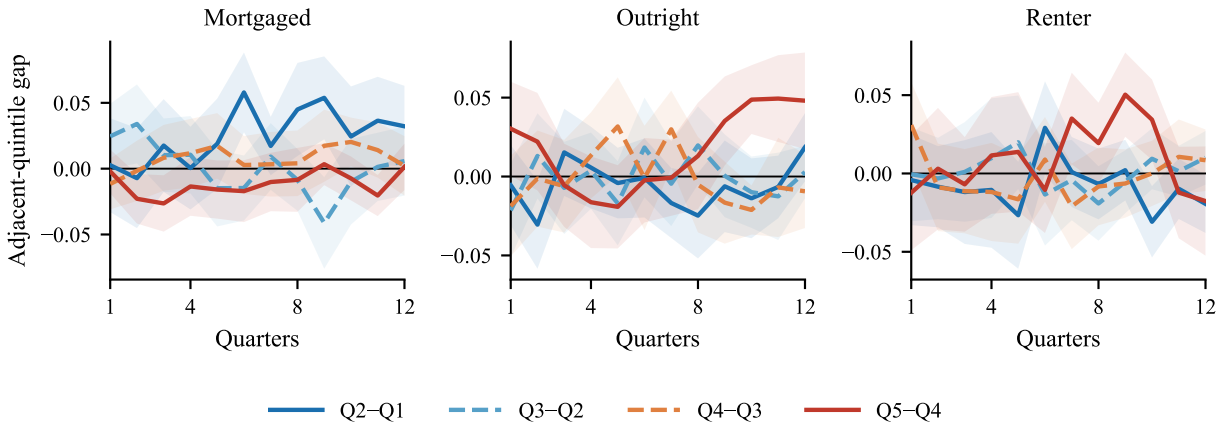


Figure E6: Standard LP Adjacent Gap: Log Non-Housing Consumption — LSAP Shock, by Tenure

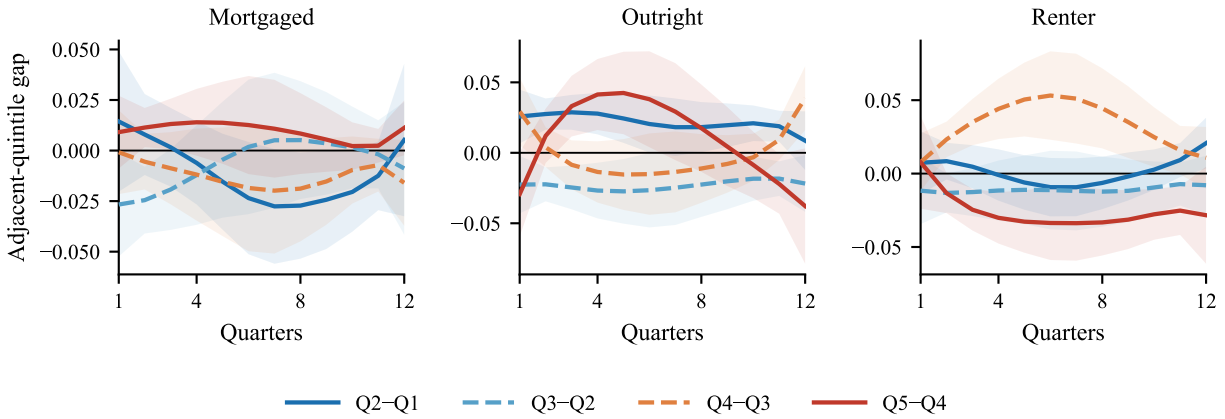


Figure E7: Smooth LP Adjacent Gap: Log Non-Housing Consumption — FFR Shock, by Tenure

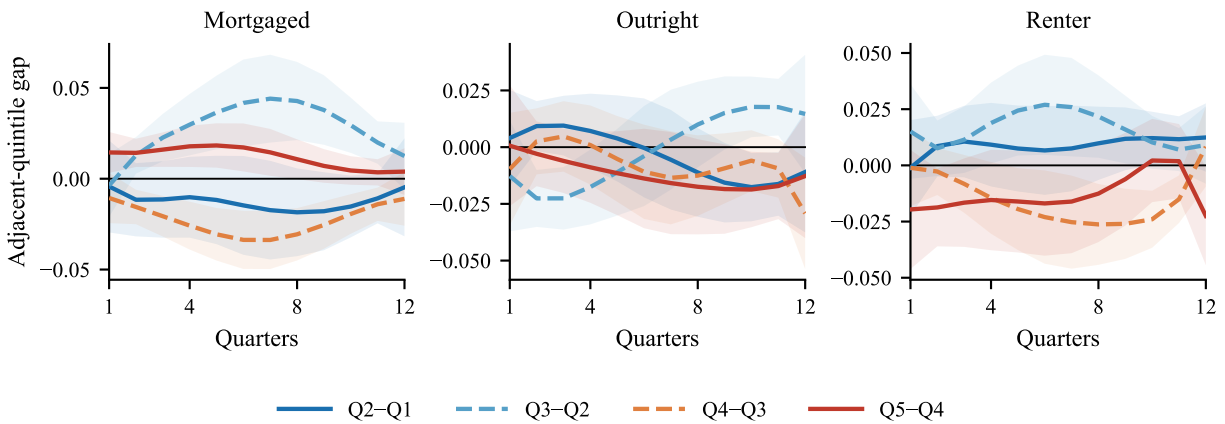


Figure E8: Smooth LP Adjacent Gap: Log Non-Housing Consumption — FG Shock, by Tenure

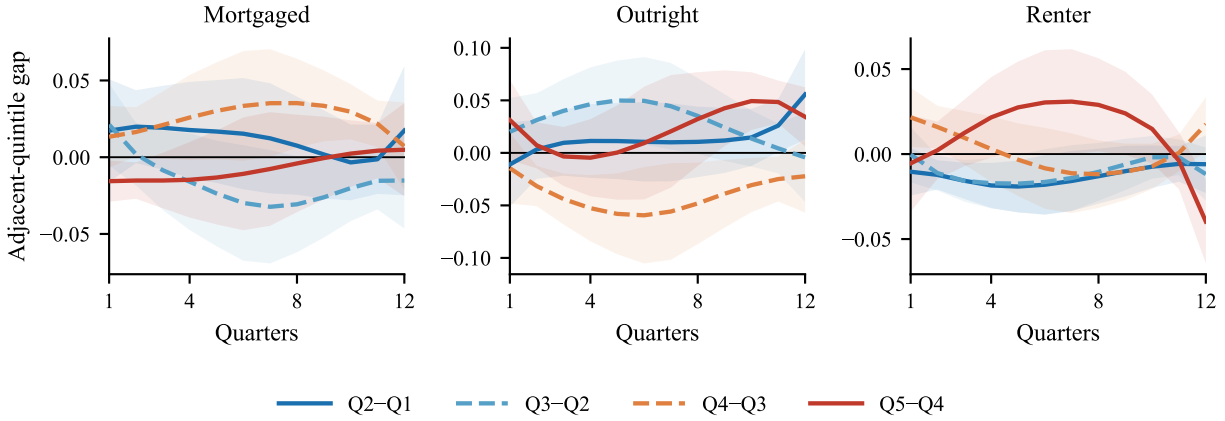


Figure E9: Smooth LP Adjacent Gap: Log Housing Expenditure — LSAP Shock, by Tenure

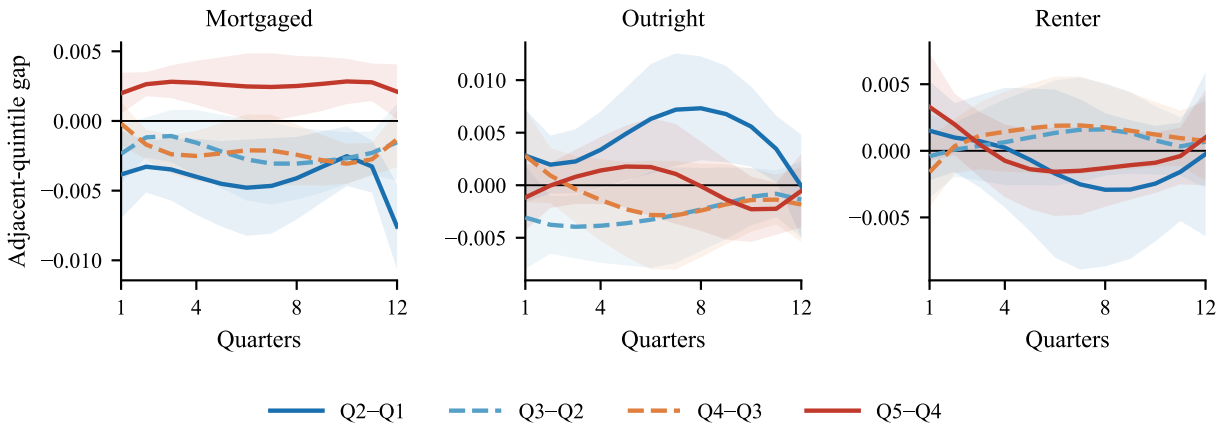


Figure E10: Smooth LP Adjacent Gap: Food Share — LSAP Shock, by Tenure

E.7 Wald Significance and Joint Tests — Tenure Estimation

Tables E3–E5 report individual-quintile Wald significance for each tenure $\times$ income cell, separately by shock. Table E6 reports the joint $Q1$ – $Q5$ difference tests underlying the mechanism discussion in Section 5.2, covering the significance levels for mortgagors, outright owners, and renters across the three shocks and each consumption variable. Table E8 tests whether the renter-minus-mortgagor difference within the same income group is statistically distinguishable across pairs of instruments, providing a direct test of the reversal in direction described in Section 5.2.

Table E3: Smooth LP Wald Test — Tenure Estimation, FFR Shock: Housing Tenure $\times$ Income Quintile (Mechanism)

Outcome	Quintile	Mortgaged	Outright	Renter
Non-Housing Cons.	Q1	(0.10)/(0.62)	(0.31)/(0.64)	***/*
	Q2	**	***	*
	Q3	(0.84)	***	***
	Q4	**	**	(0.79)
	Q5	(0.77)/(0.86)	(0.29)/(0.46)	**/(0.55)
Housing Expenditure	Q1	***/*	***/**	(0.96)/(0.98)
	Q2	(0.71)	***	**
	Q3	***	***	***
	Q4	(0.70)	***	***
	Q5	(0.86)/(0.71)	**/(0.17)	***/(0.20)
Food Share	Q1	(0.22)/(0.36)	*/(0.23)	***/**
	Q2	***	(0.23)	***
	Q3	(0.77)	***	***
	Q4	***	***	***
	Q5	***/**	**/(0.19)	(0.29)/(0.54)

Notes: cells show HAC significance for Q1–Q5 (Bootstrap significance additionally shown for Q1/Q5, as HAC/Bootstrap). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show exact p -value. HAC: Newey–West SE ($bw = h + 2$), diagonal $\chi^2(12)$ joint test, $(p_s, p_y, p_x) = (2, 3, 1)$. Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$; not computed for Q2–Q4. Horizons $h = 1, \dots, 12$. Inference (both HAC and Bootstrap) is computed from the Standard LP specification, consistent with the paper’s convention that Smooth LP is used for visualization only, not inference.

Table E4: Smooth LP Wald Test — Tenure Estimation, FG Shock: Housing Tenure $\times$ Income Quintile (Mechanism)

Outcome	Quintile	Mortgaged	Outright	Renter
Non-Housing Cons.	Q1	(0.71)/(0.82)	(0.71)/(0.82)	*/(0.23)
	Q2	(0.13)	(0.14)	(0.15)
	Q3	(0.14)	(0.11)	***
	Q4	***	***	(0.16)
	Q5	(0.95)/(0.99)	*/(0.15)	(0.15)/(0.42)
Housing Expenditure	Q1	***/*	(0.70)/(0.86)	*/(0.30)
	Q2	(0.48)	***	(0.89)
	Q3	(0.77)	(0.83)	(0.88)
	Q4	(0.77)	(0.45)	***
	Q5	(0.81)/(0.76)	(0.50)/(0.64)	(0.41)/(0.38)
Food Share	Q1	*/(0.23)	(0.70)/(0.85)	(0.11)/(0.49)
	Q2	(0.97)	(0.19)	(0.62)
	Q3	(0.88)	(0.48)	*
	Q4	(0.21)	***	(0.66)
	Q5	(0.81)/(0.83)	*/(0.26)	*/(0.33)

Notes: cells show HAC significance for Q1–Q5 (Bootstrap significance additionally shown for Q1/Q5, as HAC/Bootstrap). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show exact p -value. HAC: Newey–West SE ($bw = h + 2$), diagonal $\chi^2(12)$ joint test, $(p_s, p_y, p_x) = (2, 3, 1)$. Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$; not computed for Q2–Q4. Horizons $h = 1, \dots, 12$. Inference (both HAC and Bootstrap) is computed from the Standard LP specification, consistent with the paper’s convention that Smooth LP is used for visualization only, not inference.

Table E5: Smooth LP Wald Test — Tenure Estimation, LSAP Shock: Housing Tenure $\times$ Income Quintile (Mechanism)

Outcome	Quintile	Mortgaged	Outright	Renter
Non-Housing Cons.	Q1	***/*	**/(0.35)	(0.13)/(0.39)
	Q2	(0.59)	*	*
	Q3	(0.45)	(0.72)	(0.56)
	Q4	(0.12)	**	**
	Q5	(0.83)/(0.56)	**/(0.27)	(0.18)/(0.51)
Housing Expenditure	Q1	***/(0.22)	***/**	*/(0.42)
	Q2	(0.64)	(0.32)	(0.84)
	Q3	***	(0.14)	(0.10)
	Q4	(0.79)	**	(0.13)
	Q5	(0.83)/(0.55)	(0.18)/(0.54)	(0.13)/(0.60)
Food Share	Q1	***/**	(0.11)/(0.36)	(0.76)/(0.34)
	Q2	(0.11)	(0.49)	***
	Q3	(0.14)	(0.67)	(0.11)
	Q4	***	*	*
	Q5	*/(0.33)	***/(0.11)	(0.56)/(0.81)

Notes: cells show HAC significance for Q1–Q5 (Bootstrap significance additionally shown for Q1/Q5, as HAC/Bootstrap). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show exact p -value. HAC: Newey–West SE ($bw = h + 2$), diagonal $\chi^2(12)$ joint test, $(p_s, p_y, p_x) = (2, 3, 1)$. Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$; not computed for Q2–Q4. Horizons $h = 1, \dots, 12$. Inference (both HAC and Bootstrap) is computed from the Standard LP specification, consistent with the paper’s convention that Smooth LP is used for visualization only, not inference.

Table E6: Tenure Estimation: $Q1$ – $Q5$ Tests within Tenure Groups by Horizon Window — Housing Tenure $\times$ Income Quintile

Outcome	Shock	Horizon	Mortgaged	Outright	Renter
Non-hous.	FFR	1–4	(0.56)/(0.65)↓	(0.15)/(0.19)↑	(0.97)/(0.97)↓
		5–8	(0.25)/(0.41)↓	(0.17)/(0.34)↓	(0.98)/(0.99)↓
		9–12	*/(0.47)↓	(0.69)/(0.71)↓	(0.25)/(0.57)↓
	FG	1–4	(0.45)/(0.41)↑	(0.66)/(0.61)↓	(0.95)/(0.93)↓
		5–8	(0.75)/(0.86)↑	(0.11)/(0.19)↓	(0.55)/(0.58)↓
		9–12	(0.74)/(0.90)↑	*/(0.15)↓	(0.28)/(0.47)↑
	LSAP	1–4	(0.93)/(0.94)↑	(0.87)/(0.89)↓	(0.67)/(0.58)↓
		5–8	(0.19)/(0.43)↑	(0.74)/(0.85)↑	(0.75)/(0.76)↑
		9–12	***/*↑	***/(0.20)↑	*/(0.27)↑
Total	FFR	1–4	*/(0.20)↓	(0.62)/(0.58)↑	(0.96)/(0.95)↓
		5–8	(0.39)/(0.50)↓	(0.33)/(0.43)↑	(0.99)/(0.99)↓
		9–12	*/(0.19)↓	(0.42)/(0.47)↓	(0.18)/(0.43)↓
	FG	1–4	(0.28)/(0.21)↑	(0.62)/(0.56)↓	(0.98)/(0.97)↓
		5–8	(0.48)/(0.61)↑	(0.13)/(0.21)↓	(0.81)/(0.80)↓
		9–12	(0.30)/(0.62)↑	*/(0.19)↓	(0.22)/(0.35)↑
	LSAP	1–4	(0.41)/(0.50)↑	(0.72)/(0.75)↑	(0.70)/(0.67)↓
		5–8	(0.34)/(0.42)↑	(0.85)/(0.90)↑	(0.94)/(0.95)↓
		9–12	**/*↑	***/*↑	*/(0.27)↑
Food sh.	FFR	1–4	(0.10)/(0.14)↑	(0.35)/(0.30)↓	***/*↓
		5–8	*/(0.11)↑	(0.69)/(0.51)↑	(0.54)/(0.28)↓
		9–12	***/*↑	(0.54)/(0.56)↑	***↓
	FG	1–4	(0.89)/(0.88)↑	(0.70)/(0.58)↑	(0.31)/(0.17)↓
		5–8	(0.14)/(0.23)↑	(0.66)/(0.69)↑	(0.82)/(0.81)↓
		9–12	(0.35)/(0.55)↑	(0.56)/(0.36)↑	(0.29)/(0.30)↓
	LSAP	1–4	*/(0.17)↓	(0.57)/(0.47)↓	(0.22)/*↑
		5–8	***/*↓	(0.58)/(0.56)↑	(0.84)/(0.65)↓
		9–12	***/*↓	(0.57)/(0.49)↓	(0.78)/(0.46)↓
Housing	FFR	1–4	***/*↓	(0.48)/(0.41)↓	(0.13)/(0.23)↓
		5–8	(0.70)/(0.72)↓	(0.13)/(0.21)↑	*/(0.19)↓
		9–12	(0.63)/(0.68)↓	(0.59)/(0.48)↓	(0.46)/(0.55)↓
	FG	1–4	*//*↑	(0.24)/(0.28)↑	(0.89)/(0.80)↑
		5–8	***/*↑	(0.91)/(0.89)↓	(0.78)/(0.65)↑
		9–12	(0.34)/(0.20)↑	(0.61)/(0.66)↓	(0.11)/(0.13)↑
	LSAP	1–4	*/(0.11)↑	(0.12)/(0.23)↑	(0.93)/(0.96)↓
		5–8	(0.88)/(0.85)↑	(0.67)/(0.63)↑	(0.79)/(0.84)↓
		9–12	(0.93)/(0.91)↑	***/*↑	*/(0.22)↓

Cells show HAC/Bootstrap significance for the joint $Q1$ – $Q5$ difference test restricted to the stated window ($H_0: \beta_{Q1}^h = \beta_{Q5}^h$ for all h in the window; $(p_s, p_y, p_x) = (2, 3, 1)$; residual moving block bootstrap, $B = 5000$, $L = 12$), followed by the direction of $\delta^h \equiv \beta_{Q5}^h - \beta_{Q1}^h$ averaged over that window. $\uparrow$ denotes $\delta > 0$, i.e. $Q1$'s cumulative response lies below $Q5$'s. All three shock series are oriented so that a positive value is a monetary tightening, so the arrow has the same meaning for every instrument and no instrument-specific sign adjustment is applied. For the three consumption measures $\uparrow$ is the progressive case ($Q1$ loses more under a contraction); for the food expenditure share the reading is reversed, since a smaller rise in $Q1$'s food share indicates $Q1$ is relatively better off.

The test is reported by horizon window rather than as a single test over $h = 1, \dots, 12$ because the latter averages over the timing of the response: a distributional effect confined to one year is diluted by the eight horizons in which nothing happens. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show the exact p -value. The Mortgaged-vs-Renter test within a fixed income group is reported in Table E7.

Table E7: Mortgaged versus Renter Households within the Same Income Group, by Horizon Window — Tenure Estimation

Outcome	Shock	Window	Q1	Q2–Q4 ^a	Q5
Non-Housing Cons.	FFR	$h = 1-4$	(0.28)/(0.29)↓	*/**↓	(0.76)/(0.76)↓
		$h = 5-8$	*/(0.19)↓	**/**↓	(0.88)/(0.92)↓
		$h = 9-12$	***/**↓	(0.16)/(0.32)↓	*/(0.41)↓
	FG	$h = 1-4$	(0.60)/(0.46)↓	*/(0.39)↑	(0.33)/(0.26)↓
		$h = 5-8$	(0.75)/(0.76)↓	*/(0.36)↑	(0.50)/(0.55)↓
		$h = 9-12$	(0.62)/(0.77)↓	(0.15)/(0.49)↑	(0.37)/(0.56)↑
	LSAP	$h = 1-4$	(0.16)/(0.13)↑	(0.15)/(0.23)↑	(0.66)/(0.63)↑
		$h = 5-8$	**/(0.11)↑	(0.44)/(0.37)↓	(0.87)/(0.86)↑
		$h = 9-12$	***/**↑	(0.88)/(1.00)↓	*/(0.30)↑
Food Share	FFR	$h = 1-4$	**/**↑	(0.41)/(0.59)↑	*/(0.19)↓
		$h = 5-8$	(0.14)/(0.15)↑	(0.14)/(0.13)↑	*/**↓
		$h = 9-12$	***/***↑	(0.35)/(0.24)↑	(0.78)/(0.60)↓
	FG	$h = 1-4$	(0.11)/(0.13)↑	(1.00)/(1.00)↓	(0.93)/(0.93)↑
		$h = 5-8$	(0.36)/(0.46)↑	(1.00)/(1.00)↓	(0.67)/(0.82)↓
		$h = 9-12$	**/(0.20)↑	(0.33)/(0.76)↓	(0.12)/(0.21)↓
	LSAP	$h = 1-4$	***/***↓	(0.10)/(0.31)↓	(0.69)/(0.72)↑
		$h = 5-8$	(0.36)/(0.19)↓	***/(0.13)↓	(0.58)/(0.69)↓
		$h = 9-12$	**/**↓	*/↓	(0.98)/(0.98)↓

Cells show HAC/Bootstrap significance for the joint Mortgaged-vs-Renter test within a fixed income group ($H_0: \beta_{Mort}^h = \beta_{Rent}^h$ for all h in the window), on the same bootstrap and the same common anchor design as Table E6, reported over the same three windows. ↑ denotes $\beta_{Mort}^h < \beta_{Rent}^h$ on average over the window.

*p<0.10, **p<0.05, ***p<0.01; parentheses show the exact p -value.

^a The Q2–Q4 column reports the smallest of the three separate per-quintile p -values, Bonferroni-corrected by a factor of three; the quintiles are not pooled into a single regression.

Table E8: Instrument Contrasts of the Renter–Mortgagor Difference within the Same Income Quintile: Log Non-Housing Consumption per Capita

Quintile	LSAP – FFR			LSAP – FG			FG – FFR		
	$h=1-4$	5–8	9–12	$h=1-4$	5–8	9–12	$h=1-4$	5–8	9–12
Q1	+4.50*	+4.97**	+5.84***	+3.70	+2.91	+4.02**	+0.79	+2.06	+1.82
Q2	+3.45**	+0.50	–0.16	+1.67	–1.62	–2.54	+1.78**	+2.13	+2.38
Q3	+1.19	+1.80	+1.15	+0.00	–0.63	–0.16	+1.19	+2.43**	+1.31
Q4	–0.65	–3.01*	–1.46	–1.13	–2.49**	–1.13	+0.48	–0.52	–0.33
Q5	+2.22	+1.33	+2.90	+2.96	+1.28	+1.09	–0.74	+0.04	+1.81

Notes: $d_s^h \equiv \beta_{Renter,s}^h - \beta_{Mortgaged,s}^h$ is the renter–mortgagor difference within the stated income quintile under instrument s , as in Table 3. Each cell reports the window average of $d_A^h - d_B^h$ in percentage points per unit contractionary shock, for the contrast labelled $A - B$. A positive value means that, relative to renters, mortgagor households' consumption falls more under instrument A than under instrument B . Stars refer to the bootstrap joint test of $H_0: d_A^h = d_B^h$ for all h in the window. Over $h = 1, \dots, 12$, the Q1 contrasts give $p = 0.002$ (LSAP – FFR), $p = 0.044$ (LSAP – FG) and $p = 0.504$ (FG – FFR); with each shock rescaled to one standard deviation, $p = 0.002, 0.084$ and 0.629 .

*p<0.10, **p<0.05, ***p<0.01. Residual moving block bootstrap ($B = 5000, L = 12$) with one block index per draw shared by all six regressions of a quintile (renter and mortgagor $\times 3$ instruments). $(p_s, p_y, p_x) = (2, 3, 1)$.

F Robustness Check Details

This appendix provides the full results for the robustness checks in Section 6 of the main text. Table 4 in the main text summarizes three quantities under each specification for the two results on which this paper makes its claims: the regional result (the Northeast, LSAP, non-housing consumption) and the mechanism result (mortgagor households, LSAP, non-housing consumption). These are the magnitude of the $Q1$ – $Q5$ gap, the joint-test p -values for the three windows, and the p -value of the gap stability test. Tables F1–F4 report the full-horizon joint tests of the $Q1$ – $Q5$ difference under each check, with each cell giving the Newey–West and bootstrap significance judgments side by side; Panel A reports by region and Panel B by housing tenure. Apart from the CPI deflation, each check covers all 84 cells estimated in this paper, comprising four outcome variables in each of the regional and tenure estimations, so every result reported in this paper has corresponding robustness evidence, including those in the mechanism section. The CPI deflation does not apply to the food expenditure ratio, since the same price index cancels out in the numerator and denominator of the ratio, leaving the estimates identical before and after deflation. The subsections below first describe the three layers of evidence on which the robustness judgments rest, then report the results for each check. The three layers of evidence do not carry equal weight: the comparison of point estimates provides the most direct evidence; the joint test within the subsample is affected by the loss of power resulting from the reduced sample; and the resolution of the stability test is comparable to the magnitude of the effect being tested (see the next section). These pieces of evidence should therefore be considered jointly rather than interpreted in isolation.

Basis for the robustness judgments. The conclusions from each check rest on three mutually independent quantities. The first is the joint test within the subsample, that is, re-testing $H_0 : \delta^h = 0$ under the alternative specification. This is the most direct measure, but a reduced sample also lowers the power of the test, so an increase in the p -value cannot distinguish between the effect disappearing and precision falling. The second is the magnitude of the point estimate of the gap: if a check raises the p -value while leaving the point estimate essentially unchanged, what has changed is precision rather than the effect itself. The third is a stability test for the gap, with the null hypothesis

$$H_0 : \delta_{\text{baseline}}^h = \delta_{\text{alternative}}^h \quad \text{for all } h = 1, \dots, 12,$$

that is, whether the gap has shifted between the two samples. This differs from the null of the first test, which asks whether the gap differs from zero. The gap $\bar{\delta}$ in Table 4, as in the other robustness tables in the appendix, is re-estimated using the compressed series after the relevant quarters are dropped. In contrast, the stability test constructs the subsample by deleting the corresponding anchor periods from the original series, so that impulse responses do not span the gaps in the sample. The two quantities are therefore constructed on different bases, which accounts for their

slight differences. How that difference becomes a null distribution depends on the check. For the two quarter-exclusion checks the alternative sample is nested in the baseline, so a single simulated series is fitted twice—on all anchors and on the retained anchors—and both fits estimate the same coefficients; the difference of the two is therefore already centered on zero and is used as the null distribution directly. For `INC_RANK` the alternative panel is a separate dataset with its own fitted values, so the difference is centered on the estimated difference and is recentered to obtain the null. For the two quarter-exclusion checks, the alternative sample is obtained by dropping the corresponding anchors from the original series, and impulse responses are not computed across the resulting sample gaps. For `INC_RANK`, the two panels cover the same quarters, so the same set of block indices is applied to both to preserve their common time structure.

The power limitations of this test should be stated alongside the results. Given the sample length used here, the test can detect a uniform shift of approximately 4.0 percentage points, while the gap in the Northeast under the baseline specification is itself approximately 3.3 percentage points. The resolution of the test is therefore comparable to the magnitude of the effect being examined. It can indicate whether the gap has shifted substantially, but it is not sufficiently powerful to rule out a moderate attenuation, or even to statistically rule out the possibility that the gap has disappeared entirely. Accordingly, whenever the text below states that the stability test does not reject, this should be interpreted as indicating that no shift large enough to be detected at this resolution was observed, rather than that the gap has genuinely remained unchanged. Of the three layers of evidence, this test should therefore be given less weight than the comparison of point estimates.

F.1 Alternative Income Classification (`INC_RANK`)

Table F1 reports the joint $Q1$ – $Q5$ test results by region after replacing the baseline within-region relative ranking with the Bureau of Labor Statistics’ `INC_RANK` variable, which gives the national income percentile. In terms of direction, the differences for the Northeast, Midwest, and South run in the same direction under both classifications; the difference for the West is close to zero under the alternative classification and carries the opposite sign from the baseline, but is insignificant under both. In terms of significance, the bootstrap-corrected p -value for the Northeast is 0.011, compared with 0.014 under the baseline; the remaining three regions are likewise insignificant, yielding exactly the same significance conclusions as under the baseline. The point estimate of the difference rises from 3.29 percentage points in the baseline to 4.05 percentage points under the alternative classification, while the stability test cannot reject the null that the difference paths are the same under the two classifications, meaning that switching to the national income percentile does not relocate the effect. The timing of the effect is also preserved: the first-year window is insignificant under both classifications ($p = 0.51$ under the alternative and $p = 0.52$ under the

baseline), while the second- and third-year windows are significant under both ($p = 0.010$ and $p = 0.023$ under the alternative, against $p = 0.004$ and $p = 0.064$ under the baseline). The core finding therefore depends neither on the income classification nor, in its timing, on which of the two is used.

Table F1: Q1–Q5 Joint Test: INC_RANK Income Classification

<i>Panel A: Regional estimation</i>					
Outcome	Shock	NE	MW	S	W
Non-Housing Cons.	FFR	(0.56)/(0.86)↑	(0.56)/(0.81)↑	***/*↓	(0.84)/(0.89)↑
	FG	(0.59)/(0.63)↓	(0.63)/(0.79)↓	(0.91)/(0.94)↑	(0.20)/(0.45)↓
	LSAP	***/**↑	(0.51)/(0.57)↓	(0.83)/(0.82)↓	(0.68)/(0.71)↑
Total Cons.	FFR	(0.75)/(0.88)↓	(0.39)/(0.70)↑	***/**↓	(0.95)/(0.95)↑
	FG	(0.40)/(0.54)↓	(0.73)/(0.80)↓	(0.93)/(0.95)↓	(0.86)/(0.89)↓
	LSAP	***/**↑	(0.44)/(0.54)↓	(0.78)/(0.73)↑	(0.58)/(0.68)↓
Food Share	FFR	(0.97)/(0.95)↑	(0.10)/(0.19)↓	(0.65)/(0.57)↓	(0.40)/(0.39)↑
	FG	(0.14)/(0.40)↑	(0.46)/(0.59)↑	(0.76)/(0.67)↓	(0.95)/(0.95)↑
	LSAP	***/(0.12)↓	(0.31)/(0.29)↑	(0.75)/(0.77)↑	**/(0.21)↓
Housing Exp.	FFR	***/**↓	**/(0.28)↓	***/**↓	(0.77)/(0.77)↑
	FG	**/(0.17)↑	(0.60)/(0.58)↓	(0.40)/(0.59)↓	*/(0.47)↑
	LSAP	***/(0.12)↑	(0.64)/(0.61)↓	***/**↑	(0.52)/(0.62)↓
<i>Panel B: Tenure estimation</i>					
Outcome	Shock	Mortgaged	Outright	Renter	
Non-Housing Cons.	FFR	***/(0.16)↓	(0.21)/(0.46)↓	(0.78)/(0.94)↑	
	FG	(0.61)/(0.80)↑	***/**↓	(0.40)/(0.54)↓	
	LSAP	**/(0.28)↑	**/(0.24)↑	(0.38)/(0.59)↑	
Total Cons.	FFR	***/**↓	(0.30)/(0.42)↓	(0.67)/(0.88)↓	
	FG	(0.28)/(0.49)↑	***/*↓	(0.48)/(0.53)↓	
	LSAP	**/(0.22)↑	***/*↑	(0.33)/(0.68)↑	
Food Share	FFR	***/**↑	(0.77)/(0.62)↓	**/**↓	
	FG	(0.43)/(0.58)↑	(0.17)/(0.20)↑	(0.23)/(0.18)↓	
	LSAP	***/**↓	(0.70)/(0.61)↓	(0.61)/(0.24)↑	
Housing Exp.	FFR	***/*↓	(0.17)/(0.22)↓	***/(0.13)↓	
	FG	(0.15)/(0.12)↑	(0.88)/(0.83)↓	(0.55)/(0.55)↑	
	LSAP	(0.34)/(0.40)↑	***/*↑	(0.89)/(0.97)↑	

Notes: cells show HAC/Bootstrap significance for the joint $Q1$ – $Q5$ difference test over $h = 1, \dots, 12$, followed by the direction of δ^h averaged over the twelve horizons (↑: $\delta > 0$, i.e. $Q1$'s cumulative response lies below $Q5$'s); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A: regional estimation, one column per Census region. Panel B: tenure estimation, one column per housing tenure group. HAC: Newey–West SE, diagonal $\chi^2(12)$. Bootstrap: residual moving block bootstrap, $B = 2000$, $L = 12$. $(p_s, p_y, p_x) = (2, 3, 1)$. Income quintiles are taken from the BLS INC_RANK variable (percentile in the national income distribution) rather than from the within-region relative ranking. Every household in the estimation sample has an INC_RANK record, so no fallback assignment is needed: the households that lacked one were the incomplete income reporters that the income-variable correction removes. The Northeast result is almost identical to the baseline; the other three regions are insignificant under both classifications.

F.2 Excluding NBER Recession Quarters

Table F2 reports the joint $Q1$ – $Q5$ test results after excluding the 2001 recession (2001Q1–2001Q4) and the 2007–2009 recession (2007Q4–2009Q2), reducing the anchor periods available for estimation from 96 to 85. This is the only check in which the joint test no longer holds at any horizon: the bootstrap p -value for the Northeast rises from 0.014 to 0.421 over the full horizon.

The point estimate of the difference falls from 3.29 percentage points in the baseline to 2.29 percentage points, while the standard error rises by approximately 70% over the same comparison. The stability test cannot reject the null that the difference paths in the two samples are the same. Across all 84 cells, this test rejects 18 times for this exclusion and 10 times when extreme shock quarters are excluded, indicating that the failure to reject here is not simply a consequence of low power; at this resolution, however, it also cannot rule out some attenuation of the magnitude observed in this case. By window, all three tests fall well short of the 10% threshold. The full-horizon test must use one degree of freedom for each of the 12 horizons, and the estimates indicate that the first four carry almost no signal, so this cost is difficult to absorb once the sample is reduced. The problem goes further: even when considering only the second-year window, in which the effect actually emerges, the reduced sample is far from sufficient to detect it. Excluding recession quarters does not merely reduce precision; it removes from the sample the variation on which identification of this effect depends, since the excluded quarters are precisely those in which the unconventional instruments were used on a large scale.

After excluding recession quarters, the point estimate of the distributional effect falls, but the stability test cannot reject equality with the baseline; the change lies in whether the effect can still be distinguished statistically.

Table F2: Q1–Q5 Joint Test: Excluding Recession Quarters

<i>Panel A: Regional estimation</i>					
Outcome	Shock	NE	MW	S	W
Non-Housing Cons.	FFR	***/(0.19)↑	***/**↑	(0.62)/(0.54)↑	***/*↑
	FG	(0.92)/(0.84)↓	(0.28)/(0.33)↑	(0.11)/(0.15)↑	(0.67)/(0.65)↓
	LSAP	(0.50)/(0.42)↑	**/(0.12)↓	(0.91)/(0.87)↓	**/(0.11)↓
Total Cons.	FFR	(0.10)/(0.25)↑	***/**↑	(0.65)/(0.52)↓	***/**↑
	FG	(0.95)/(0.89)↑	(0.62)/(0.54)↑	(0.20)/(0.18)↑	(0.81)/(0.76)↑
	LSAP	(0.81)/(0.76)↑	***/*↓	(0.78)/(0.59)↓	**/(0.14)↓
Food Share	FFR	(0.83)/(0.74)↓	(0.11)/(0.36)↑	(0.29)/(0.37)↑	(0.52)/(0.59)↑
	FG	***/(0.12)↑	(0.62)/(0.68)↑	(1.00)/(0.98)↑	(0.92)/(0.91)↓
	LSAP	*/(0.25)↓	(0.19)/(0.33)↑	*/(0.28)↓	(0.25)/(0.40)↓
Housing Exp.	FFR	(0.49)/(0.29)↑	**/(0.25)↑	(0.50)/(0.51)↓	***/**↑
	FG	(0.46)/(0.48)↑	(0.38)/(0.31)↑	(0.58)/(0.55)↓	(0.68)/(0.72)↑
	LSAP	(0.15)/(0.36)↓	***/**↓	(0.18)/(0.20)↑	**/(0.23)↓
<i>Panel B: Tenure estimation</i>					
Outcome	Shock	Mortgaged	Outright	Renter	
Non-Housing Cons.	FFR	***/(0.27)↑	***/**↑	(0.86)/(0.82)↓	
	FG	(0.27)/(0.55)↓	(0.51)/(0.60)↓	**/**↓	
	LSAP	(0.18)/(0.64)↑	(0.84)/(0.74)↓	(0.69)/(0.69)↑	
Total Cons.	FFR	***/(0.18)↑	***/**↑	(0.43)/(0.36)↓	
	FG	(0.20)/(0.50)↑	(0.80)/(0.75)↑	*/(0.13)↓	
	LSAP	(0.32)/(0.59)↑	(0.86)/(0.68)↓	(0.91)/(0.86)↓	
Food Share	FFR	(0.25)/(0.55)↑	***/*↓	(0.85)/(0.73)↓	
	FG	(0.24)/(0.51)↑	(0.53)/(0.62)↑	(0.83)/(0.74)↑	
	LSAP	**/(0.35)↓	*/(0.21)↑	*/*↓	
Housing Exp.	FFR	*/(0.19)↑	***/**↑	***/**↓	
	FG	(0.18)/(0.29)↑	(0.33)/(0.36)↑	(0.90)/(0.85)↑	
	LSAP	(0.80)/(0.68)↓	*/(0.22)↓	(0.60)/(0.64)↓	

Notes: cells show HAC/Bootstrap significance for the joint Q1–Q5 difference test over $h = 1, \dots, 12$, followed by the direction of δ^h averaged over the twelve horizons (↑: $\delta > 0$, i.e. Q1's cumulative response lies below Q5's); *p<0.10, **p<0.05, ***p<0.01. Panel A: regional estimation, one column per Census region. Panel B: tenure estimation, one column per housing tenure group. HAC: Newey–West SE, diagonal $\chi^2(12)$. Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$. $(p_s, p_y, p_x) = (2, 3, 1)$. Drops 2001Q1–Q4 and 2007Q4–2009Q2 (96→85 anchor periods).

F.3 Excluding Quarters with Extreme Shocks

Table F3 reports the joint $Q1$ – $Q5$ test results after excluding the 15 quarters in which any shock series exceeds two standard deviations in absolute value, reducing the anchor periods available for estimation to 81. The bootstrap p -value for the Northeast rises from 0.014 to 0.083, weaker but still significant at the 10% level; the third-year window is 0.042 and the second-year window 0.087.

Unlike the exclusion of recession quarters, this exclusion does not shrink the point estimate. The $Q1$ – $Q5$ gap per unit shock rises from 3.29 percentage points in the baseline to 4.39 percentage points, and the stability test again cannot reject that the two are the same ($p = 0.372$). The rise in the p -value has a clear mechanical explanation: the excluded quarters are precisely those with the largest shock variance, concentrated mainly around 2001 and during the 2007–2009 financial crisis. Because observations with greater regressor variance provide more information for identifying the coefficient, removing these observations increases the standard errors by approximately 100%.

This check therefore does not show that the distributional effect varies with shock size; it shows that identifying the effect requires a sample that contains policy shocks of sufficient magnitude.

Table F3: Q1–Q5 Joint Test: Excluding Extreme Shock Quarters

<i>Panel A: Regional estimation</i>					
Outcome	Shock	NE	MW	S	W
Non-Housing Cons.	FFR	(0.42)/(0.50)↑	***/**↑	(0.15)/(0.14)↑	(0.19)/(0.46)↑
	FG	(0.47)/(0.33)↑	(0.82)/(0.79)↓	(0.63)/(0.43)↑	(0.32)/(0.26)↓
	LSAP	**/*↑	(0.58)/(0.59)↑	(0.62)/(0.59)↑	(0.30)/(0.39)↑
Total Cons.	FFR	(0.73)/(0.58)↑	***/*↑	(0.19)/*↓	**/(0.24)↑
	FG	(0.93)/(0.81)↓	(0.99)/(0.98)↓	(0.72)/(0.41)↑	(0.95)/(0.79)↓
	LSAP	(0.24)/(0.24)↑	(0.72)/(0.60)↑	(0.80)/(0.67)↑	(0.33)/(0.38)↑
Food Share	FFR	*/(0.12)↓	(0.37)/(0.53)↑	**/(0.19)↓	*/(0.24)↑
	FG	(0.46)/(0.54)↑	(0.12)/(0.15)↑	(0.13)/*↑	(0.56)/(0.29)↑
	LSAP	(0.19)/(0.48)↓	(0.96)/(0.95)↑	(0.44)/(0.54)↑	(0.78)/(0.75)↓
Housing Exp.	FFR	(0.47)/(0.32)↓	(0.23)/(0.32)↑	**/**↓	***/*↑
	FG	(0.92)/(0.83)↓	(0.51)/(0.24)↑	(0.94)/(0.75)↑	(0.86)/(0.61)↑
	LSAP	(0.38)/(0.34)↓	(0.99)/(0.90)↑	(0.42)/(0.22)↓	(0.75)/(0.70)↑
<i>Panel B: Tenure estimation</i>					
Outcome	Shock	Mortgaged	Outright	Renter	
Non-Housing Cons.	FFR	**/(0.16)↑	(0.19)/(0.37)↑	(0.18)/(0.11)↑	
	FG	(0.20)/(0.28)↑	(0.60)/(0.47)↓	(0.53)/(0.44)↓	
	LSAP	(0.68)/(0.80)↑	**/(0.22)↑	(0.24)/(0.19)↑	
Total Cons.	FFR	(0.17)/(0.24)↑	(0.45)/(0.48)↑	(0.54)/(0.17)↑	
	FG	(0.31)/(0.17)↑	(0.87)/(0.67)↓	(0.91)/(0.77)↓	
	LSAP	(0.85)/(0.81)↑	(0.37)/(0.41)↑	(0.63)/(0.50)↑	
Food Share	FFR	(0.59)/(0.59)↑	(0.84)/(0.78)↓	(0.55)/(0.28)↓	
	FG	(0.84)/(0.71)↑	(0.55)/(0.37)↓	(0.47)/(0.37)↑	
	LSAP	(0.59)/(0.72)↓	**/(0.16)↓	(0.80)/(0.58)↓	
Housing Exp.	FFR	(0.66)/(0.60)↑	(0.89)/(0.86)↓	(0.97)/(0.73)↓	
	FG	(0.16)/**↑	(0.69)/(0.64)↓	(1.00)/(0.98)↑	
	LSAP	(0.21)/(0.15)↑	(0.90)/(0.81)↑	(0.82)/(0.71)↓	

Notes: cells show HAC/Bootstrap significance for the joint $Q1$ – $Q5$ difference test over $h = 1, \dots, 12$, followed by the direction of δ^h averaged over the twelve horizons (↑: $\delta > 0$, i.e. $Q1$'s cumulative response lies below $Q5$'s); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A: regional estimation, one column per Census region. Panel B: tenure estimation, one column per housing tenure group. HAC: Newey–West SE, diagonal $\chi^2(12)$. Bootstrap: residual moving block bootstrap, $B = 5000$, $L = 12$. $(p_s, p_y, p_x) = (2, 3, 1)$. Drops the 15 quarters in which any shock exceeds two standard deviations in absolute value (96→81 anchor periods).

F.4 CPI-Deflated Consumption

Table F4 reports the joint $Q1$ – $Q5$ test results after deflating non-housing consumption, total consumption, and housing expenditure to real values using CPIAUCSL (FRED, monthly data averaged to quarterly). If the regressions for the two quintiles used exactly the same controls, a common price deflator would cancel out in the income difference. The specification includes each cell’s own lagged dependent variable, however, so deflation does not affect the estimates for the two quintiles identically. This check is therefore not mechanical; rather, it directly tests whether the distributional effect could be driven by price factors. The results bear this out: the bootstrap p -value for the Northeast under the LSAP shock moves from 0.0140 in nominal terms to 0.0132 in real terms, and total consumption from 0.0120 to 0.0116; the conclusion for the South under the FFR shock is likewise unchanged (total consumption 0.040 nominal against 0.037 real). It should be noted that the assumption of symmetric price movements across income groups is not uncontested. Cravino et al. (2020) find that price stickiness varies across the income distribution, implying that the effective price indices faced by different income groups respond differently to monetary policy. This paper uses only the national consumer price index and therefore cannot construct income-group-specific price indices. Consequently, the influence of this channel on the results cannot be ruled out.

Table F4: Q1–Q5 Joint Test: CPI-Deflated (Real) Consumption

<i>Panel A: Regional estimation</i>					
Outcome	Shock	NE	MW	S	W
Non-Housing Cons. (real)	FFR	(0.52)/(0.80)	*/(0.30)	*/(0.11)	(0.16)/(0.46)
	FG	(0.36)/(0.54)	(0.89)/(0.91)	(0.32)/(0.58)	(0.29)/(0.47)
	LSAP	***/**	(0.45)/(0.49)	(0.62)/(0.61)	(0.23)/(0.35)
Total Cons. (real)	FFR	(0.57)/(0.68)	(0.12)/(0.45)	***/**	(0.21)/(0.41)
	FG	(0.70)/(0.75)	(0.97)/(0.96)	(0.56)/(0.76)	(0.98)/(0.98)
	LSAP	***/**	(0.14)/(0.31)	(0.85)/(0.74)	*/(0.24)
Housing Exp. (real)	FFR	***/**	(0.13)/(0.43)	***/**	(0.31)/(0.44)
	FG	(0.13)/(0.24)	(0.88)/(0.78)	*/(0.17)	*/(0.28)
	LSAP	***/(0.16)	(0.16)/(0.29)	***/**	*/(0.38)
<i>Panel B: Tenure estimation</i>					
Outcome	Shock	Mortgaged	Outright	Renter	
Non-Housing Cons. (real)	FFR	(0.13)/(0.60)	(0.22)/(0.42)	(0.90)/(0.96)	
	FG	(0.79)/(0.92)	*/(0.18)	(0.72)/(0.75)	
	LSAP	*/(0.25)	(0.11)/(0.48)	(0.37)/(0.51)	
Total Cons. (real)	FFR	***/(0.22)	(0.49)/(0.52)	(0.86)/(0.92)	
	FG	(0.25)/(0.58)	*/(0.22)	(0.82)/(0.77)	
	LSAP	*/(0.25)	*/(0.27)	(0.41)/(0.61)	
Housing Exp. (real)	FFR	*/(0.19)	(0.33)/(0.35)	*/(0.29)	
	FG	***/**	(0.70)/(0.69)	(0.61)/(0.37)	
	LSAP	(0.55)/(0.59)	***/(0.14)	(0.26)/(0.60)	

Notes: cells show HAC/Bootstrap significance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; parentheses show exact p -value) for the joint Q1–Q5 difference test on CPI-deflated outcomes, over all twelve horizons. Outcomes deflated by CPIAUCSL (FRED, monthly $\rightarrow$ quarterly average, log difference from sample-start base). Panel A: regional estimation, one column per Census region. Panel B: tenure estimation, one column per housing tenure group. The food share is not reported: a common deflator cancels from the ratio, so its estimates are unchanged. $(p_s, p_y, p_x) = (2, 3, 1)$, residual moving block bootstrap $B = 5000$, $L = 12$. A common deflator would cancel from the gap only if both quintiles were projected on identical controls; because the specification includes each cell's own lagged outcomes, deflation moves the two estimates slightly differently, so the check is informative rather than mechanical. Northeast/LSAP is essentially unchanged (bootstrap p -value 0.0132 real vs. 0.0140 nominal for non-housing; 0.0116 real vs. 0.0120 nominal for total). South/FFR total consumption is likewise unchanged (bootstrap p -value 0.037 real vs. 0.040 nominal); South/FFR non-housing consumption does not pass in either version. Housing-expenditure cells reported in the main text (bootstrap p -value for the window cited there): South/FFR, third-year window 0.002 real vs. 0.002 nominal; Northeast/FFR, first-year window 0.006 real vs. 0.005 nominal; South/LSAP, full horizon 0.027 real vs. 0.028 nominal.